

60. MRAM

This MCU incorporates code MRAM and option-setting memory. The code MRAM stores instructions and operands. For option-setting memory, see [section 7, Option-Setting Memory](#).

60.1 Overview

[Table 60.1](#) lists the specification of MRAM, and [Figure 60.1](#) shows a block diagram of the MRAM-related modules.

For the I/O pins used in serial programming mode, see [Table 60.39](#).

The CPSEQ (Code MRAM programming sequencer) controls programming of the code MRAM. EPSEQ (Extra MRAM programming sequencer) controls programming of extra MRAM which includes option-setting memory. The MACI (MRAM application command interface) controls EPSEQ according to the specified MACI commands.

For the configuration of the code MRAM, see [section 60.2. Structure of Memory](#).

Table 60.1 Specifications of MRAM (1 of 2)

Item	Code MRAM	Option-setting memory in extra MRAM
Memory capacity	User area: 1 MB max	See section 7, Option-Setting Memory .
Read cycle	See section 60.5.2. MRCFREQ : Code MRAM Frequency Notifications Register	See section 60.5.3. MREFREQ : Extra MRAM Frequency Notifications Register
Programming method	Write page buffer	MACI command
Protection	Protects against erroneous rewriting of the MRAM	
Dual bank function	Not available	Not available
Block swap function	Not available	Not available
Background operations (BGOs)	- The code MRAM can be read while option-setting memory is being programmed - Option-setting memory can be read while the code MRAM is being programmed.	
Units of programming	1 to 32 bytes	16 bytes
MACI command	Not available	Program: 16 bytes Forced stop Status clear Configuration set: 16 bytes Increment counter: 1 bit Read counter: 8 bytes
Security function	- Protect against illicit tampering with or reading out of data in MRAM - OFS area protection - Startup area select setting protection - BTFLG, BTSIZE, and MSUACR registers are protected by the FSPR bit. - Permanent block protect setting protection - User area is permanently protected from programming operation by the permanent block protect function. - MRAM protection for TrustZone - Protection for MRAM area (program) - Protection for MRAM area (read) - Protection for register - Extra MRAM program mode entry protection - Anti-rollback counter - HUK protection	
Safety function	Software protection - MACI command protection by MENTRYR register - User area protection by MRCPC0 and MRCPC1 registers - User area protection by the block protection setting. Error protection - Error is detected when unintended commands or prohibited settings occur. The MACI command is not accepted after an error detection. Boot area protection - The startup area select function allows the customer to safely update the startup program. The size of the startup area can be selected from 8 KB, 16 KB, or 32 KB. ECC support for double-bit error correction and triple-bit error detection.	

Table 60.1 Specifications of MRAM (2 of 2)

Item	Code MRAM	Option-setting memory in extra MRAM
Interrupt request	• MRAM_MRCRD (Code MRAM read error) Enabled by INTENBTC and INTENBDC bits. • MRAM_MRERD (Extra MRAM read error) Enabled by INTENBTE and INTENBDE bits. • MRAM_MRCPR (Code MRAM sequencer error) Enabled by MRCAEIE bits. • MRAM_MREPR (Extra MRAM sequencer error) Enabled by CMDLKIEE and MREAEIE bits. • MRAM_ENDOFPE (Extra MRAM sequencer ready (processing end)) Enabled by MRDYIE bit.	

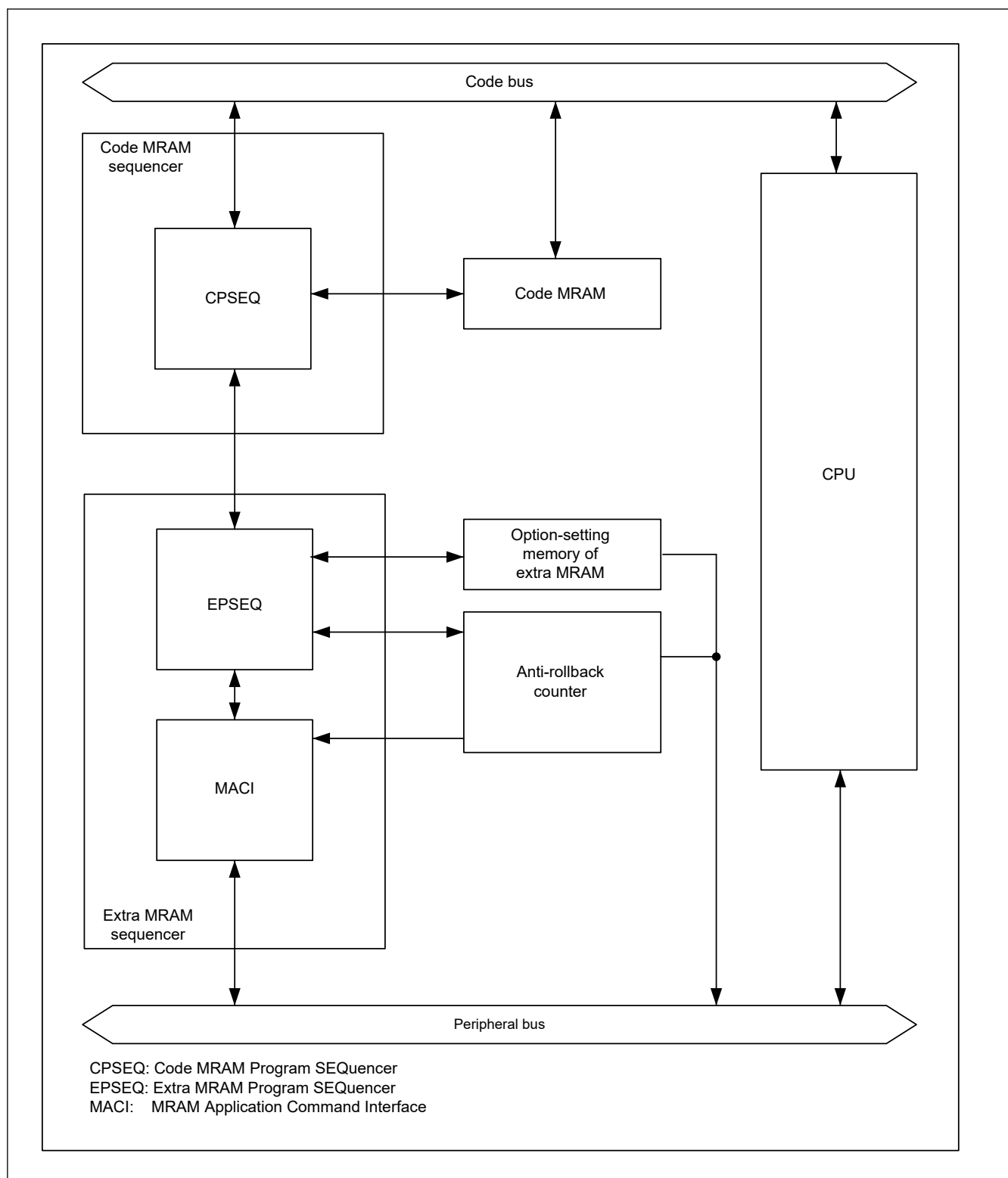
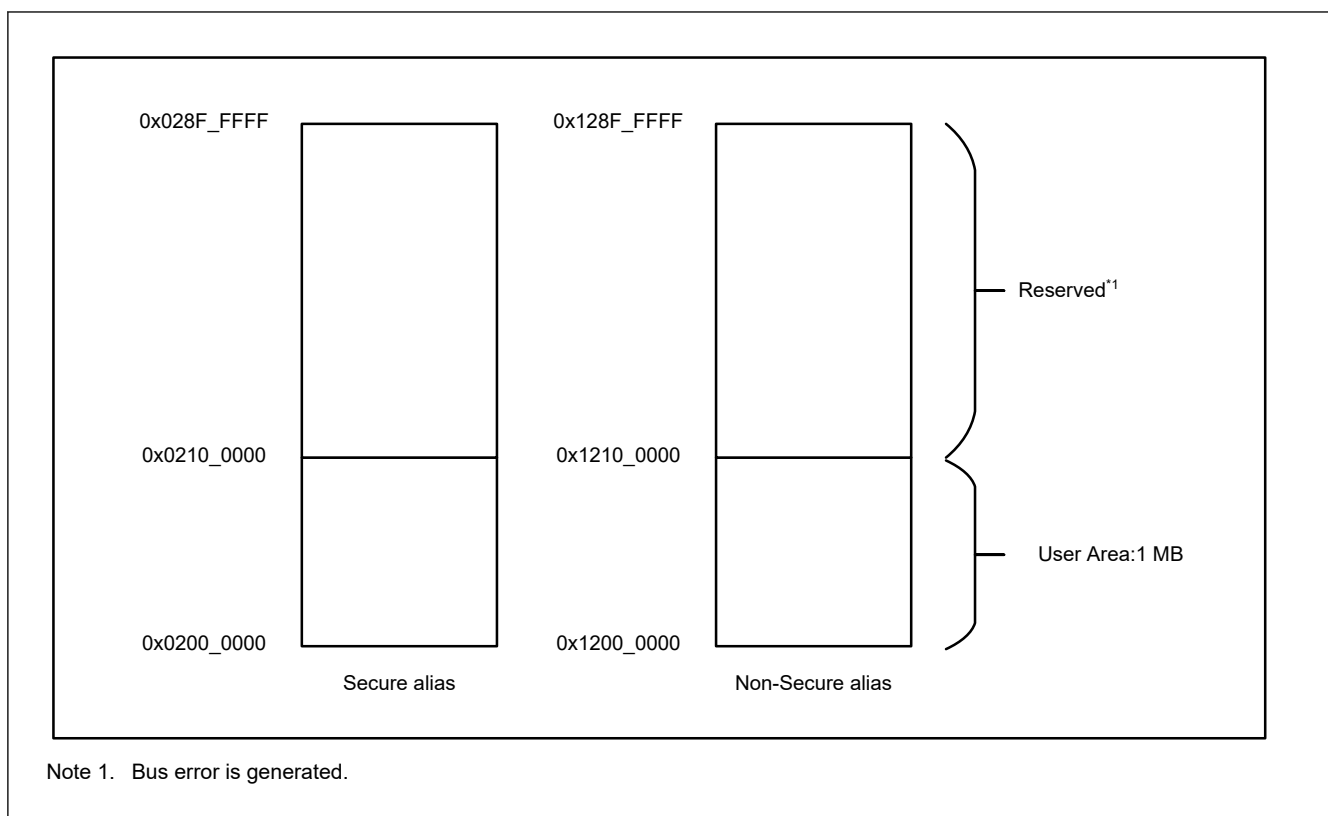


Figure 60.1 Block diagram of MRAM-related module

60.2 Structure of Memory

Figure 60.2 shows the memory map of the code MRAM.

**Figure 60.2** Map of the code MRAM**Table 60.2** Read and programming address by product for the code MRAM

Product	Address
1 MB product	Secure alias: 0x0200_0000 to 0x020F_FFFF Non-Secure alias: 0x1200_0000 to 0x120F_FFFF

60.3 Address Space

Using the hardware interface with MRAM requires access to all registers of the hardware, which is for issuing MACI commands. The base address is shown in [Table 60.3](#). [Table 60.4](#) provides information about the hardware interface.

Table 60.3 Base address

Base address symbol	Address	Secure alias
BASE_MC	0x0200_0000 or 0x1200_0000	Secure or Non-Secure
BASE_MCS	0x0200_0000	Secure
BASE_MCN	0x1200_0000	Non-Secure
BASE_MR	0x4012_0000	Secure
	0x5012_0000	Non-Secure

Table 60.4 Information on the hardware interface area

Area	Address	Capacity
Area containing the various registers of the hardware	See section 60.5. Register Descriptions .	See section 60.5. Register Descriptions .
MACI command-issuing area	BASE_MR	4 bytes

For the address information of the extra MRAM, see [section 5, Address Space](#).

60.4 MRAM Read and Program Function

60.4.1 Prefetch Buffer

Prefetch buffer speeds up CPU instruction fetches from bus master to code MRAM.

Table 60.5 Prefetch buffer overview

Function	Implementation
Target region	0x000_0000 to 0x00F_FFFF (1 MB)
Target bus master	CPU0 and CPU1 instruction fetch
Capacity	64 bytes/CPU
Associativity	Full associative
	256 bits per entry (256 bits aligned data), 2 entries
Access cycle	Hit: 0 wait
	Miss: See section 60.5.2. MRCFREQ : Code MRAM Frequency Notifications Register .

60.4.2 Programming Procedure

Program data for code MRAM is entered to the program buffer by bus access and is written to code MRAM in 32-bytes units. Renesas recommends the following to avoid system bus stall while processing code MRAM program.

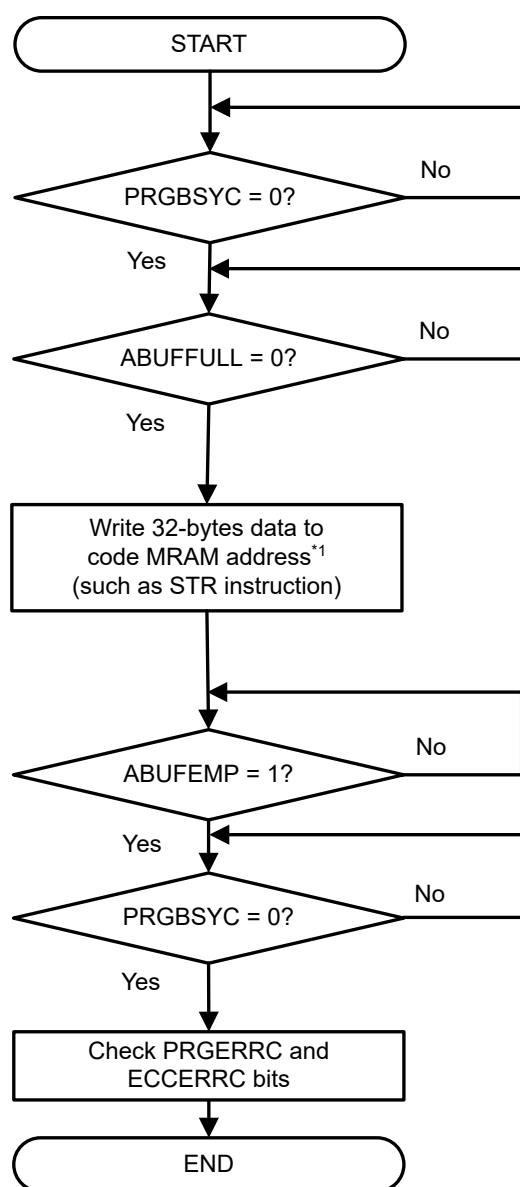
- Only one bus master can perform code MRAM write access
- Program data is 32 bytes or less in the same 32-bytes boundary.

The programming procedures of code MRAM are shown in [Figure 60.3](#) and [Figure 60.4](#). Before writing to code MRAM address, confirm that other program processing is not performed by PRGBSYC and ABUFFULL bits in the MRCPS register. The write data to the code MRAM address is stored in the 32-bytes program data buffer, and is programmed to the code MRAM in either of the following conditions.

- Program data buffer is full ([Figure 60.3](#))
- Written code MRAM address is out of 32-bytes boundary from stored data in the program data buffer
- MRCFL bit in MRCFLR register is set to 1 ([Figure 60.4](#)).

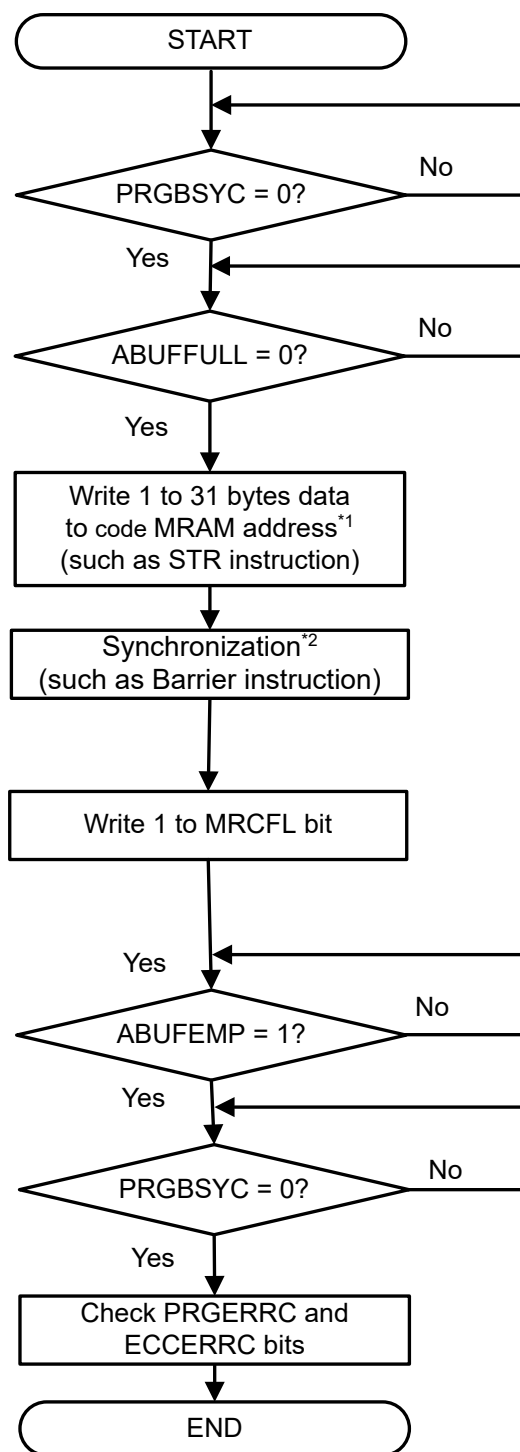
Program processing error status can be confirmed by the PRGERRC and ECCERRC bits in the MRCPS register.

See [section 60.7. MACI Commands](#) for programming procedure of extra MRAM.



Note 1. Program address starts from 32-bytes boundary and continuous address.

Figure 60.3 Code MRAM programming procedure (32-bytes programming)



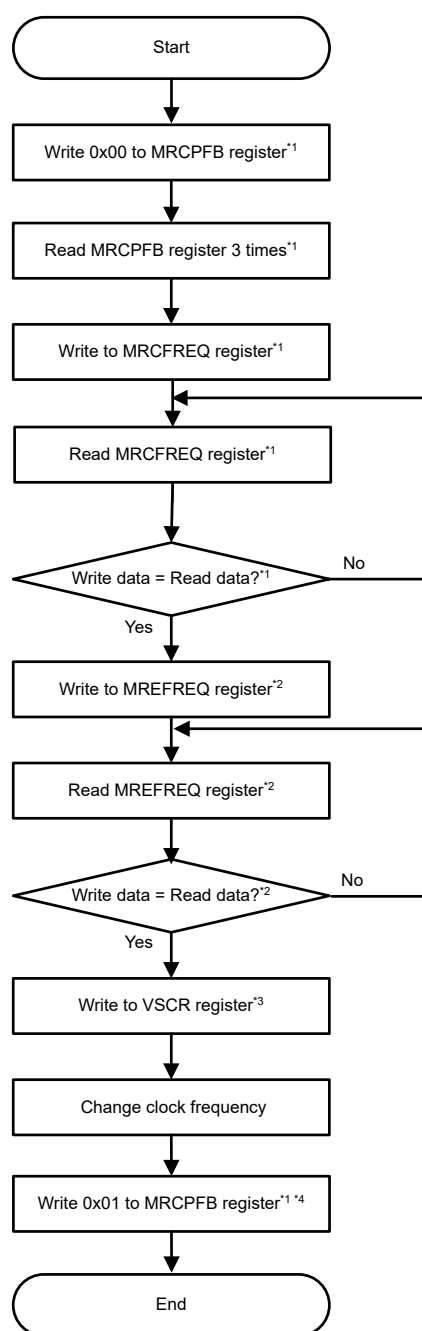
Note 1. Program address is continuous address.

Note 2. See [section 60.14. Usage Notes](#) for details.

Figure 60.4 Code MRAM programming procedure (smaller than 32-bytes programming)

60.4.3 Frequency Change Procedure

The frequency setting is not reflected even if the register is written while programming or reading MRAM. To change the clock frequency, check that the register value is set to the write value before changing it. The frequency change procedures for code MRAM and extra MRAM are shown in [Figure 60.5](#) and [Figure 60.6](#).



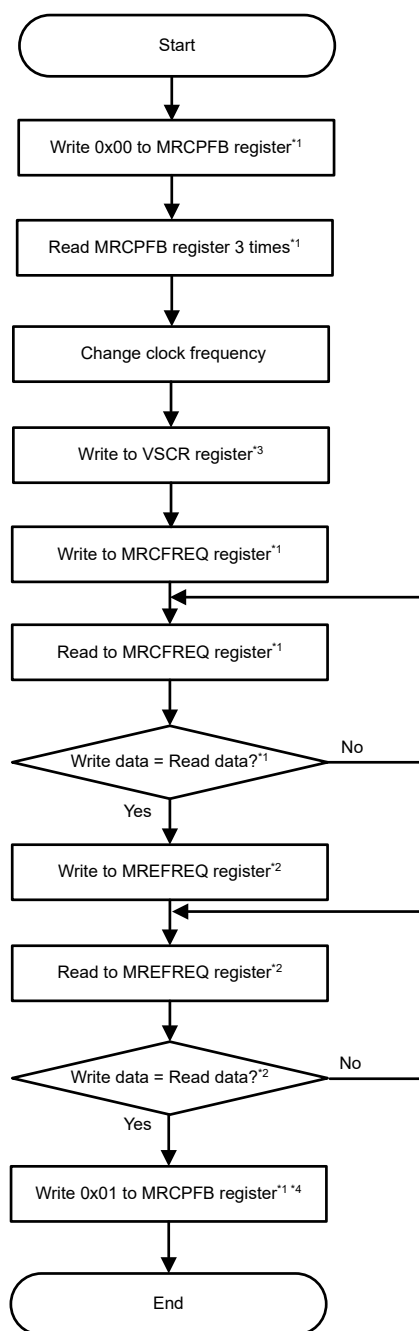
Note 1. Only when changing MRCFREQ register.

Note 2. Only when changing MREFREQ register.

Note 3. Only when changing the voltage.

Note 4. When the value of MRCFREQ register is 0x65 or higher and prefetch buffer function is enabled to operate the device.

Figure 60.5 Frequency up procedure



Note 1. Only when changing MRCFREQ register.

Note 2. Only when changing MREFREQ register.

Note 3. Only when changing the voltage.

Note 4. When the value of MRCFREQ register is 0x65 or higher, and prefetch buffer function is enabled to operate the device.

Figure 60.6 Frequency down procedure

60.5 Register Descriptions

60.5.1 MRCPFB : Code MRAM Prefetch Buffer Enable Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x000

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MPFB EN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MPFBEN	Code MRAM Prefetch Buffer Enable 0: Prefetch buffer disable 1: Prefetch buffer enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

MPFBEN bit (Code MRAM Prefetch Buffer Enable)

This bit is used to enable or disable code MRAM prefetch buffer.

It is prohibited to enable prefetch buffer when the MRCFREQ register value is 0x064 or less.

60.5.2 MRCFREQ : Code MRAM Frequency Notifications Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MRCMHZ[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0

Bit	Symbol	Function	R/W
9:0	MRCMHZ[9:0]	Code MRAM Frequency Notifications These bits are used to set the operating frequency of the code MRAM. Set the MRICLK frequency. Number of read access waits to code MRAM. MRCMHZ[9:0] ≤ 0x064 (100 MHz): non wait access 0x064 (100 MHz) < MRCMHZ[9:0] ≤ 0x0C8 (200 MHz): 1 wait access 0x0C8 (200 MHz) < MRCMHZ[9:0] ≤ 0x0FA (250 MHz): 2 wait access Other setting is reserved.	R/W ^{*1}
23:10	—	These bits are read as 0. The write value should be 0.	R/W
31:24	KEY[7:0]	Key Code	R/W ^{*2}

Note: S-TYPE-3, P-TYPE-2

Note 1. Writing to these bits is only possible when 32 bits are written, and the value written to the KEY bits is 0x1E.

Note 2. Written values are not retained by these bits (always reads 0x00).

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCFREQ register sets read and program frequency of code MRAM. For a faster clock frequency, set MRCMHZ[9:0] before changing the clock frequency. The ratio of code MRAM to extra MRAM frequency is limited to 5 or less. To slow

down the clock frequency, set MRCMHZ[9:0] after changing the clock frequency. For information on the frequency setting, see [section 60.4.3. Frequency Change Procedure](#).

MRCMHZ[9:0] bits (Code MRAM Frequency Notifications)

These bits specify the operating frequency of the code MRAM. Specifically, convert the frequency represented in MHz into a binary number and set it in these bits. If the operating frequency is 32.768 KHz or less, set the MCMHZ bits to 0x000.

Example:

Frequency is 249.9 MHz (MRCMHZ = 0x0FA)

Round up the first decimal place of 250 MHz to a whole number (= 250) and convert it into a binary number.

Frequency is 0.5 MHz (MRCMHZ = 0x001)

Round up the first decimal place of 0.5 MHz to a whole number (= 1) and convert it into a binary number.

Frequency is 32.768 kHz (MRCMHZ = 0x000)

KEY[7:0] bits (Key Code)

These bits control permission and prohibition of writing to the MCMHZ bits.

60.5.3 MREFREQ : Extra MRAM Frequency Notifications Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	MREMHZ[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0

Bit	Symbol	Function	R/W
7:0	MREMHZ[7:0]	Extra MRAM Frequency Notifications These bits are used to set the operating frequency of the extra MRAM. Set the MRPCLK frequency. Number of read access waits to extra MRAM. MREMHZ[7:0] ≤ 0x64 (100 MHz): non wait access 0x64 (100 MHz) < MREMHZ[7:0] ≤ 0x7D (125 MHz): 1 wait access Other setting is reserved.	R/W ¹
23:8	—	These bits are read as 0. The write value should be 0.	R/W
31:24	KEY[7:0]	Key Code	R/W ²

Note: S-TYPE-3, P-TYPE-2

Note 1. Writing to these bits is only possible when 32 bits are written, and the value written to the KEY bits is 0xE1.

Note 2. Written values are not retained by these bits (always reads 0x00).

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

MREFREQ sets the read and program frequency of the extra MRAM. For a faster clock frequency, set MREMHZ[7:0] before changing the clock frequency. To slow down the clock frequency, set MREMHZ[7:0] after changing the clock frequency. For information on the frequency setting, see [section 60.4.3. Frequency Change Procedure](#).

MREMHZ[7:0] bits (Extra MRAM Frequency Notifications)

These bits specify the operating frequency of the extra MRAM. Specifically, convert the frequency represented in MHz into a binary number and set it in these bits. If the operating frequency is 32.768 kHz or less, set the MREMHZ[7:0] bits to 0x000.

Example:

Frequency is 124.9 MHz (MREMHZ = 0x7D)

Round up the first decimal place of 124.9 MHz to a whole number (= 125) and convert it into a binary number.

Frequency is 0.5 MHz (MREMHZ = 0x001)

Round up the first decimal place of 0.5 MHz to a whole number (= 1) and convert it into a binary number.

Frequency is 32.768 kHz (MREMHZ = 0x000)

KEY[7:0] bits (Key Code)

These bits control permission and prohibition of writing to the MREMHZ[7:0] bits.

60.5.4 MRCDECC : Code MRAM ECC Decoder Control Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x010

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	ECCSEL ELC	DECDI SC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DECDISC	Code MRAM ECC Decoder Disable 0: ECC decoder enable 1: ECC decoder disable	R/W ^{*1}
1	ECCSEL	Code MRAM ECC Bits Select 0: ECC bits unselect 1: ECC bits select	R/W ^{*1}
7:2	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ^{*2}

Note: S-TYPE-3, P-TYPE-2

Note 1. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0x5A.

Note 2. Written values are not retained by these bits (always reads 0x00).

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCDECC register is used to test the ECC decoder circuits for the code MRAM. For information on the test flow of the ECC decoder circuits, see [section 60.9.4. Fault Correction and Detection](#).

DECDISC bit (Code MRAM ECC Decoder Disable)

This bit enables or disables the ECC decoder circuit in the code MRAM.

[Setting condition]

- 1 is written to while write enabling conditions are satisfied.

[Clearing condition]

- 0 is written to while write enabling conditions are satisfied.

ECCSEL bit (Code MRAM ECC Bits Select)

The ECC bits of code MRAM is output to the [144:128] bits and [16:0] bits of the read data when ECCSEL is set to 1. A value of 0 is output for the [255:145] and [127:17] bits of the read data. When a DEC error occurs, the corrected data is output. Read data is not guaranteed when a TED error is detected.

[Setting condition]

- 1 is written to while write enabling conditions are satisfied.

[Clearing condition]

- 0 is written to while write enabling conditions are satisfied.

KEY[7:0] bits (Key Code)

These bits control permission and prohibition of writing to the DECDISC and ECCSELC bits.

60.5.5 MRCRAEINT : Code MRAM Read Access Error Interrupt Enable Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x014

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	INTENBTC	INTENBDC
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
0	INTENBDC	Code MRAM DEC Error Interrupt Enable 0: Generation of an MRAM_MRCRD interrupt request is disabled when MRCRAES.DECERRC is set to 1 1: Generation of an MRAM_MRCRD interrupt request is enabled when MRCRAES.DECERRC is set to 1	R/W
1	INTENBTC	Code MRAM TED Error Interrupt Enable 0: Generation of an MRAM_MRCRD interrupt request is disabled when MRCRAES.TEDERRC is set to 1 1: Generation of an MRAM_MRCRD interrupt request is enabled when MRCRAES.TEDERRC is set to 1	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCRAEINT register enables or disables generation of a code MRAM read access error (MRAM_MRCRD) interrupt request.

INTENBDC bit (Code MRAM DEC Error Interrupt Enable)

This bit enables or disables generation of an MRAM_MRCRD interrupt request when a code MRAM read access violation occurs. This violation leads to the DECERRC bit in the MRCRAES register being set to 1.

INTENBTC bit (Code MRAM TED Error Interrupt Enable)

This bit enables or disables generation of an MRAM_MRCRD interrupt request when a code MRAM read access violation occurs. This violation leads to the TEDERRC bit in the MRCRAES register being set to 1.

60.5.6 MRCRAES : Code MRAM Read Access Error Status Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x018

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEDE RRC	DECE RRC
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DECERRC	Code MRAM DEC Error Detected 0: DEC error undetected 1: DEC error detected	R/W ^{*1}
1	TEDERRC	Code MRAM TED Error Detected 0: TED error undetected 1: TED error detected	R/W ^{*1}
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. Only 0 can be written to clear the flag after 1 is read. These bits should be checked after 0 is written and rewritten to 0 if 1 is read.

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

DECERRC bit (Code MRAM DEC Error Detected)

This bit indicates the result of a DEC error check for read data. When accessing reads from the debugger, DECERRC is not set to 1 even if a DEC error occurs. To resolve the DEC error, read 32 bytes of data from the address indicated by the MRCRDEA register and write it back to the same address.

[Setting condition]

- Error detected
Error is detected to read data of bus masters (this bit is not set when error is detected to prefetched data).

[Clearing condition]

- 0 is written after reading 1 from this bit.

TEDERRC bit (Code MRAM TED Error Detected)

This bit indicates the result of a TED error check for read data. When accessing reads from the debugger, TEDERRC is not set to 1 even if a TED error occurs. To resolve the TED error, write 32 bytes of data to the address indicated by the MRCRTEA register.

[Setting condition]

- Error detected
Error is detected to read data of bus masters (this bit is not set when error is detected to prefetched data).

[Clearing condition]

- 0 is written after reading 1 from this bit.

60.5.7 MRCRTEA : Code MRAM TED Error Address Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x01C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MRCRTEA[26:11]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRCRTEA[10:0]												—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	—	These bits are read as 0.	R
31:5	MRCRTEA[26:0]	Code MRAM Read Access TED Error Address Indicates the address when a TED error occurs.	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCRTEA register stores the address when a TED error occurs in the code MRAM readout. The address of an ECC error is stored when the MRCRAES.TEDERRC bit is set to 1. MRCRTEA bits do not capture the error address when a TED error occurs until the MRCRAES.TEDERRC bit is cleared to 0.

MRCRTEA bits (Code MRAM Read Access TED Error Address)

[Setting condition]

- The TEDERRC bit in the MRCRAES register changes from 0 to 1.

60.5.8 MRCRDEA : Code MRAM DEC Error Address Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MRCRDEA[26:11]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRCRDEA[10:0]												—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	—	These bits are read as 0.	R
31:5	MRCRDEA[26:0]	Code MRAM Read Access DEC Error Address Indicates the address when a DEC error occurs.	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCRDEA register stores the address when a DEC error occurs in the code MRAM readout. The address of an ECC error is stored when the MRCRAES.DECERRC bit is set to 1. MRCRDEA bits do not capture the error address when a DEC error occurs until the MRCRAES.DECERRC bit is cleared to 0.

MRCRDEA[26:0] bits (Code MRAM Read Access DEC Error Address)

[Setting condition]

- The DECERRC bit in the MRCRAES register changes from 0 to 1.

60.5.9 MRERAEINT : Extra MRAM Read Access Error Interrupt Enable Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x034

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	INTEN BTE	INTEN BDE
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
0	INTENBDE	Extra MRAM DEC Error Interrupt Enable 0: Generation of an MRAM_MRERD interrupt request is disabled when MERAES.DECERRE is set to 1 1: Generation of an MRAM_MRERD interrupt request is enabled when MERAES.DECERRE is set to 1	R/W
1	INTENBTE	Extra MRAM TED Error Interrupt Enable 0: Generation of an MRAM_MRERD interrupt request is disabled when MERAES.TEDERRE is set to 1 1: Generation of an MRAM_MRERD interrupt request is enabled when MERAES.TEDERRE is set to 1	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRERAIEINT register enables or disables generation of an extra MRAM read access error (MRAM_MRERD) interrupt request.

INTENBDE bit (Extra MRAM DEC Error Interrupt Enable)

This bit enables or disables generation of an MRAM_MRERD interrupt request when an extra MRAM read access violation occurs. This violation leads to the DECERRE bit in the MRERAES register being set to 1.

INTENBTE bit (Extra MRAM TED Error Interrupt Enable)

This bit enables or disables generation of an MRAM_MRERD interrupt request when an extra MRAM read access violation occurs. This violation leads to the TEDERRE bit in the MRERAES register being set to 1.

60.5.10 MRERAES : Extra MRAM Read Access Error Status Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x038

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TEDE RRE	DECE RRE

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DECERRE	Extra MRAM DEC Error Detected 0: DEC error undetected 1: DEC error detected	R/W ¹
1	TEDERRE	Extra MRAM TED Error Detected 0: TED error undetected 1: TED error detected	R/W ¹
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. Only 0 can be written to clear the flag after 1 is read. These bits should be checked after 0 is written and rewritten to 0 if 1 is read.

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

DECERRE bit (Extra MRAM DEC Error Detected)

This bit indicates the result of a DEC error check for read data. When accessing reads from the debugger, DECERRE is not set to 1 even if a DEC error occurs. To resolve the DEC error, read 16 bytes of data from the address indicated by the MRERDEA register and write it back to the same address.

[Setting condition]

- Error detected
Error is detected to read data of bus masters (this bit is not set when error is detected to prefetched data).

[Clearing condition]

- 0 is written after reading 1 from this bit.

TEDERRE bit (Extra MRAM TED Error Detected)

This bit indicates the result of a TED error check for read data. When accessing reads from the debugger, TEDERRE is not set to 1 even if a TED error occurs. To resolve the TED error, write 16 bytes of data to the address indicated by the MRERTEA register.

[Setting condition]

- Error detected
Error is detected to read data of bus masters (this bit is not set when error is detected to prefetched data).

[Clearing condition]

- 0 is written after reading 1 from this bit.

60.5.11 MRERTEA : Extra MRAM TED Error Address Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x03C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MRERTEA[27:12]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRERTEA[11:0]												—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0.	R
31:4	MRERTEA[27:0]	Extra MRAM Read Access TED Error Address Indicates the address when a TED error occurs.	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRERTEA register stores the address when a TED error occurs in the extra MRAM readout. The address of an ECC error is stored when the MRERAES.TEDERRE bit is set to 1. MRERTEA bits do not capture the error address when a TED error occurs until the MRERAES.TEDERRE bit is cleared to 0.

MRERTEA bits (Extra MRAM Read Access TED Error Address)

[Setting condition]

- The TEDERRE bit in the MRERAES register changes from 0 to 1.

60.5.12 MRERDEA : Extra MRAM DEC Error Address Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MRERDEA[27:12]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRERDEA[11:0]												—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0.	R
31:4	MRERDEA[27:0]	Extra MRAM Read Access DEC Error Address Indicates the address when a DEC error occurs.	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRERDEA register stores the address when a DEC error occurs in the extra MRAM readout. The address of the ECC error is stored when the MRERAES.DECERRE bit is set to 1. The MRERDEA bits do not capture the error address when a DEC error occurs until the MRERAES.DECERRE bit is cleared to 0.

MRERDEA bits (Extra MRAM Read Access DEC Error Address)

[Setting condition]

- The DECERRE bit in the MRERAES register changes from 0 to 1.

60.5.13 MSAR : MRAM Security Attribution Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x100

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRCP SEQS A	MREP SEQS A	MRCP SA	—	MACIT RSA	MACI CMRS A	MACI CMISA	—	—	—	—	MPFB ENSA	MRCF REQS A	MRCE CCSA	MREF REQS A	MREE CCSA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MREECCSA	Extra MRAM ECC Register Security Attribution 0: Secure 1: Non-Secure	R/W
1	MREFREQSA	MREFREQ Register Security Attribution 0: Secure 1: Non-Secure	R/W
2	MRCECCSA	Code MRAM ECC Register Security Attribution 0: Secure 1: Non-Secure	R/W
3	MRCFREQSA	MRCFREQ Register Security Attribution 0: Secure 1: Non-Secure	R/W

Bit	Symbol	Function	R/W
4	MPFBENSA	MRCPFB Register Security Attribution 0: Secure 1: Non-Secure	R/W
8:5	—	These bits are read as 0. The write value should be 0.	R/W
9	MACICMISA	MACI command Issuing Security Attribution 0: Secure 1: Non-Secure	R/W
10	MACICMRSA	MACI Command Registers Security Attribution 0: Secure 1: Non-Secure	R/W
11	MACITRSA	MACI Transfer Security Attribution 0: Secure 1: Non-Secure	R/W
12	—	This bit is read as 0. The write value should be 0.	R/W
13	MRCPSA	Code MRAM Program Register Security Attribution 0: Secure 1: Non-Secure	R/W
14	MREPSEQSA	EPSEQ Area Register Security Attribution 0: Secure 1: Non-Secure	R/W
15	MRCPSEQSA	CPSEQ Area Register Security Attribution 0: Secure 1: Non-Secure	R/W

Note: S-TYPE-1, P-TYPE-1

Write access is invalid when `PRCR_S.PRC4 = 0`. For details, see [section 13, Register Write Protection Function](#).

MREECCSA bit (Extra MRAM ECC Register Security Attribution)

This bit sets the security attribute of MREDECC, MRERAEINT, MRERAES, MRERTEA, and MRERDEA registers.

MREFREQSA bit (MREFREQ Register Security Attribution)

This bit sets the security attribute of the MREFREQ register.

MRCECCSA bit (Code MRAM ECC Register Security Attribution)

This bit sets the security attribute of MRCDECC, MRCRAEINT, MRCRAES, MRCRTEA, and MRCRDEA registers.

MRCFREQSA bit (MRCFREQ Register Security Attribution)

This bit sets the security attribute of the MRCFREQ register.

MPFBENSA bit (MRCPFB Register Security Attribution)

This bit sets the security attribute of the MCPFB register.

MACICMISA bit (MACI command Issuing Security Attribution)

This bit sets the security attribute of the MRAM access area.

MACICMRSA bit (MACI Command Registers Security Attribution)

This bit sets the security attribute of MASTAT, MREPAEINT, MRDYIE, MSADDR, MSTATR, MENTRYR, MSUINTR, MCMDR, and MSUASMON registers.

MACITRSA bit (MACI Transfer Security Attribution)

This bit sets the security attribute of MCTRCNTR, MCTRLSR, and MCTRSTATR registers.

MRCPSA bit (Code MRAM Program Register Security Attribution)

This bit sets the security attribute of MRCPC0, MRCBPROT0, MRCPS, MRCPEAINT, and MRCPEA registers.

MREPSEQSA bit (EPSEQ Area Register Security Attribution)

This bit sets the security attribute of MRPSC and MREECC registers.

MRCPSEQSA bit (CPSEQ Area Register Security Attribution)

This bit sets the security attribute of the MRCEECC register.

60.5.14 MREZS : Extra MRAM Zeroization Status Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x400

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	WHUK EXE	WHUK ZF
Value after reset:	1	1	1	1	1	1	0	1/0 ^{*1}

Bit	Symbol	Function	R/W
0	WHUKZF	W-HUK Zero Flag Status 0: W-HUK is zeroized 1: W-HUK is not zeroized	R
1	WHUKEXE	W-HUK Zeroization Executing Status 0: W-HUK zeroization is not executing 1: W-HUK zeroization is executing	R
7:2	—	These bits are read as 0.	R

Note: S-TYPE-6, P-TYPE-2

Note 1. The initial value is changed by the status of W-HUK.

WHUKZF bit (W-HUK Zero Flag Status)

See [section 60.8.6. HUK Protection](#) for the update conditions of WHUKZF.

[Setting condition]

- W-HUK zeroing not executed.

[Clearing conditions]

- After W-HUK zeroing is complete.

WHUKEXE bit (W-HUK Zeroization Executing Status)

[Setting condition]

- W-HUK zeroization is started.

[Clearing condition]

- W-HUK zeroization is finished.

60.5.15 MREZC : Extra MRAM Zeroization Control Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x404

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	WHUKZE[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	WHUKZE[2:0]	W-HUK Zeroization Execute 0XX, X0X, X X 0: W-HUK zeroization not executes 1 1 1: W-HUK zeroization executes	R/W ^{*1,2}
7:3	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ^{*3}

Note: S-TYPE-6, P-TYPE-2

Note 1. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0x3C.

Note 2. MREZC register cannot be written in the following conditions.

- When the W-HUK pseudo-zeroing operation in [Table 60.31](#) has already been completed and pseudo-zeroing is to be started again by register writing without any system resets.
- If the operation of W-HUK zeroing in [Table 60.31](#) has already been completed.

Note 3. Written values are not retained by these bits (always reads 0x00).

The MREZC register is used to trigger zeroization of W-HUK by secure software.

WHUKZE[2:0] bits (W-HUK Zeroization Execute)

0 cannot be written to these bits.

[Setting condition]

- 1 is written to while write enabling conditions are satisfied.

[Clearing condition]

- After W-HUK zeroing is complete.

60.5.16 MASTAT : Extra MRAM Access Status Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2010

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CMDL K	MREA E	—	—	—

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	MREAE	Extra MRAM Access Violation Flag 0: No extra MRAM access violation has occurred 1: An extra MRAM access violation has occurred	R/W ^{*1}
4	CMDLK	Command Lock Flag 0: The MRAM sequencer is not in the command-locked state 1: The MRAM sequencer is in the command-locked state	R
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. Only 0 can be written to clear the flag after 1 is read.

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MASTAT register indicates whether an extra MRAM access violation has occurred. If any of the CMDLK and MREAE flags is set to 1, the extra MRAM sequencer enters the command-locked state (see [section 60.9.2. Error Protection](#)). To release it from the command-locked state, the Status Clear or Forced Stop command must be issued to the extra MRAM sequencer.

MREAE flag (Extra MRAM Access Violation Flag)

This flag indicates whether an extra MRAM access violation occurred. When this flag is set to 1, the ILGLERR bit in the MSTATR register is set to 1, placing the extra MRAM sequencer in the command-locked state.

[Setting conditions]

In general, commands are issued in extra MRAM program mode as follows.

- The setting of the MSADDR register is a reserved portion of the extra MRAM area.
- The Configuration Set command is issued while the setting of the MSADDR register is out of BASE_MC + 0x00C9 F020 to BASE_MC + 0x00C9 F0A0 and BASE_MC + 0x00C9 F0C0 to BASE_MC + 0x00C9 F7FF.
- MACI command is issued to extra MRAM with the wrong security attribution.

[Clearing conditions]

- When 0 is written to this flag after 1 is set
- When the extra MRAM sequencer starts to process the Status Clear or Forced Stop command.

CMDLK flag (Command Lock Flag)

This flag indicates that the extra MRAM sequencer is in the command-locked state.

[Setting condition]

- The extra MRAM sequencer detects an error and enters the command-locked state.

[Clearing condition]

- When an extra MRAM sequencer starts to process the Status Clear or Forced Stop command.

60.5.17 MPAEINT : Extra MRAM Access Error Interrupt Enable Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2014

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CMDLKIE	MREAEIE	—	—	—
Value after reset:	0	0	0	1	1	0	0	0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	MREAEIE	Extra MRAM Access Violation Interrupt Enable 0: Generation of an MRAM_MREPR interrupt request is disabled when MASTAT.MREAE is set to 1 1: Generation of an MRAM_MREPR interrupt request is enabled when MASTAT.MREAE is set to 1	R/W
4	CMDLKIE	Command Lock Interrupt Enable 0: Generation of an MRAM_MREPR interrupt request is disabled when MASTAT.CMDLK is set to 1 1: Generation of an MRAM_MREPR interrupt request is enabled when MASTAT.CMDLK is set to 1	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MPAEINT register enables or disables generation of an extra MRAM access error (MRAM_MREPR) interrupt request.

MREAEIE bit (Extra MRAM Access Violation Interrupt Enable)

This bit enables or disables generation of an MRAM_MREPR interrupt request when an extra MRAM access violation occurs. This violation leads to the MREAE bit in the MASTAT register being set to 1.

CMDLKIE bit (Command Lock Interrupt Enable)

This bit enables or disables generation of an MRAM_MREPR interrupt request when an extra MRAM sequencer enters the command-locked state. This violation leads to the CMDLK bit in the MASTAT register being set to 1.

60.5.18 MRDYIE : Extra MRAM Ready Interrupt Enable Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2018

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MRDY IE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MRDYIE	Extra MRAM Ready Interrupt Enable 0: Generation of an MRAM_ENDOFPE interrupt request is disabled 1: Generation of an MRAM_ENDOFPE interrupt request is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRDYIE register enables or disables generation of an extra MRAM ready (MRAM_ENDOFPE) interrupt.

MRDYIE bit (Extra MRAM Ready Interrupt Enable)

This bit is used to enable or disable generation of an MRAM_ENDOFPE interrupt request when the MRDY bit in the MSTATR register is changed from 0 to 1. This change occurs on completion of processing by the extra MRAM sequencer of Program and Configuration set command.

60.5.19 MSADDR : MACI Command Start Address Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2030

Bit position:	31	0
Bit field:	MSADDR[31:0]	
Value after reset:	0 0	

Bit	Symbol	Function	R/W
31:0	MSADDR[31:0]	Start Address for MACI Command Processing [Command]: [Address boundary] Program (extra MRAM): 16 bytes Configuration set (extra MRAM): 16 bytes	R/W ^{*1}

Note: S-TYPE-3, P-TYPE-2

Note 1. These bits can be written when the MRDY bit in the MSTATR register is 1. Writing to these bits are ignored when the MRDY bit is 0. Bits [3:0] are read-only.

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MSADDR register specifies the address where the target area for command processing starts when the MACI command for Program or Configuration set is issued.

The MSADDR value is initialized when the SUINIT bit in the MSUINITR register is set to 1. It is also initialized by a reset.

MSADDR[31:0] bits (Start Address for MACI Command Processing)

These bits specify the start address for MACI command processing. Bits [31:29] and bits [27:25] are ignored in MACI command processing for extra MRAM. Bit [28] is used for the security alias for extra MRAM sequencer command. When bit [28] is 0, the extra MRAM sequencer can update the secure region with secure access. When bit [28] is 1, the extra MRAM sequencer can update the non-security region with secure and non-secure access. Bits corresponding to address bits of lower order than the corresponding boundary listed above are also ignored.

See the User's Manual for information on the addresses of the extra MRAM.

See [Table 60.15](#) and [Table 60.16](#) for information on the addresses of Program and Configuration set commands.

60.5.20 MCNTSELR : MRAM Counter Select Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2048

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	CNTSEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CNTSEL[2:0]	Counter select 0 0 0: Anti-rollback counter is not selected. 0 0 1: ARC_SEC is selected. Counter size is 64 bits. 0 1 0: ARC_OEMBL is selected. Counter size is 64 bits. 0 1 1: Anti-rollback counter is not selected. 1 x x: ARC_NSEC is selected. Counter configuration is 64 or 256 bits. See Table 60.6 , Table 60.7 , and Table 60.8 for details of the counter select condition.	R/W ^{*1}
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6, P-TYPE-2

Note 1. This bit can be written when MSTATR.MRDY = 1. Writing to this bit is ignored when MSTATR.MRDY = 0.

CNTSEL[2:0] bits (Counter select)

The MCNTSELR register specifies 3 types of anti-rollback counter (ARC_SEC, ARC_NSEC or ARC_OEMBL) in Increment Counter or Read Counter command.

The MCNTSELR value is initialized when the SUINIT bit in MSUINITR is set to 1.

[Setting condition]

- This register is written to in 8-bit access while MSTATR.MRDY = 1.

[Clearing condition]

- This register is written to in 8-bit access while MSTATR.MRDY = 1.

Before using the ARC_NSEC counter, it is necessary to set whether the counter is to be used with 4 lines × 64 bits or 1 line × 256 bits using the program command. The bits to be set are CNF_ARCNS0 and CNF_ARCNS1. If the CNF_ARCNS1 bit is set to 1, the ARC_NSEC counter cannot be used. Once the counter type is decided, it cannot be changed.

Table 60.6 Select condition of ARC_NSEC (64 bits) in increment/read counter command

MCNTSELR[2:0]	Target counter	Counter configuration
100b	ARC_NSEC[63:0]	Counter number is 4. Counter size is 64 bit.
101b	ARC_NSEC[127:64]	
110b	ARC_NSEC[191:128]	
111b	ARC_NSEC[255:192]	

Table 60.7 Select condition of ARC_NSEC (256 bits) in increment counter command

MCNTSEL _R [2:0]	Target counter	Counter configuration
100b	ARC_NSEC[255:0]	Counter number is 1. Counter size is 256 bit.
101b		
110b		
111b		

Table 60.8 Select condition of ARC_NSEC (256 bits) in read counter command

MCNTSEL _R [2:0]	Target counter	Counter configuration
100b	ARC_NSEC[63:0]	Counter number is 1. Counter size is 256 bit.
101b	ARC_NSEC[127:64]	
110b	ARC_NSEC[191:128]	
111b	ARC_NSEC[255:192]	

60.5.21 MCNTDTR_n : MRAM Counter Data Register (n = 0, 1)

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x204C + 0x04 × n

Bit position: 31

0

Bit field:

CNTRDAT[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CNTRDAT[31:0]	Counter Read Data Reading data of 32 bits (LSB side of counter values) is output to MCNTDTR0. Reading data of 32 bits (MSB side of counter values) is output to MCNTDTR1.	R

Note: S-TYPE-6, P-TYPE-2

The MCNTDTR0 and MCNTDTR1 registers indicate anti-rollback counter values (64 bits) that read by the Read Counter command.

CNTRDAT[31:0] bits (Counter Read Data)

[Setting condition]

- MACI has accepted Read Counter command.

[Clearing condition]

- MACI has accepted the extra MRAM sequencer command except for the Read Counter command
- The extra MRAM sequencer enters the command-locked state
- W-HUK zeroization started.

60.5.22 MCTRCNTR : MRAM Configuration Update Transfer Control Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2060

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

Key[7:0]

—

—

—

—

—

—

—

—

—

TRTR
G

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	TRTRG	Transfer Start Trigger 0: Configuration update transfer processing does not start 1: Configuration update transfer processing starts	R/W ^{*1} *2 *3
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	Key[7:0]	Key code These bits enable or disable TRTRG bit modification. Data written to these bits are not stored.	R/W ^{*1}

Note: S-TYPE-3, P-TYPE-2

Note 1. Written data is not stored in these bits.

Note 2. This bit can be written when MCTRSTATR register is 0x00. Writing to this bit is ignored when MCTRSTATR register is other than 0x00 or when W-HUK zeroing is in progress.

Note 3. Writing to this bit is enabled only when this register is accessed in 16-bit size and 0x45 is written to the Key[7:0] bits.

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MTRCNTR register specifies the start trigger and setting information in the configuration update transfer processing.

TRTRG bit (Transfer Start Trigger)

Starts the configuration update transfer processing.

[Setting condition]

- 1 is written to the TRTRG bit while write enabling conditions are satisfied.

60.5.23 MCTRLSR : MRAM Configuration Update Transfer List Select Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2064

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	TRLIST[2:0]		

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
2:0	TRLIST[2:0]	Configuration Update Transfer List 0 0 1: Transfer list 1 is selected Others: Transfer list 1 is not selected	R/W ^{*1}
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note 1. This bit can be written when the TRBUSY bit in the MCTRSTATR register is 0. Writing to this bit when TRBUSY = 1 is ignored.

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MCTRLSR register specifies the target transfer list in the configuration update transfer.

TRLIST[2:0] bits (Configuration Update Transfer List)

Transfer list 1 includes the coefficient value for DAC12.

[Setting condition]

- 1 is written to the TRLIST[2:0] bits while write enabling conditions are satisfied.

[Clearing condition]

- 0 is written to the TRLIST[2:0] bits while write enabling conditions are satisfied.

60.5.24 MCTRSTATR : MRAM Configuration Update Transfer Status Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x206C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	TRMD	—	TRBUSY

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	TRBUSY	Transfer Busy Status 0: Configuration update transfer is not processing 1: Configuration update transfer is processing	R
1	—	This bit is read as 0.	R
2	TRMD	Transfer Mode Setting Status 0: MENTRYR register is a data transferable setting 1: MENTRYR register is not a data transferable setting	R
7:3	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

TRBUSY bit (Transfer Busy Status)

Indicates status of the configuration update transfer processing.

[Setting condition]

- 1 is written to the TRTRG bit while write enabling conditions are satisfied.

[Clearing condition]

- The configuration update transfer processing is finished.

TRMD bit (Transfer Mode Setting Status)

Indicates the MENTRYR register setting that the configuration update transfer can be started.

[Setting condition]

- MENTRYR is 0x0080.

[Clearing condition]

- MENTRYR is 0x0000.

60.5.25 MSTATR : Extra MRAM Status Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2080

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	ILGCO MERR	—	SECE RR	OTER R	TZFE RR	—	—	—

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRDY	ILGLE RR	—	PRGE RR	—	—	—	—	—	—	CFGS ETER R	—	—	—	—	—

Value after reset: 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
4:0	—	These bits are read as 0.	R
5	CFGSETERR	Configuration Set Error Flag 0: Configuration set has been completed successfully 1: An error has occurred during configuration set	R
11:6	—	These bits are read as 0.	R
12	PRGERR	Programming Error Flag 0: Programming has been completed successfully 1: An error has occurred during programming	R
13	—	These bits are read as 0.	R
14	ILGLERR	Illegal Command Error Flag 0: The extra MRAM sequencer has not detected an illegal MACI command or illegal extra MRAM access 1: The extra MRAM sequencer has detected an illegal MACI command or illegal extra MRAM access	R
15	MRDY	Extra MRAM Ready Flag 0: Program, Configuration set, Increment Counter, or Read Counter command processing is in progress 1: None of the above is in progress	R
18:16	—	These bits are read as 0.	R
19	TZFERR	TrustZone Filter Error 0: A TrustZone filter error has not been detected 1: A TrustZone filter error has been detected	R
20	OTERR	Other Error 0: An error has not been detected 1: An error has been detected	R
21	SECERR	Security Error 0: A write protection error against MSUASMON.FSPR bit has not been detected 1: A write protection error against MSUASMON.FSPR bit has been detected	R
22	—	This bit is read as 0.	R
23	ILGCOMERR	Illegal Command Error 0: An illegal MACI command error has not been detected 1: An illegal MACI command error has been detected	R
31:24	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MSTATR register indicates the state of the extra MRAM sequencer.

CFGSETERR flag (Configuration Set Error Flag)

This flag indicates the result of an extra MRAM configuration setting. If this flag is 1, the extra MRAM sequencer is in the command-locked state.

[Setting condition]

- A TED error occurs when an extra MRAM sequencer reads data not to update from the configuration area during configuration setting.

[Clearing condition]

- The extra MRAM sequencer starts processing of the Status Clear or Forced Stop command.

PRGERR flag (Programming Error Flag)

This flag indicates the result of an extra MRAM programming. If this flag is 1, the extra MRAM sequencer is in the command-locked state.

[Setting condition]

- An error occurs during programming.

[Clearing condition]

- The extra MRAM sequencer starts processing of the Status Clear or Forced Stop command.

ILGLERR flag (Illegal Command Error Flag)

This flag indicates that the extra MRAM sequencer has detected an illegal MACI command or extra MRAM access. If this flag is 1, the extra MRAM sequencer is in the command-locked state.

[Setting Conditions] (See [section 60.9.2. Error Protection](#))

- The extra MRAM sequencer has detected an illegal command
- The extra MRAM sequencer has detected illegal extra MRAM access.

[Clearing condition]

- The extra MRAM sequencer starts processing of the Status Clear or Forced Stop command.

MRDY flag (Extra MRAM Ready Flag)

This flag indicates the command processing state of the extra MRAM sequencer.

[Setting conditions]

- The extra MRAM sequencer completes command processing
- The extra MRAM sequencer has received the Forced Stop command and ended command processing
- W-HUK zeroing in processing.

[Clearing conditions]

- The extra MRAM sequencer has received the MACI command
- For Program and Configuration Set commands, the first write access to the MACI command-issuing area
- For other commands, the last write access to the MACI command issuing area.

TZFERR bit (TrustZone Filter Error)

This bit indicates a TrustZone protection error in the MACI command.

See [Table 60.36](#). When this bit is 1, the extra MRAM sequencer enters the command-locked state.

[Setting condition]

- An error has been detected.

[Clearing condition]

- Status Clear or Forced Stop command processing is completed.

OTERR bit (Other Error)

This bit indicates that an MACI command has been issued when the condition of accepting commands is not satisfied.

See [Table 60.36](#). When this bit is 1, the extra MRAM sequencer enters the command-locked state.

[Setting condition]

- An error has been detected.

[Clearing condition]

- Status Clear or Forced Stop command processing is completed.

SECERR bit (Security Error)

This bit indicates that the write protection by the MSUASMON.FSPR bit is violated.

See [Table 60.36](#). When this bit is 1, the extra MRAM sequencer enters the command-locked state.

[Setting condition]

- An error has been detected.

[Clearing condition]

- Status Clear or Forced Stop command processing is completed.

ILGCOMERR bit (Illegal Command Error)

This bit indicates that the extra MRAM sequencer has detected an illegal MACI command.

See [Table 60.36](#). When this bit is 1, the extra MRAM sequencer enters the command-locked state.

[Setting condition]

- An error has been detected.

[Clearing condition]

- Status Clear or Forced Stop command processing is completed.

60.5.26 MENTRYR : Extra MRAM Program Mode Entry Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2084

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								MENTRY	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.	R/W
7	MENTRY	Extra MRAM Program Mode Entry 0: Extra MRAM is in read mode 1: Extra MRAM is in program mode	R/W ^{*1 *2 *3}
15:8	KEY[7:0]	Key Code	R/W ^{*4}

Note: S-TYPE-3, P-TYPE-2

Note 1. These bits can be written when the MRDY bit in the MSTATR register is 1. Writing to these bits are ignored when the MRDY bit is 0.

Note 2. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0xAA.

Note 3. The MENTRY bit cannot be set to 1 during configuration update, and the MENTRYR register cannot be written to during W-HUK zeroing.

Note 4. Written values are not retained by these bits (always reads 0x00).

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MENTRYR register is used to specify the extra MRAM program mode. To specify an extra MRAM program mode so that the extra MRAM sequencer can receive MACI commands, set the MENTRY bit to 1 to place the extra MRAM sequencer in program mode.

The MENTRYR register is initialized when the SUINIT bit in MSUINITR is set to 1. It is also initialized by a reset.

MENTRY bit (Extra MRAM Program Mode Entry)

This bit specifies the extra MRAM program mode.

[Setting condition]

- Write 1 to the MENTRY bit while writing to MENTRYR is enabled and MENTRYR is 0x00.

[Clearing conditions]

- Write 8 bits to MENTRYR while the MRDY bit is 1
- A value other than 0xAA is specified in the KEY[7:0] bits and 16 bits are written to MENTRYR while the MRDY bit is 1
- Write 0 to the MENTRY bit while writing to MENTRYR is enabled

- Write to MENTRYR while writing is enabled and its value is other than 0x00.

KEY[7:0] bits (Key Code)

These bits control permission and prohibition of writing to the MENTRY bit.

60.5.27 MSUINTR : Extra MRAM Sequencer Setup Initialization Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x208C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	SUINIT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SUINIT	Setup Initialization 0: The MSADDR, MENTRYR, and MCNTSELR extra MRAM sequencer setup registers keep their current values 1: The MSADDR, MENTRYR, and MCNTSELR extra MRAM sequencer setup registers are initialized	R/W ^{*1 *2}
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ^{*3}

Note: S-TYPE-3, P-TYPE-2

Note 1. This bit can be written when the MRDY bit in the MSTATR register is 1. Writing to this bit is ignored when the MRDY bit is 0.

Note 2. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0x2D.

Note 3. Written values are not retained by these bits (always reads 0x00).

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MSUINTR register is used for initialization of the extra MRAM sequencer setup.

SUINIT bit (Setup Initialization)

This bit initializes the following extra MRAM sequencer setup registers.

- MSADDR
- MENTRYR
- MCNTSELR

KEY[7:0] bits (Key Code)

These bits control permission and prohibition of writing to the SUINIT bit.

60.5.28 MCMDR : MACI Command Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x20A0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CMDR[7:0]								PCMDR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	PCMDR[7:0]	Pre-command Flag The command just before the latest command is stored.	R

Bit	Symbol	Function	R/W
15:8	CMDR[7:0]	Command Flag The latest command is stored.	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MCMDR register records the two most recent commands accepted by the extra MRAM sequencer.

PCMDR[7:0] flags (Pre-command Flag)

These flags indicate the command received immediately before the latest command received by the extra MRAM sequencer.

CMDR[7:0] flags (Command Flag)

These flags indicate the latest command received by the extra MRAM sequencer.

Table 60.9 States of MCMDR after receiving commands

Command	CMDR	PCMDR
Program	0xE8	Previous Command
Status Clear	0x50	Previous Command
Forced Stop	0xB3	Previous Command
Configuration set	0x40	Previous Command
Increment Counter	0xD0	0x35
Read Counter	0xD0	0x39

60.5.29 MSUASMON : MRAM Startup Area Select Monitor Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x20DC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	BTFLG	BTSIZE[1:0]	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1/0	1/0	1/0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	FSPR	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1/0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	—	These bits are read as 0.	R
15	FSPR	Protection flag of programming to set boot flag and startup area control 0: Protected state 1: Non-protected state	R
28:16	—	These bits are read as 0.	R
30:29	BTSIZE[1:0]	Size of startup area select for boot swap 0 0: Reserved 0 1: Startup area size is set to 32 KB 1 0: Startup area size is set to 16 KB 1 1: Startup area size is set to 8 KB	R
31	BTFLG	Flag of startup area select for boot swap 0: The startup area is the alternate block (block 1) 1: The startup area is the default block (block 0)	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

FSPR flag (Protection flag of programming to set boot flag and startup area control)

This flag indicates the protection state against the Configuration Set command for the BTFLG bit, BTSIZE[1:0] bits, and the MSUACR register.

In response to a reset or Program command, the MACI transfers data from the extra MRAM to this register. The data corresponding to the FSPR flag is POFSPS[13] bit. See [section 60.8.1. Permanent OFS Protect Setting](#) for details.

BTSIZE[1:0] bits (Size of startup area select for boot swap)

These bits indicate the size of the startup area.

In response to a reset or Configuration Set command, the MACI transfers data from the extra MRAM to this register.

BTFLG flag (Flag of startup area select for boot swap)

This flag indicates whether the address of the startup area is exchanged for the boot swap function or not.

In response to a reset or Configuration Set command, the MACI transfers data from the extra MRAM to this register.

60.5.30 MSUACR : MRAM Startup Area Control Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x20E8

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	SAS[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	SAS[1:0]	Startup Area Select 0 X: Startup area is selected by BTFLG bit 1 0: Startup area is temporarily switched to the default area 1 1: Startup area is temporarily switched to the alternate area	R/W ^{*1}
7:2	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ^{*2}

Note: S-TYPE-6, P-TYPE-2

Note 1. The following described the write condition of this bit (these conditions are required at the same time).

1. The access size to this register is 16 bits.
2. The value of KEY [7:0] is 0x66.
3. The FSPR bit is 1.

Note 2. Written values are not retained by these bits (always reads 0x00).

The MSUACR register sets the startup area for the boot swap function.

SAS[1:0] bits (Startup Area Select)

These bits are used to select the startup area. Three methods are capable for changing the startup area.

KEY[7:0] bits (Key Code)

These bits control permission and prohibition of writing to the SAS[1:0] bits.

60.5.31 MRPSC : MRAM Program Speed Control Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x2800

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MHSP EN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MHSPEN	MRAM High-Speed Program Mode Enable 0: Normal program mode 1: High-speed program mode	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

MHSPEN bit (MRAM High-Speed Program Mode Enable)

This bit enables or disables high-speed program mode of the code MRAM or extra MRAM. After setting this bit to 1 and programming the code MRAM or extra MRAM, the mode is changed to high-speed program mode. To change to normal program mode, set this bit to 0, read the MRPSC register once, and program or read the MRAM. See [section 60.6.2. MRAM High-speed Program Mode](#) for details of mode changing.

60.5.32 MRCPC0 : Code MRAM Program Control Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x3000

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	MRCP NEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MRCPNEN	Code MRAM Program Enable for Non-secure 0: Code MRAM non-secure aliases are not programmable 1: Code MRAM non-secure aliases are programmable	R/W ¹
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ²

Note: S-TYPE-3, P-TYPE-2

Note 1. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0x86.

Note 2. Written values are not retained by these bits (always reads 0x00).

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

MRCPNEN bit (Code MRAM Program Enable for Non-secure)

The MRCPNEN bit enables programming of non-secure alias.

Setting the NVMWE bit in the DBGNVMCR register to 1 allows programming to a non-secure alias that is not protected by the block protection of code MRAM, regardless of the value of MRCPNEN.

60.5.33 MRCPC1 : Code MRAM Program Control Register for Secure

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x3004

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]							—	—	—	—	—	—	—	—	MRCPSEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MRCPSEN	Code MRAM Program Enable for Secure 0: Code MRAM secure aliases are not programmable 1: Code MRAM secure aliases are programmable	R/W ^{*1}
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ^{*2}

Note: S-TYPE-6, P-TYPE-2

Note 1. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0x68.

Note 2. Written values are not retained by these bits (always reads 0x00).

MRCPSEN bit (Code MRAM Program Enable for Secure)

The MRCPSEN bit enables programming of secure alias.

Setting the NVMWE bit in the DBGNVMCR register to 1 allows programming to a secure alias that is not protected by the block protection of the code MRAM, regardless of the value of MRCPSEN.

60.5.34 MRCBPROT0 : Code MRAM Block Protection Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x3008

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]							—	—	—	—	—	—	—	—	BPCN0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BPCN0	Code MRAM Block Protection Cancel for Non-secure 0: Code MRAM block protection is enabled 1: Code MRAM block protection is disabled	R/W ^{*1 *2}
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ^{*3}

Note: S-TYPE-3, P-TYPE-2

Note 1. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0x78.

Note 2. Writing to this bit is ignored when the MRCPNEN bit in the MRCPC0 register is 0x0.

Note 3. Written values are not retained by these bits (always reads 0x00).

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCBPROT0 register is used to disable block protect function for non-secure. When the block protect setting is locked by permanent block setting, it cannot be disabled by this register.

BPCN0 bit (Code MRAM Block Protection Cancel for Non-secure)

This bit disables block protect for non-secure function.

[Setting condition]

- 1 is written to BPCN0 while the write enabling conditions are satisfied.

[Clearing conditions]

One of the following conditions is satisfied.

- 0 is written to BPCN0 while the write enabling conditions are satisfied
- 0 is written to the MRCPC0.MRCPNEN bit.

60.5.35 MRCBPROT1 : Code MRAM Block Protection Register for Secure

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x300C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	BPCN1
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BPCN1	Code MRAM Block Protection Cancel for Secure 0: Code MRAM block protection is enabled 1: Code MRAM block protection is disabled	R/W*1 *2
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W*3

Note: S-TYPE-6, P-TYPE-2

Note 1. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0xB1.

Note 2. Writing to this bit is ignored when the MRCPC1.MRCPSEN bit in the MRCPC1 register is 0x0.

Note 3. Written values are not retained by these bits (always reads 0x00).

The MRCBPROT1 register is used to disable block protect function for secure. When the block protect setting is locked by permanent block setting, it cannot be disabled by this register.

BPCN1 bit (Code MRAM Block Protection Cancel for Secure)

This bit disables block protect for secure function.

[Setting condition]

- 1 is written to BPCN1 while the write enabling conditions are satisfied.

[Clearing conditions]

One of the following conditions is satisfied.

- 0 is written to BPCN1 while the write enabling conditions are satisfied
- 0 is written to the MRCPC1.MRCPSEN bit.

60.5.36 MRCPS : Code MRAM Program Status Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x3010

Bit position:	7	6	5	4	3	2	1	0
Bit field:	PRGB SYC	ABUF FULL	ABUF EMP	—	—	—	ECCE RRC	PRGE RRC
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PRGERRC	Programming Error 0: Programming has been completed successfully 1: An error has occurred during programming	R/W*1

Bit	Symbol	Function	R/W
1	ECCERRC	ECC Error 0: Programming has been completed successfully 1: An error has occurred during programming	R/W ^{*1}
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	ABUFEMP	Address Buffer Empty 0: Address buffer is not empty, and code MRAM write data can flush 1: Address buffer is empty, and code MRAM write data cannot flush	R
6	ABUFFULL	Address Buffer Full 0: Address buffer is not full, and code MRAM write transaction is acceptable 1: Address buffer is full, and code MRAM write transaction is not acceptable	R
7	PRGBSYC	Code MRAM Program Busy 0: Code MRAM is not in program status 1: Code MRAM is in program status	R

Note: S-TYPE-3, P-TYPE-2

Note 1. Only 0 can be written to clear the flag after 1 is read. These bits should be checked after 0 is written and rewritten to 0 if 1 is read.

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCPS register indicates the code MRAM program status.

PRGERRC bit (Programming Error)

This bit indicates the result of code MRAM programming.

This bit can only be clear. Setting it is ignored.

When the DBGSTOPNVMER bit in the DBGSTOPPCR register is set to 1, the PRGERRCbit is not set to 1 even if a program error occurs.

[Setting condition]

- An error has occurred during programming and the ECCERRC bit is 0.

[Clearing condition]

- 0 is written after reading 1 from this bit.

ECCERRC bit (ECC Error)

This bit indicates the occurrence of an uncorrectable error when programming data from the code bus is less than 32 bytes. Programming data to the code MRAM is generated from data in the target address and data from the code bus. If a TED error occurs when reading data from the target address, programming data to the code MRAM cannot be correctly generated if the data from the code bus is less than 32 bytes. To perform programming to an address with ECCERRC set to 1, all 32 bytes data must be input from the code bus.

This bit can only be clear. Setting it is ignored.

When the DBGSTOPNVMER bit in the DBGSTOPPCR register is set to 1, the ECCERRC bit is not set to 1 even if an ECC error occurs.

[Setting condition]

- Uncorrectable error occurred when programming data from the code bus is less than 32 bytes and the PRGERRC bit is 0.

[Clearing condition]

- 0 is written after reading 1 from this bit.

ABUFEMP bit (Address Buffer Empty)

If the address buffer for the code MRAM programming is empty, the ABUFEMP bit is set to 1. When the ABUFEMP bit is 1, data in the program data buffer cannot be flushed because the MRCFL bit in the MRCFLR register cannot be written.

ABUFFFULL bit (Address Buffer Full)

If the address buffer for the code MRAM programming is full, the ABUFFFULL bit is set to 1. When the ABUFFFULL bit is 1, a new write transaction cannot be accepted until the code MRAM programming starts or finishes.

PRGBSYC bit (Code MRAM Program Busy)

If CPSEQ executes the code MRAM programming processing, the PRGBSYC bit is set to 1. When the PRGBSYC bit is 1, a new read transaction to the code MRAM cannot be accepted until the code MRAM programming finishes.

60.5.37 MRCPAEINT : Code MRAM Program Access Error Interrupt Enable Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x3014

Bit position:	7	6	5	4	3	2	1	0
Bit field:	MRCA EIE	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.	R/W
7	MRCAEIE	Code MRAM Program Access Error Interrupt Enable 0: Generation of an MRAM_MRCPR interrupt request is disabled when the ECCERRC or PRGERRC bits are 1 1: Generation of an MRAM_MRCPR interrupt request is enabled when the ECCERRC or PRGERRC bits are 1	R/W

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCPAEINT register enables or disables generation of a code MRAM program error (MRAM_MRCPR) interrupt request.

MRCAEIE bit (Code MRAM Program Access Error Interrupt Enable)

The MRCAEIE bit enables or disables MRAM_MRCPR interrupt request when a code MRAM program access error occurs and the ECCERRC or PRGERRC bits in MRCPS becomes 1.

60.5.38 MRCPEA : Code MRAM Program Error Address Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x3018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MCPEA[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MCPEA[10:0]												—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	—	These bits are read as 0.	R
31:5	MCPEA[26:0]	Code MRAM Program Error Address Indicates the address when a program error occurs.	R

Note: S-TYPE-3, P-TYPE-2

Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

The MRCPEA register stores the error address when a program or ECC error occurs.

MCPEA[26:0] bits (Code MRAM Program Error Address)

[Setting condition]

- When a program or ECC error occurs while the ECCERRC bit is 0 and the PROGERRC bit is 0, the error address is stored in the MRCPEA register.

60.5.39 MRCFLR : Code MRAM Flush Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x3030

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	MRCFL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MRCFL	Flush Write Data Buffer for Code MRAM 0: Not execute to flush program data buffer 1: Execute to flush program data buffer	R/W ^{*1 *2}
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ^{*3}

Note: S-TYPE-5, P-TYPE-2

Note 1. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0xC3.

Note 2. When the ABUFEMP bit in the MRCPS register is 1, it is not possible to write 1 to this bit.

Note 3. Written values are not retained by these bits (always reads 0x00).

The MRCFLR register is used for programming data less than 32 bytes.

MRCFL bit (Flush Write Data Buffer for Code MRAM)

This bit is used for starting code MRAM programming operation.

[Setting condition]

- 1 is written to while write enabling conditions are satisfied.

[Clearing condition]

- When the ABUFEMP bit in the MRCPS register is set to 1.

60.5.40 MRCEECC : Code MRAM ECC Encoder Control Register

Base address: MRAM = 0x4013_C000
MRAM_NS = 0x5013_C000

Offset address: 0x3804

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ECCB YPC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ECCBYPC	Code MRAM ECC Encoder Outputs Bypass Enable 0: ECC encoder output is programmed to the ECC bits of the code MRAM 1: Bits [144:128] and bits [16:0] of written data are programmed to the ECC bits of the code MRAM	R/W ^{*1}

Bit	Symbol	Function	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code	R/W ²

Note: S-TYPE-3, P-TYPE-2

Note 1. Writing to these bits is only possible when 16 bits are written, and the value written to the KEY bits is 0xCD.

Note 2. Written values are not retained by these bits (always reads 0x00).

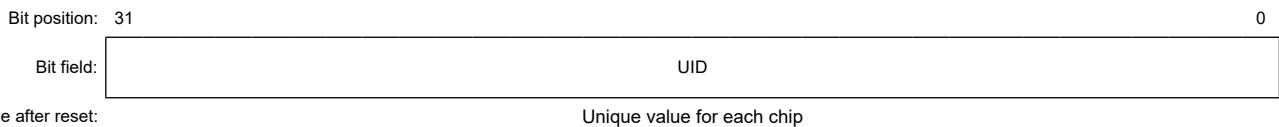
Associated SA: MSAR register. See [section 60.5.13. MSAR : MRAM Security Attribution Register](#).

ECCBYPC bit (Code MRAM ECC Encoder Outputs Bypass Enable)

The ECCBYPC bit is used to test the ECC decoder circuits for the code MRAM. For information on the test flow of the ECC decoder circuits, see [section 60.9.4. Fault Correction and Detection](#).

60.5.41 UIDRn : Unique ID Registers n (n = 0 to 3)

Address: 0x02F0_7B00 + 4 × n



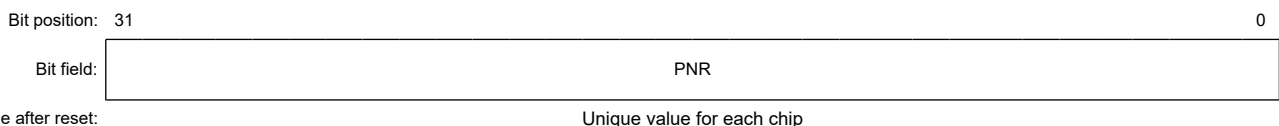
Bit	Symbol	Function	R/W
31:0	UID	Unique ID	R

Note: S-TYPE-5, P-TYPE-5

The UIDRn is a read-only register that stores a 16-byte ID code (unique ID) for identifying the individual MCU. The UIDRn register should be read in 32-bit units. When reading by the signature request command of the serial programming interface, data is read in order from the data with the large address. That is, the data in 0x02F0_7B0F is read first, and in 0x02F0_7B00 is read last.

60.5.42 PNRn : Part Numbering Register n (n = 0 to 3)

Address: 0x02C1_EC38 + 4 × n



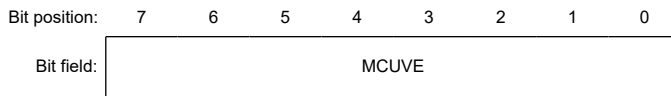
Bit	Symbol	Function	R/W
31:0	PNR	Part Number	R

Note: S-TYPE-5, P-TYPE-5

The PNRn is a read-only register that stores a 16-byte part numbering. The PNRn register should be read in 32-bit units. Each byte corresponds to the ASCII code representation of the product part number as detailed in [Table 1.14](#). The first character (R, 0x52 in ASCII code) of the part number is stored in the byte with the smallest address (0x02C1_EC38). When reading by the signature request command of the serial programming interface, the data is read in order from the data with the small address. That is, the data in 0x02C1_EC38 is read first, and in 0x02C1_EC47 is read last.

60.5.43 MCUVER : MCU Version Register

Address: 0x02C1_EC48



Value after reset: Values depend on the chip

Bit	Symbol	Function	R/W
7:0	MCUVE	MCU Version	R

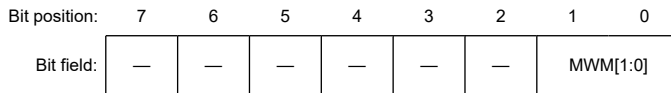
Note: S-TYPE-5, P-TYPE-5

The MCUVER is a read-only register that stores the MCU version. The MCUVER register should be read in 8-bit units.

60.5.44 MWMCR : OTP Write Mode Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xAB4



Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	MWM[1:0]	OTP Write Mode [1:0] 0 0: OTP normal speed write mode 0 1: OTP high speed write mode 0 1 0: OTP high speed write mode 1 1 1: Prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The MWMCR register controls the OTP write mode when writing data to OTP.

MWM[1:0] bits (OTP Write Mode [1:0])

1. MWM[1:0] = 00: OTP is set to OTP normal speed write mode.
You can set MWM[1:0] to 00 when $1.62\text{ V} \leq VCC \leq 3.63\text{ V}$.
Refer to the write period of OTP normal speed write mode to [section 70, Electrical Characteristics](#).
2. MWM[1:0] = 01: OTP is set to OTP high speed write mode 0.
You can set MWM[1:0] to 01 when $2.5\text{ V} \leq VCC \leq 3.63\text{ V}$.
Refer to the write period of OTP high speed write mode 0 to [section 70, Electrical Characteristics](#).
3. MWM[1:0] = 10: OTP is set to OTP high speed write mode 1.
You can set MWM[1:0] to 10 when $3.0\text{ V} \leq VCC \leq 3.63\text{ V}$.
Refer to the write period of OTP high speed write mode 1 to [section 70, Electrical Characteristics](#).
While writing OTP in high speed mode 1:
 - Make CPUCLK0, NPUCCLK and ICLK frequencies the same
 - Stop ESWM operation
 - Do not call Open process in the API of security engine.

60.6 Operating Mode

60.6.1 MRAM Read/Program Mode

Code MRAM and extra MRAM have modes for reading and programming as shown in [Figure 60.7](#).

Transition from code MRAM read mode to code MRAM program mode is initiated by a software request generated by either of following conditions.

Programming of code MRAM starts automatically after transition to code MRAM program mode. When programming of code MRAM is completed, mode is transit from code MRAM program mode to code MRAM read mode automatically.

- Program data buffer is full
- Written code MRAM address is out of 32 bytes boundary from stored data in a program data buffer
- The MRCFL bit in the MRCFLR register is set to 1.

Transitions between extra MRAM read mode and extra MRAM program mode are initiated by changing the value of the MENTRYR register or by W-HUK zeroization. See [section 60.8.6. HUK Protection](#). MRAM is in extra MRAM read mode when the MENTRYR register is 0x0000 and W-HUK zeroization is not executing. MRAM is in extra MRAM program mode in other cases. Programming of extra MRAM can be started using the MACI command in extra MRAM program mode or by starting W-HUK zeroization.

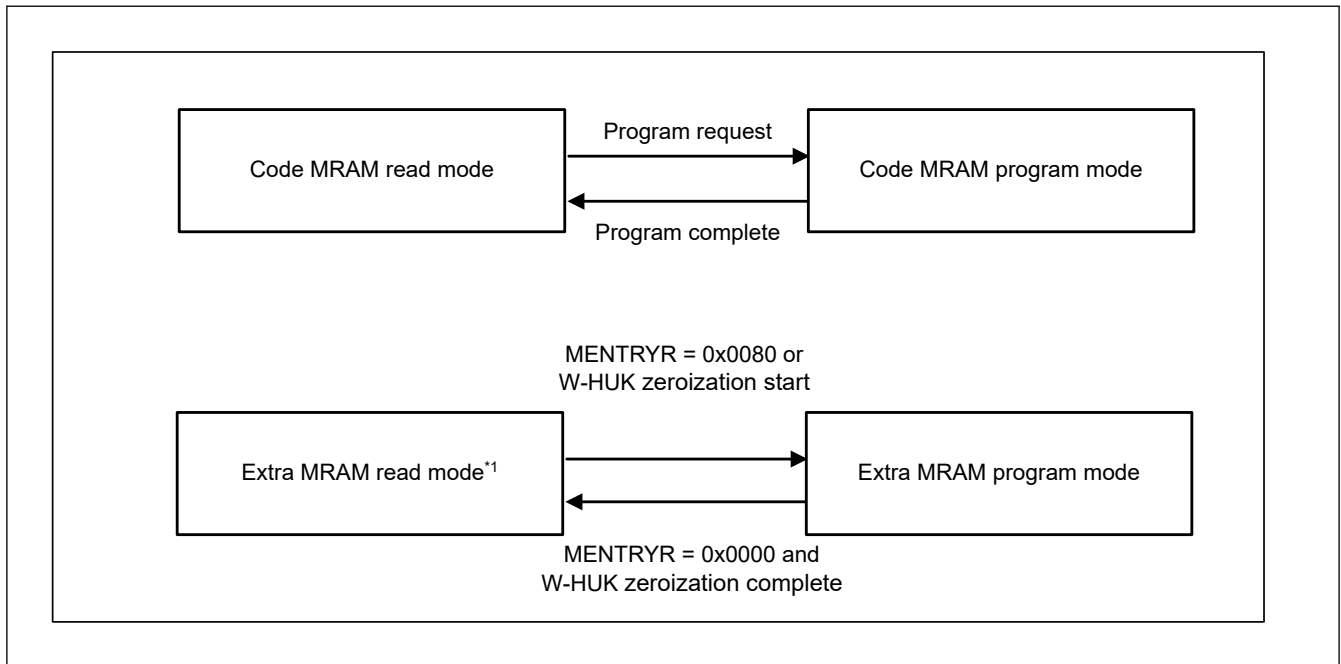


Figure 60.7 MRAM read/program mode transition

60.6.2 MRAM High-speed Program Mode

MRAM has normal program mode and high-speed program mode as shown in [Figure 60.8](#), and the same mode setting is applied for both code MRAM program mode and extra program mode.

After setting the MHSPEN bit of the MRPSC register to 1 in MHSPM disabled status in normal program mode, the status changes to MHSPM start ready. When code MRAM or extra MRAM is programmed in MHSPM start ready status once, MRAM program mode changes to MHSPM enabled status in high-speed program mode.

To return to normal program mode from MHSPM enabled status, set the MHSPEN bit to 0, read the MHSPEN bit once, and transition into MHSPM end ready status at first. When both code MRAM and extra MRAM are accessed in MHSPM end ready status, MRAM program mode changes to MHSPM disabled status in normal program mode.

When the MCU transitions to Software Standby mode, the MRAM program mode changes to normal program mode while retaining the MHSPEN bit. If the MHSPEN bit is set to 1, programming the code MRAM or extra MRAM once after returning from Software Standby mode changes the MRAM program mode to high-speed program mode.

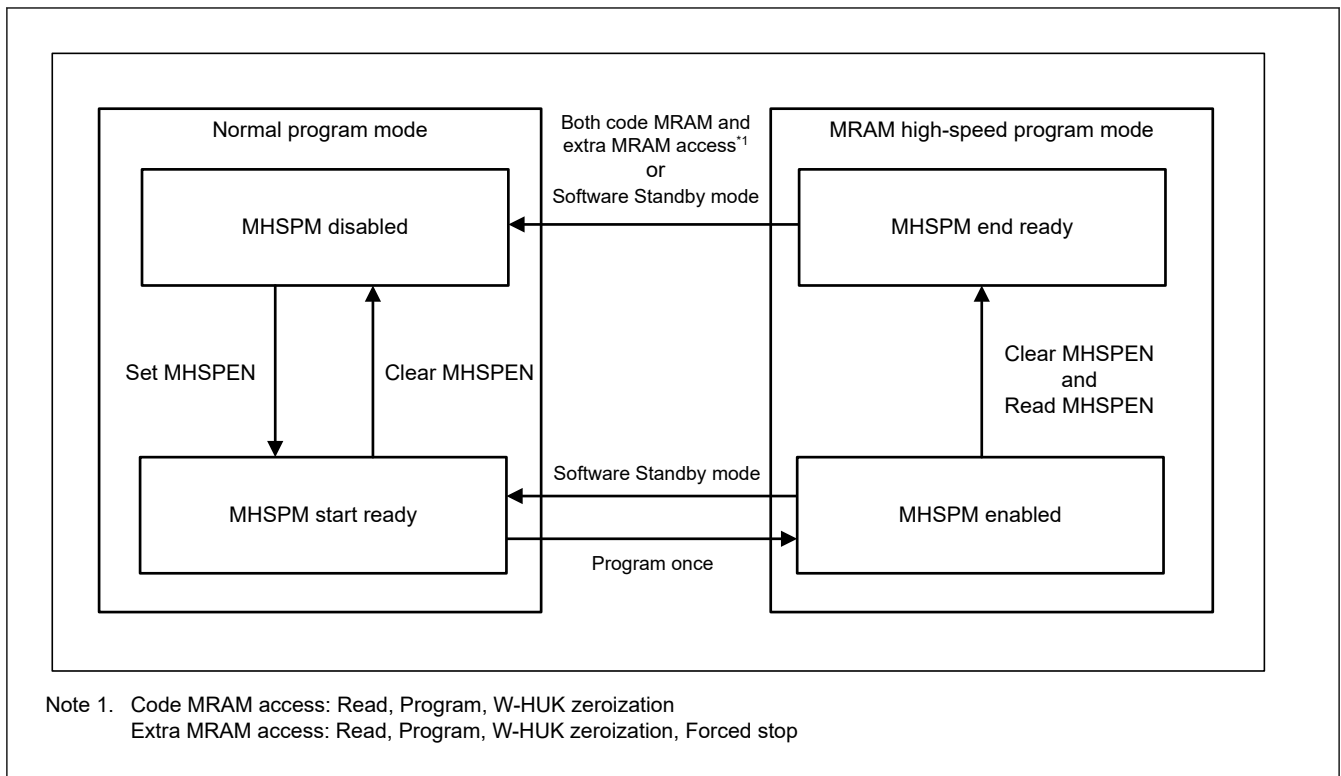


Figure 60.8 High-speed program mode transition

60.6.3 Debugger Support Functions

Precondition of functions

For non-secure debugging, the debugger can read and program to the Code MRAM area with non-secure attributes. Registers set to the secure attribute cannot be read and written by non-secure access. It is assumed that the MRCFREQ register and MREFREQ register are set to the appropriate values in user software when user software is running. For the debugger download, or break setting is enabled before user software startup, the chip startup frequency must be set to the initial register values of MRCFREQ and MREFREQ registers.

Program function

Code MRAM can be programmed by setting the NVMWE bit in the DBGNVMCR register to 1 without setting the MRCPC0 and MRCPC1 registers. However, it cannot be programmed in areas protected by block protection. If the block protection needs to be cleared, execute the Configuration program command in serial programming mode to do so. To avoid interfering with the operation of the debugger, when the DBGSTOPNVMER bit in the DBGSTOPPCR register is set to 1, it masks the setting of the ECCERRC and PRGERRC bits in the MRCPS register, which is an error notification register during code MRAM programming. If the error bits were already set before the DBGSTOPNVMER bit is set, the value is held until the register is cleared.

Debugging by the debugger targets the code and data areas, so errors in the program caused by MACI commands other than in the data area of the extra MRAM are not masked.

Read function

The TEDERRC and DECERRC bits of the MRCRAES register and the TEDERRC and DECERRC bits of the MRERAES register are not set if an ECC error occurs during access from the debugger. DBGSTOPNVMER bit is not used for masking ECC error in reading from the debugger.

60.7 MACI Commands

60.7.1 Programmable Area with MACI Commands

Table 60.10 shows the MRAM and OTP areas in extra MRAM that can be programmed with the MACI command. These areas store setting data for security, safety, and system configuration. Each bit (both data and ECC) in the OTP area can only

be programmed once from 1 to 0. There are two types of OTP areas with ECC and without ECC as shown in [Table 60.10](#). A unit of 16-bytes data can be programmed only once to the OTP area with ECC. Reprogramming of different data cannot be retried because some bits of ECC may need to change from 0 to 1. Reliability of OTP area without ECC is secured by redundant data configuration. A unit of 16-bytes data can be programmed multiple times to OTP area without ECC when there is no bit changing from 0 to 1.

Table 60.10 Programmable area with MACI commands

Address (Offset address: BASE_MC)	Area name	Memory type	ECC
0x00C9_F020 to 0x00C9_F0AF	Configuration	MRAM	Use
0x00C9_F0C0 to 0x00C9_F7FF	Configuration	MRAM	Use
0x00E0_7400 to 0x00E0_74BF	Hash of OEM_ROOT_PKn (n = 0 to 3)	OTP	Use
0x00E0_7600 to 0x00E0_760F	FSBL setting	OTP	Use
0x00E0_7610 to 0x00E0_769F	Start address of measurement report Start address of Code Certificate	OTP	Use
0x00E0_76A0 to 0x00E0_76FF	General purpose OTP	OTP	Use
0x00E1_7700 to 0x00E1_777F	Permanent Block Protect Setting for Secure	OTP	Do not use
0x00E1_7780 to 0x00E1_77FF	Permanent Block Protect Setting	OTP	Do not use
0x00E1_7900 to 0x00E1_790F	OFS Permanent Lock Setting	OTP	Do not use
0x00E1_7910 to 0x00E1_791F	REVOKE	OTP	Do not use
0x00E1_7920 to 0x00E1_792F	Zeroization HUK in OTP Enable	OTP	Do not use
0x00E1_7930 to 0x00E1_793F	Anti-Rollback Counter Setting	OTP	Do not use
0x00E1_7950 to 0x00E1_79FF	Reserved	OTP	Do not use
—	Anti-Rollback Counter	OTP	Do not use

60.7.2 List of MACI Commands

Table 60.11 List of MACI commands

MACI command	Function
Program	This command is used to program extra MRAM area except for the configuration area. The program unit is 16 bytes.
Status Clear	This command initializes the ILGLERR, PRGERR, ILGCOMERR, SECERR, OTERR, TZFERR, and CFGPRGERR bits in the MSTATR register and the CMDLK and MREAE bits in the MASTAT register, including the extra MRAM sequencer released from command-locked state.
Forced Stop	This command forcibly stops processing of MACI commands and initializes the MSTATR and MASTAT register.
Configuration set	This command is used to program configuration area for setting security function and safety function Units of setting are 16 bytes.
Increment Counter (Secure access only)	The anti-rollback counter value can be incremented. The target counter is selected by MCNTSELR register. Update data unit is 1 bit.
Read Counter (Secure access only)	The anti-rollback counter value can be read. The target counter is selected by the MCNTSELR register. Reading data is output to MCNTDATAR0 and MCNTDATAR1 registers. Read unit is 64 bits.

The MACI commands are issued by writing to the MACI command-issuing area (see [Table 60.4](#)). When write access as shown in [Table 60.12](#) proceeds in the specified state, the extra MRAM sequencer executes the processing corresponding to the given command (see [section 60.7.3. Relationship Between the Extra MRAM Sequencer State and MACI Commands](#)).

Table 60.12 MACI command formats

MACI commands	Number of write access	Write data to the MACI command-issuing area			
		1st access	2nd access	3rd to (N+2)th access	(N+3)th access
Program 16-byte programming: N = 8	N + 3	0xE8	0x08 (= N)	WD ₁ to WD _N	0xD0
Status Clear	1	0x50	—	—	—
Forced Stop	1	0xB3	—	—	—
Configuration set N = 8 (16 Bytes)	N + 3	0x40	0x08 (= N)	WD ₁ to WD _n	0xD0
Increment Counter	2	0x35	0xD0	—	—
Read Counter	2	0x39	0xD0	—	—

Note: WDN (N = 1, 2,...): Nth 16-bit data to be programmed.

The extra MRAM sequencer clears the MSTATR.MRDY bit to 0 at the start of processing of a command other than the Status Clear command. The extra MRAM sequencer also sets this bit to 1 on completion of command processing.

If the setting of the MRDYIE.MRDYIE bit is 1, MRAM_ENDOFPE interrupt is generated when the MSTATR.MRDY bit is set to 1.

60.7.3 Relationship Between the Extra MRAM Sequencer State and MACI Commands

The MACI commands are accepted according to the mode/state of the extra MRAM sequencer. MACI commands should be issued after the transition to the extra MRAM program mode and checking the state of the extra MRAM sequencer. Use the MSTATR and MASTAT registers to check the state of the extra MRAM sequencer. In addition, the occurrence of errors in general can be checked by reading the CMDLK bit in the MASTAT register. Its value is the logical OR of the ILGLERR, ILGCOMERR, SECERR, OTERR, TZFERR, PRGERR, and CFGPRGERR bits in the MSTATR register. [Table 60.13](#) lists the available commands in each operating mode.

Table 60.13 Operating mode and available commands

Operation mode	MENTRYR	Available commands
Extra MRAM read mode	0x0000	None
Extra MRAM program mode (W-HUK zeroization is not executed)	0x0080	Program Status Clear Forced Stop Configuration Set Increment Counter Read Counter
Extra MRAM program mode (W-HUK zeroization is executed)	0x0000/0x0080 ^{*1}	None

Note 1. MRAM is in extra MRAM program mode when W-HUK zeroization is executed regardless of MENTRYR value. See [section 60.6.1. MRAM Read/Program Mode](#).

[Table 60.14](#) shows the state of the extra MRAM sequencer and acceptable MACI commands. An appropriate mode is assumed to have been set before the commands are executed.

Table 60.14 Acceptable MACI commands and the state of the extra MRAM sequencer

	Program, Configuration set, Increment Counter, Read Counter, Forced Stop command processing	Command-locked state (MRDY = 1)	Command-locked state (MRDY = 0)	W-HUK zeroization is executed	Other state
MRDY bit	0	1	0	0/1	1
CMDLK bit	0	1	1	0/1	0
Program	X	X	X	X	O
Status Clear	X	O	X	X	O
Forced Stop	O	O	O	X	O
Configuration set	X	X	X	X	O
Increment Counter	X	X	X	X	O
Read Counter	X	X	X	X	O

Remarks:

O: Acceptable

X: Not acceptable (places the sequencer in the command-locked state)

60.7.4 Usage of MACI Commands

This section gives an overview of the usage of MACI commands.

60.7.4.1 Overview of Command Usage in Extra MRAM Program Mode

[Figure 60.9](#) shows an overview of MACI command usage in extra MRAM program mode. See [Table 60.13](#) for the available commands in extra MRAM program mode.

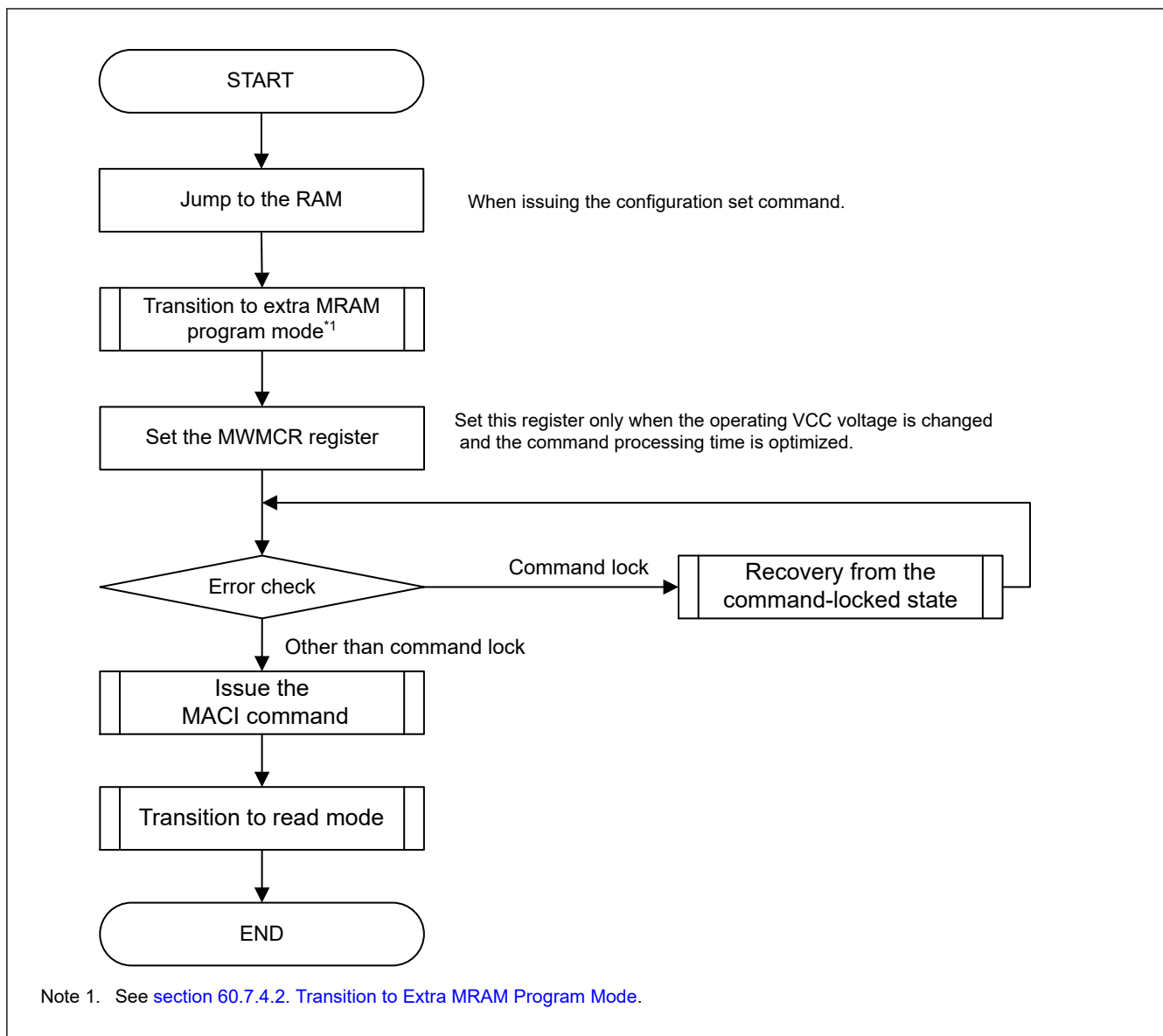


Figure 60.9 Overview of command usage in extra MRAM program mode

60.7.4.2 Transition to Extra MRAM Program Mode

To issue the MACI commands for the extra MRAM, a transition to extra MRAM program mode is required. To cause the transition to extra MRAM program mode, set the MENTRY bit in the MENTRYR register to 1.

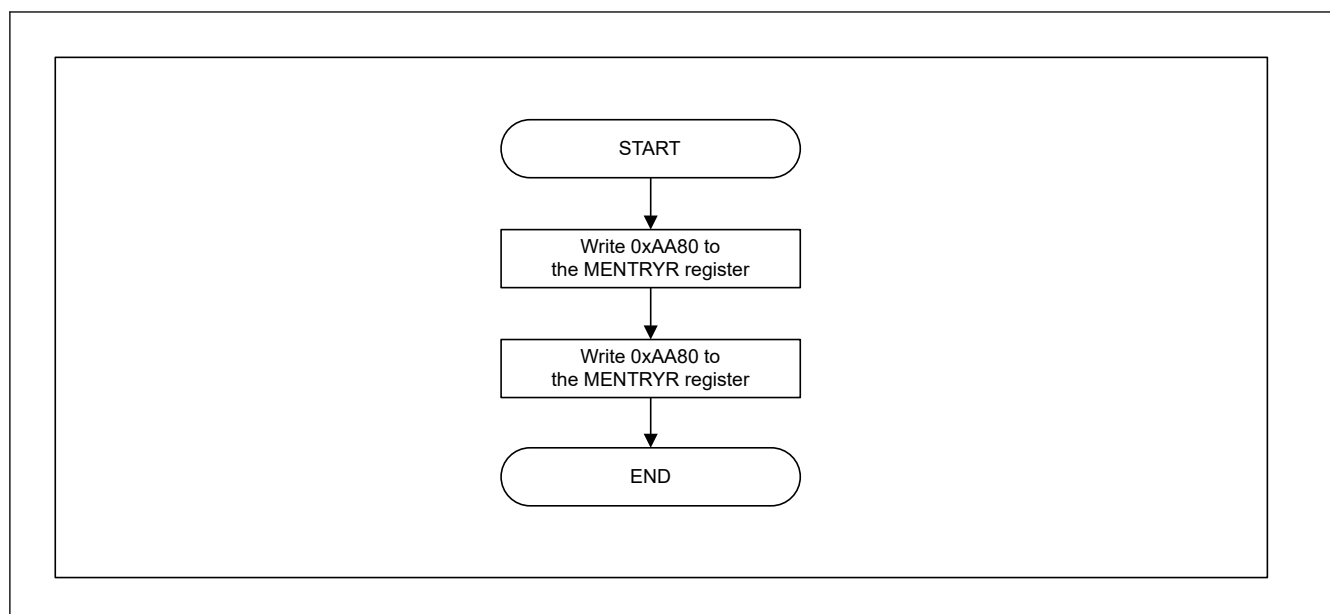


Figure 60.10 Procedure for transition to extra MRAM program mode

60.7.4.3 Transition to Read Mode

To read the extra MRAM, the MRDY bit in the MSTATR register must be set to 1 or a transition to read mode is required. To cause a transition to read mode, set the MENTRYR register to 0x0000. The transition to read mode should be made while processing by the extra MRAM sequencer is completed and the operation is in a state other than command-locked.

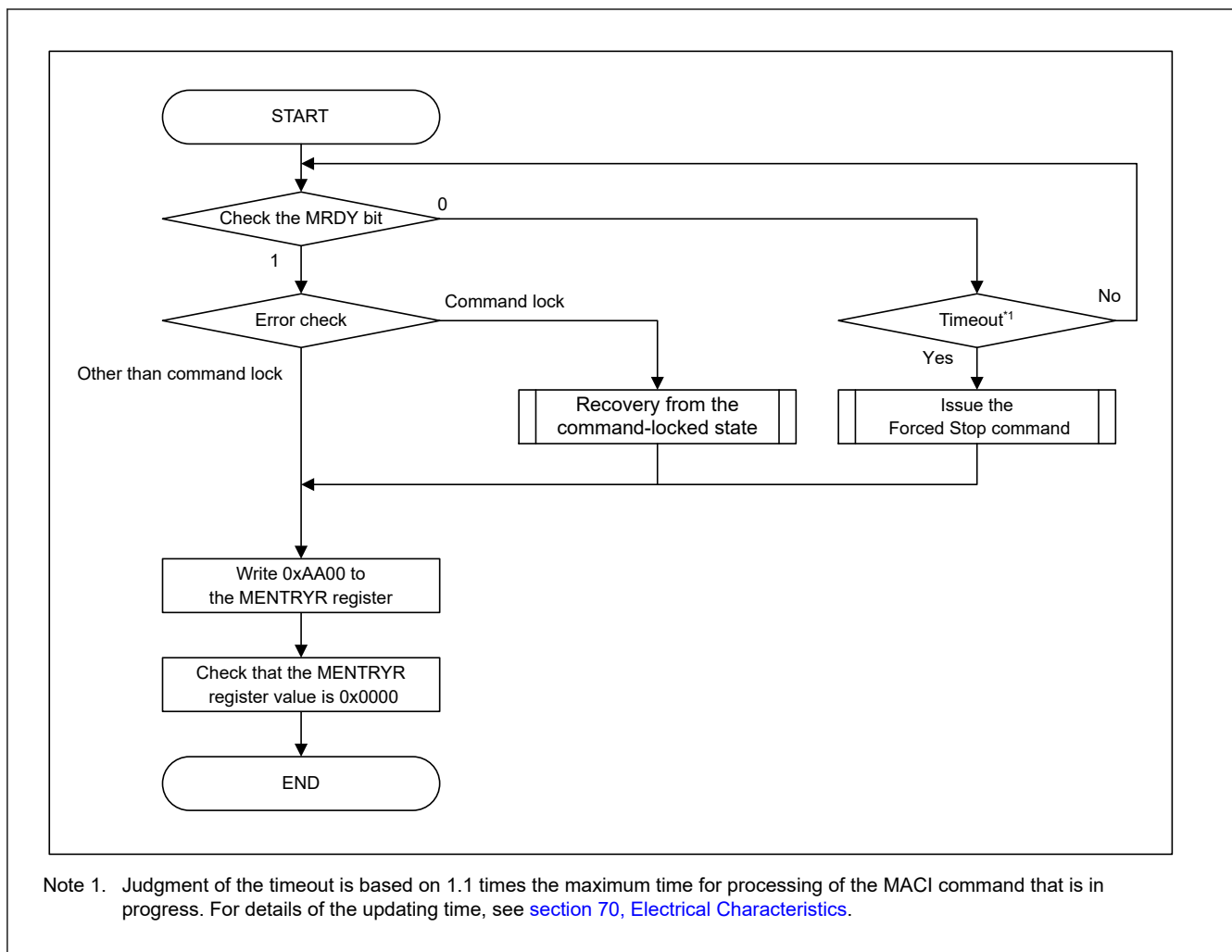


Figure 60.11 Procedure for transition to read mode

60.7.4.4 Recovery from the Command-Locked State

When the extra MRAM sequencer enters the command-locked state, MACI commands cannot be accepted. To release the sequencer from the command-locked state, use the Status Clear command, Forced Stop command.

If processing is not completed in time beyond the maximum programming/erasure time specified in the electrical characteristics, this can be considered a timeout, and the extra MRAM sequencer should be stopped by the Forced Stop command.

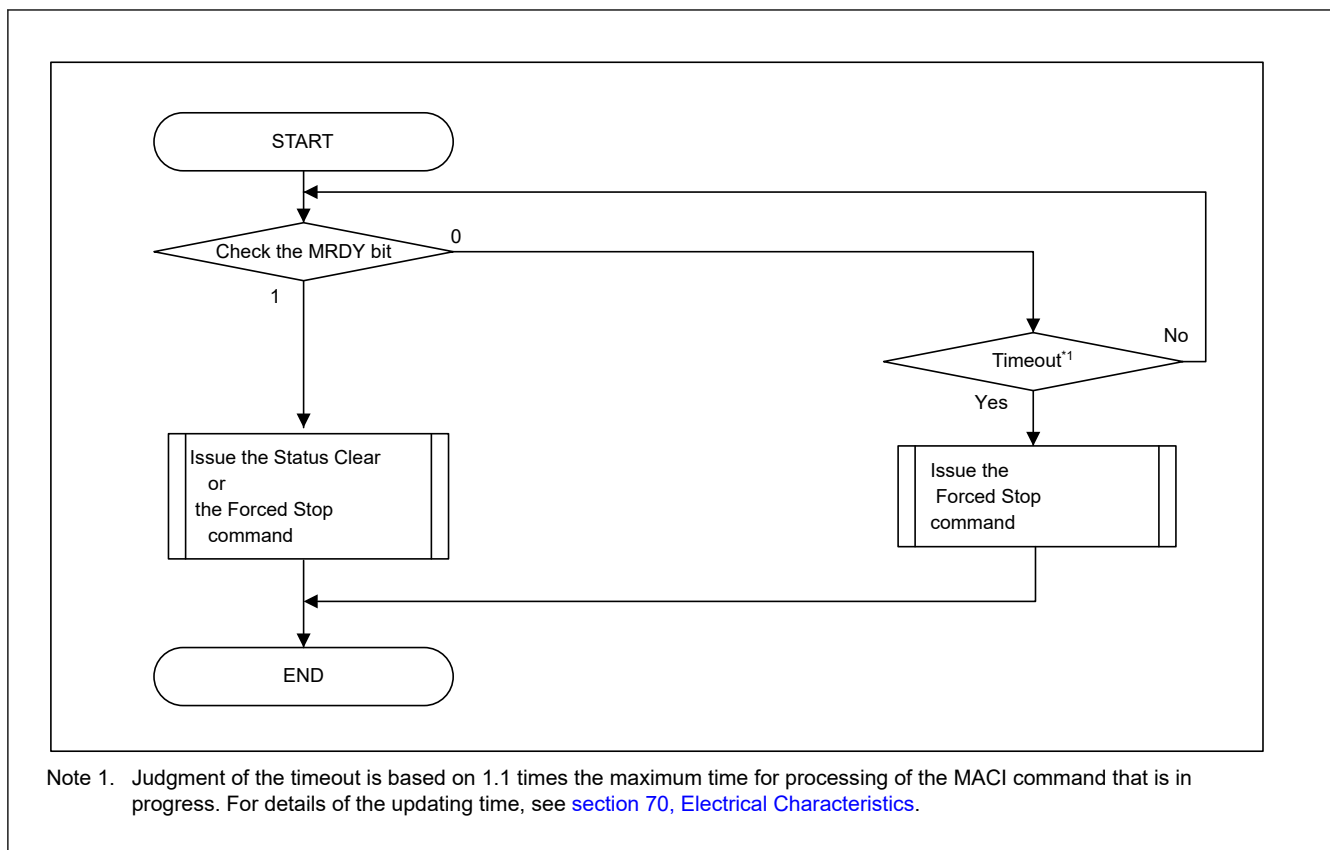
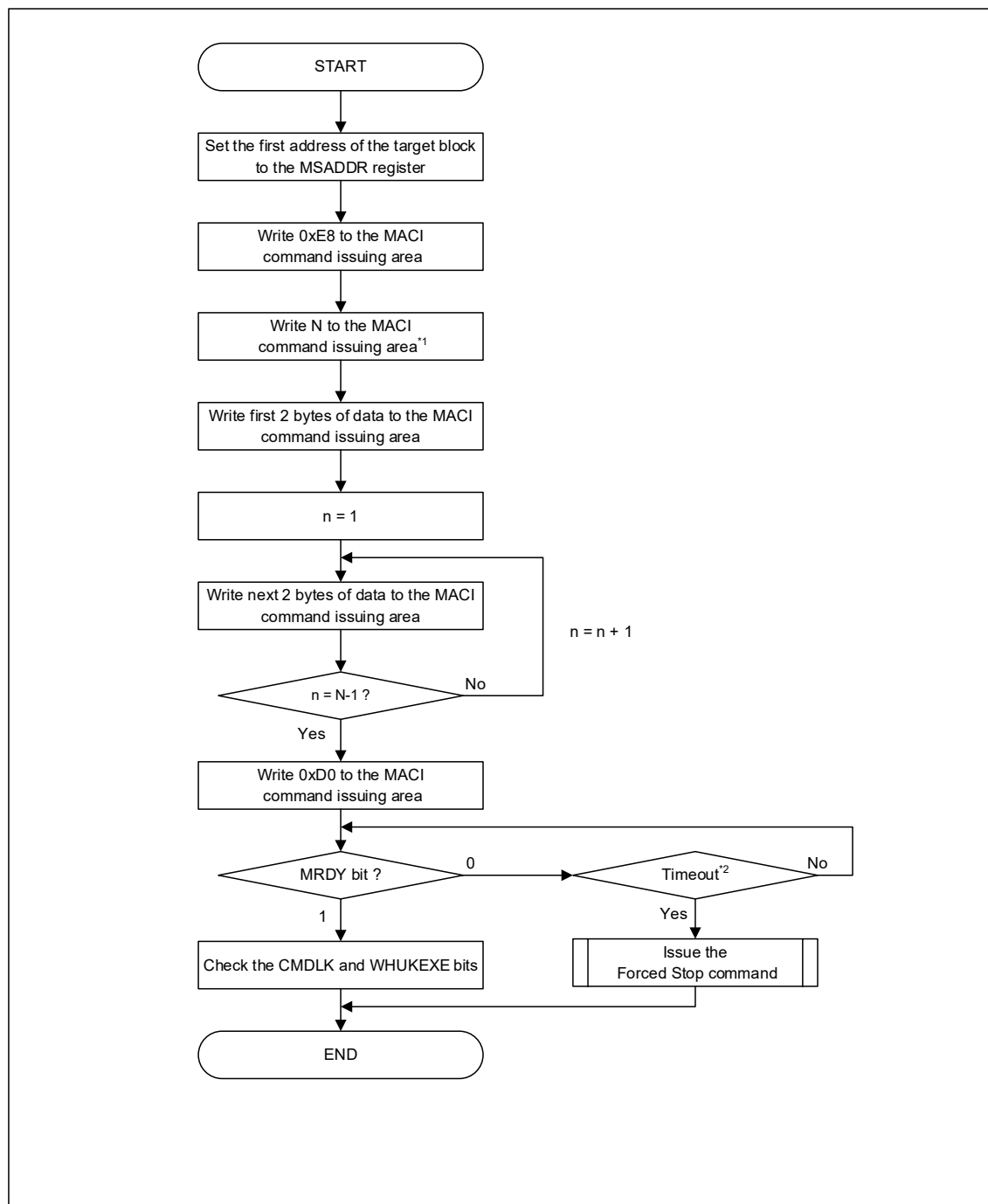


Figure 60.12 Recovery from the command-locked state

60.7.4.5 Program Command

The Program command is used for programming to the Extra MRAM area, excluding the configuration area. Before issuing the Program command, set the first address of the target block in the MSADDR register.

Writing 0xD0 to the MACI command issuing area at the final access of the MACI command issuing starts the program command processing.



Note 1. N

0x08: 16-bytes programming

Note 2. Judgment of the timeout is based on 1.1 times the maximum time of programming. For details of the updating time, see [section 70, Electrical Characteristics](#).

Figure 60.13 Usage of the Program command

The correspondence between the possible target data and the address value set in the MSADDR register is shown in [Table 60.15](#). For details on the MSADDR register, see [section 60.5.19. MSADDR : MACI Command Start Address Register](#).

Table 60.15 Address used by Program command

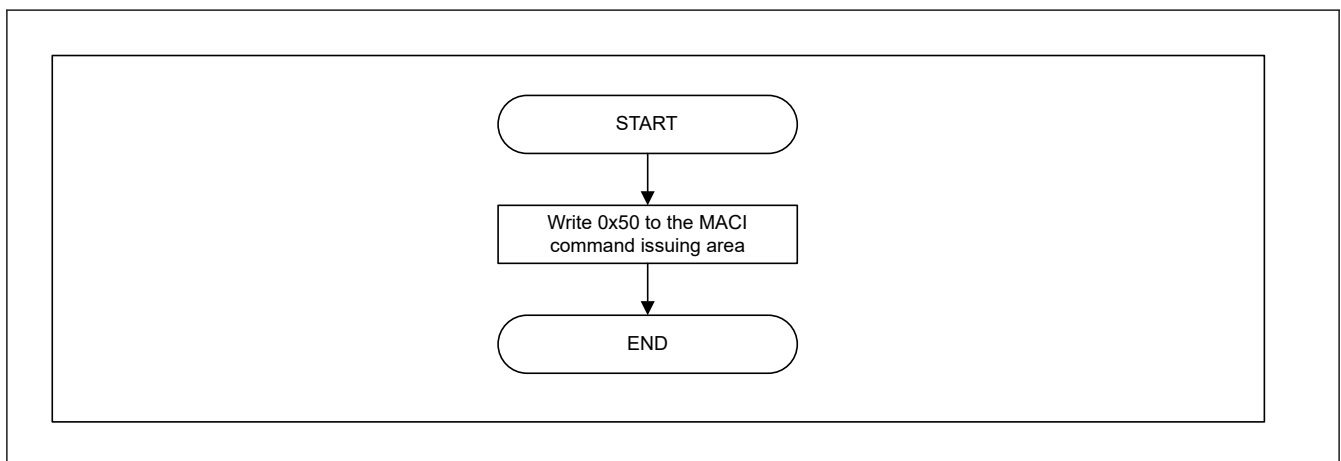
Address	MSADDR register value	Setting data	Timing when the setting is enabled
BASE_MC + 0x00E0_7600	BASE_MCS + 0x00E0_7600	FSBL setting	At a reset
BASE_MC + 0x00E0_7610	BASE_MCS + 0x00E0_7610 to BASE_MCS + 0x00E0_7690	Start address of measurement report Start address of code certificate	—
BASE_MC + 0x00E0_76A0	BASE_MCS + 0x00E0_76A0 to BASE_MCS + 0x00E0_76F0	General purpose OTP	—
BASE_MC + 0x00E1_7700	BASE_MCS + 0x00E1_7700 to BASE_MCS + 0x00E1_7770	Permanent Block Protect Setting for Secure (PBPS_SEC)* ¹	When a reset or command is executed
BASE_MC + 0x00E1_7780	BASE_MCN + 0x00E1_7780 to BASE_MCN + 0x00E1_77F0	Permanent Block Protect Setting (PBPS)* ²	When a reset or command is executed
BASE_MC + 0x00E1_7900	BASE_MCS + 0x00E1_7900	Permanent OFS Protect Setting (POFSPS)	When a reset or command is executed
BASE_MC + 0x00E1_7910	BASE_MCS + 0x00E1_7910	REVOKE	When a reset or command is executed
BASE_MC + 0x00E1_7920	BASE_MCS + 0x00E1_7920	Zeroization HUK Enable	When a reset or command is executed
BASE_MC + 0x00E1_7930	BASE_MCS + 0x00E1_7930	Anti-rollback Counter Setting	—
BASE_MC + 0x00E1_7950	BASE_MCS + 0x00E1_7950 to BASE_MCS + 0x00E1_79F0	Reserved	—

Note 1. The PBPS_SEC[n] bit can be programmed to 0 when the BPS_SEC[4n], BPS_SEC[4n + 1], BPS_SEC[4n + 2], and BPS_SEC[4n + 3] bits are all 0.

Note 2. The PBPS[n] bit can be programmed to 0 when the BPS[4n], BPS[4n + 1], BPS[4n + 2], and BPS[4n + 3] bits are all 0.

60.7.4.6 Status Clear Command

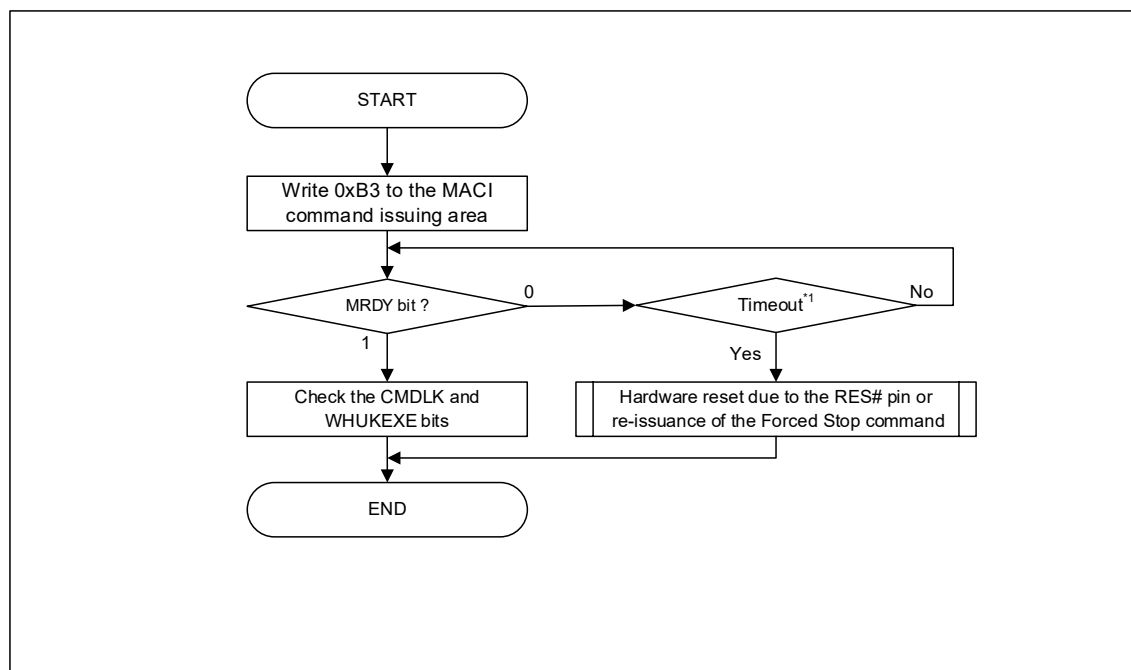
The Status Clear command is used to clear the command-locked state (see [section 60.7.4.4. Recovery from the Command-Locked State](#)). To clear the ILGLERR, ILGCOMERR, SECERR, OTERR, PRGERR, TZFERR, and CFGPRGERR bits in the MSTATR register in the command-locked state, the Status Clear command is available.

**Figure 60.14 Usage of the Status Clear command**

60.7.4.7 Forced Stop Command

The Forced Stop command forcibly ends command processing by the extra MRAM sequencer. Although this command halts command processing, values from the programming that was in progress are not guaranteed. Furthermore, resumption of processing is not possible. Processing of programming that was halted by the Forced Stop command is also defined as one from the perspective of programming endurance.

Executing the Forced Stop command also initializes part of the MACI, the whole extra MRAM sequencer, the MSTATR register, and the MASTAT register. Accordingly, this command can be used in the procedure for recovery from the command-locked state and in processing in response to a timeout of the extra MRAM sequencer (see [section 60.7.4.4. Recovery from the Command-Locked State](#)).

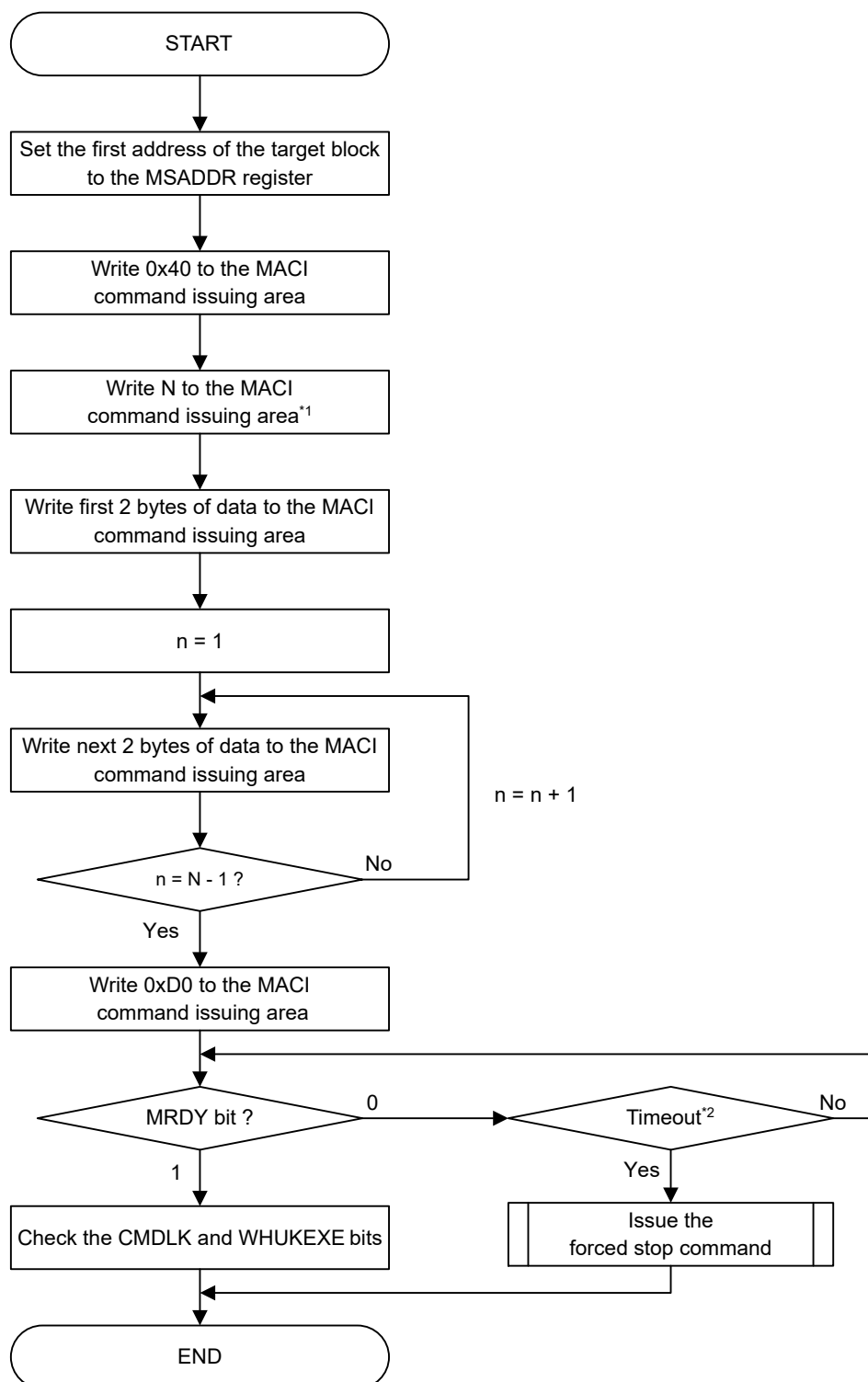


Note 1. Judgment of the timeout is based on 1.1 times the maximum time of the Forced Stop command latency. For details of the updating time, see [section 70, Electrical Characteristics](#).

Figure 60.15 Usage of the Forced Stop command

60.7.4.8 Configuration Set Command

The Configuration Set command is used for security function and safety function. Before issuing the Configuration Set command, set the specified address (shown in [Table 60.16](#)) in the MSADDR register. Writing 0xD0 to the MACI command issuing area in the final access (for issuing the MACI command) starts MACI processing of the Configuration Set command.



Note 1. N

0x08: 16-bytes programming

Note 2. Judgment of the timeout is based on 1.1 times the maximum time of the command.
See the Electrical Characteristics of the User's Manual about those time.

Figure 60.16 Usage of the Configuration Set command

The correspondence between the possible target data for configuration setting and the address value set in the MSADDR register is shown in [Table 60.16](#). For details on the MSADDR register, see [section 60.5.19. MSADDR : MACI Command Start Address Register](#).

Table 60.16 Address used by the Configuration Set command

Address	MSADDR register value	Setting data	Operation of additional writing		Timing when the setting is enabled
			POFSPS.POFSPS [13] bit is 1	POFSPS.POFSPS [13] bit is 0	
BASE_MC + 0x00C9_F040	BASE_MCS + 0x00C9_F040	Option Function Select Register 0 (OFS0)	Writable ^{*1}	Writable ^{*1}	At a reset
BASE_MC + 0x00C9_F044	BASE_MCS + 0x00C9_F040	Option Function Select Register 2 (OFS2)	Writable ^{*1}	Writable ^{*1}	At a reset
BASE_MC + 0x00C9_F074	BASE_MCS + 0x00C9_F070	Startup Area Setting Register (SAS)	Writable ^{*1}	Not writable ^{*1*2}	When a reset or command is executed
BASE_MC + 0x00C9_F0C0	BASE_MCS + 0x00C9_F0C0	Option Function Select Register 1 Secure (OFS1_SEC)	Writable ^{*1}	Writable ^{*1}	At a reset
BASE_MC + 0x00C9_F0C4	BASE_MCS + 0x00C9_F0C0	Option Function Select Register 3 Secure (OFS3_SEC)	Writable ^{*1}	Writable ^{*1}	At a reset
BASE_MC + 0x00C9_F120	BASE_MCS + 0x00C9_F120	Option Function Select Register 1 Secure (OFS1_SEL)	Writable ^{*1}	Writable ^{*1}	At a reset
BASE_MC + 0x00C9_F124	BASE_MCS + 0x00C9_F120	Option Function Select Register 3 Secure (OFS3_SEL)	Writable ^{*1}	Writable ^{*1}	At a reset
BASE_MC + 0x00C9_F200	BASE_MCS + 0x00C9_F200 BASE_MCS + 0x00C9_F210 BASE_MCS + 0x00C9_F220 BASE_MCS + 0x00C9_F230	Block Protect Setting Register Secure (BPS_SEC)	Writable ^{*3}	Writable ^{*3}	When a reset or command is executed
BASE_MC + 0x00C9_F4C0	BASE_MCN + 0x00C9_F4C0	Option Function Select Register 1 (OFS1)	Writable	Writable	At a reset
BASE_MC + 0x00C9_F4C4	BASE_MCN + 0x00C9_F4C0	Option Function Select Register 3 (OFS3)	Writable	Writable	At a reset
BASE_MC + 0x00C9_F600	BASE_MCN + 0x00C9_F600 BASE_MCN + 0x00C9_F610 BASE_MCN + 0x00C9_F620 BASE_MCN + 0x00C9_F630	Block Protect Setting Register (BPS)	Writable ^{*4}	Writable ^{*4}	When a reset or command is executed

Note 1. This area can be protected by the POFSPS.POFSPS bit. If the POFSPS.POFSPS bit corresponding to the update area is 0, it cannot be updated by the Configuration Set command.

Note 2. The POFSPS.POFSPS[13] bit cannot be restored to 1 using the Program command once it is set to 0. Therefore, setting the startup area select flags and startup size select flags again becomes impossible. When the Configuration Set command is issued to the address of BASE_MCS + 0x00C9_F074, the command is locked. Exercise extra caution when handling the POFSPS.POFSPS[13] bit.

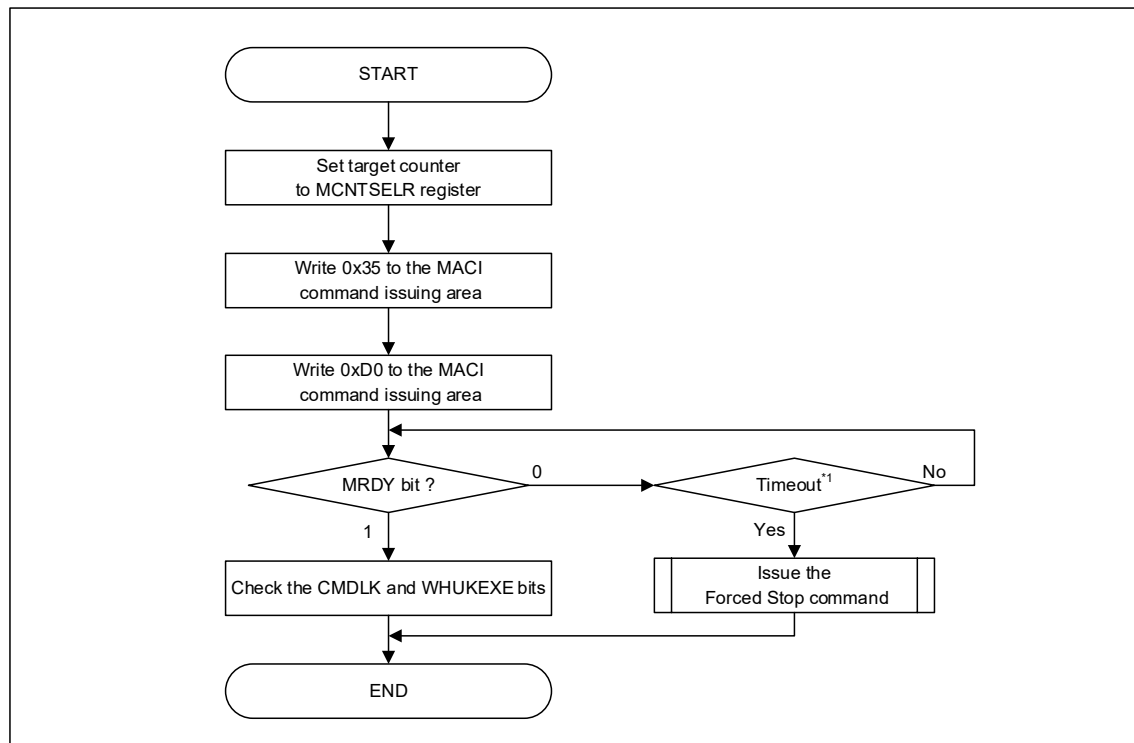
Note 3. Once the PBPS_SEC[n] bit is set to 0, the BPS_SEC[4n], BPS_SEC[4n + 1], BPS_SEC[4n + 2], and BPS_SEC[4n + 3] bits cannot be restored to 1 using the Configuration Set command.

Note 4. Once the PBPS[n] bit is set to 0, the BPS[4n], BPS[4n + 1], BPS[4n + 2], and BPS[4n + 3] bits cannot be restored to 1 using the Configuration Set command.

60.7.4.9 Increment Counter Command

The Increment Counter command is used for security function and safety function, and for updating three types of anti-rollback counter (ARC_SEC, ARC_NSEC, and ARC_OEMBL). The counter update is executed for increment processing and not for decrement processing.

Writing 0xD0 at the 2nd write access of the extra MRAM sequencer command triggers MACI to start the Increment Counter command processing. Completion of command processing can be confirmed by the MRDY bit of the MSTATR register.



Note 1. Judgment of the timeout is based on 1.1 times the maximum time of the command.
See the Electrical Characteristics of the User's Manual about those time.

Figure 60.17 Usage of the Increment Counter command

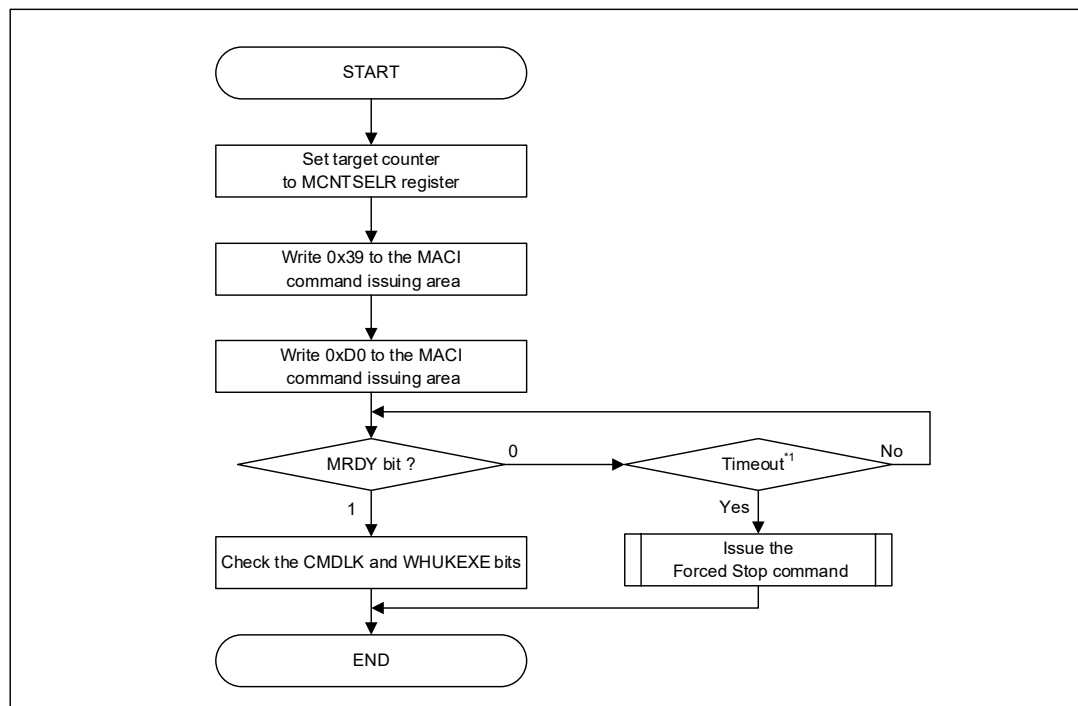
The anti-rollback counter value is not increased by the following conditions using the Increment Counter command.

- The target counter is protected by ARCSEC_LK, ARCNSC_LK0 to 3, or ARCOEMBL_LK.
- The previous command is not the Read Counter command against ARC_OEMBL when the target counter is ARC_OEMBL.
- The update value of ARC_OEMBL is equal to or higher than the OEM boot loader version (OEM BL Ver.) output from the security engine.
- The target counter value is maximum (error does not occur).
- The target counter is ARC_NSEC when CNF_ARCNSC0 to 1 is an invalid setting.

60.7.4.10 Read Counter Command

The Read Counter command is used for security function and safety function, and for reading three types of anti-rollback counter (ARC_SEC, ARC_NSEC, and ARC_OEMBL). The extra MRAM sequencer reads 64-bits data, and stores to MCNTDATAR0 and MCNTDATAR1 registers.

Writing 0xD0 at the 2nd write access of the extra MRAM sequencer command triggers MACI to start the Read Counter command processing. Completion of command processing can be confirmed by the MRDY bit of the MSTATR register.



Note 1. Judgment of the timeout is based on 1.1 times the maximum time of the command.
See the Electrical Characteristics of the User's Manual about those time.

Figure 60.18 Usage of the Read Counter command

60.8 Security Function

The extra MRAM sequencer supports the following security functions in a security-enabled product:

- Permanent OFS protect setting
- Security flag for startup area and startup size
- Permanent block protect setting
- MRAM protection for TrustZone
- Extra MRAM configuration area protection
- Anti-rollback counter
- HUK protection
- OEM_ROOT_KEY protection.

60.8.1 Permanent OFS Protect Setting

OFS area in the configuration area can be permanent protected by POFSPS bits in the OTP area as shown in [Table 60.17](#).
See the User's Manual: Section xx, Option-Setting Memory.

Table 60.17 Protection condition of OFS area in configuration area (1 of 2)

Protect address	MSADDR register value	Protection size	Protect bit	Remarks
BASE_MC + 0x00C9_F040	BASE_MCS + 0x00C9_F040	4 Bytes	POFSPS[0]	—
BASE_MC + 0x00C9_F044	BASE_MCS + 0x00C9_F040	4 Bytes	POFSPS[1]	—

Table 60.17 Protection condition of OFS area in configuration area (2 of 2)

Protect address	MSADDR register value	Protection size	Protect bit	Remarks
BASE_MC + 0x00C9_F048	BASE_MCS + 0x00C9_F040	4 Bytes	POFSPS[2]	—
BASE_MC + 0x00C9_F04C	BASE_MCS + 0x00C9_F040	4 Bytes	POFSPS[3]	—
⋮	⋮	⋮	⋮	⋮
BASE_MC + 0x00C9_F06C	BASE_MCS + 0x00C9_F060	4 Bytes	POFSPS[11]	—
BASE_MC + 0x00C9_F070	BASE_MCS + 0x00C9_F070	4 Bytes	POFSPS[12], POFSPS[13]	When POFSPS[13] is 0, programming in the protect area with the program command generates SECERR.
BASE_MC + 0x00C9_F074	BASE_MCS + 0x00C9_F070	4 Bytes	POFSPS[13]	
BASE_MC + 0x00C9_F078	BASE_MCS + 0x00C9_F070	4 Bytes	POFSPS[13], POFSPS[14]	
BASE_MC + 0x00C9_F07C	BASE_MCS + 0x00C9_F070	4 Bytes	POFSPS[13], POFSPS[15]	
BASE_MC + 0x00C9_F080	BASE_MCS + 0x00C9_F080	4 Bytes	POFSPS[16]	—
⋮	⋮	⋮	⋮	⋮
BASE_MC + 0x00C9_F0B0	BASE_MCS + 0x00C9_F0B0	4 Bytes	POFSPS[28]	For the target address, even if 0 is written to the POFSPS bit, the protection is disabled.
BASE_MC + 0x00C9_F0B4	BASE_MCS + 0x00C9_F0B0	4 Bytes	POFSPS[29]	
BASE_MC + 0x00C9_F0B8	BASE_MCS + 0x00C9_F0B0	4 Bytes	POFSPS[30]	
BASE_MC + 0x00C9_F0BC	BASE_MCS + 0x00C9_F0B0	4 Bytes	POFSPS[31]	
BASE_MC + 0x00C9_F0C0	BASE_MCS + 0x00C9_F0C0	4 Bytes	POFSPS[32]	—
⋮	⋮	⋮	⋮	⋮
BASE_MC + 0x00C9_F170	BASE_MCS + 0x00C9_F170	4 Bytes	POFSPS[76]	—
BASE_MC + 0x00C9_F174	BASE_MCS + 0x00C9_F170	4 Bytes	POFSPS[77]	—
BASE_MC + 0x00C9_F178	BASE_MCS + 0x00C9_F170	4 Bytes	POFSPS[78]	—
BASE_MC + 0x00C9_F17C	BASE_MCS + 0x00C9_F170	4 Bytes	POFSPS[79]	—

60.8.2 Security Flag for Startup Area and Startup Size Select

The security flag (POFSPS.POFSPS[13]) for the startup area and startup size are located in the OTP area as the [section 60.7.4.5. Program Command](#). The POFSPS[13] bit can be monitored as the FSPR bit in the MSUASMON register.

When the POFSPS[13] bit is 0, issuing the Configuration set command to change the BTFLG and BTSIZE bits causes the command-locked state. Also, when the POFSPS[13] bit is 0, it is not possible to write the startup area select bits (SAS[1:0]) in the MSUACR register. The POFSPS[13] bit provides the protection. See [section 7, Option-Setting Memory](#).

60.8.3 Permanent Block Protect Setting

The permanent block protect setting is the clear protection for block protection setting. User area cannot be permanently updated by the MACI command when the permanent block protect setting is enabled. See [section 60.9.1.2. Protection by Block Protect Setting](#) in details of the block protect setting.

The block protect setting and the permanent block protect setting have the write/clear protection against the Configuration set command and Program command for each other. The extra MRAM sequencer does not detect an error when the Program command is issued to the write/clear protected settings.

[Figure 60.19](#), [Table 60.18](#), [Table 60.19](#), [Table 60.20](#), [Table 60.21](#) and [Table 60.22](#) show the write/clear protection against the block protect setting (BPS[4n]), (BPS[4n+1]), (BPS[4n+2]), (BPS[4n+3]) and the permanent protect setting (PBPS[n]). [Figure 60.20](#), [Table 60.23](#), [Table 60.24](#), [Table 60.25](#), [Table 60.26](#) and [Table 60.27](#) show the write/clear protection against the block protect setting for secure (BPS_SEC[4n]), (BPS_SEC[4n+1]), (BPS_SEC[4n+2]), (BPS_SEC[4n+3]) and permanent protect setting for secure (PBPS_SEC[n]).

Effective permanent block protect setting (PBPS or PBPS_SEC) depends on Code MRAM boundary setting.

For details of permanent block protect setting (PBPS or PBPS_SEC), see the User's Manual: Section xx, Option-Setting Memory.

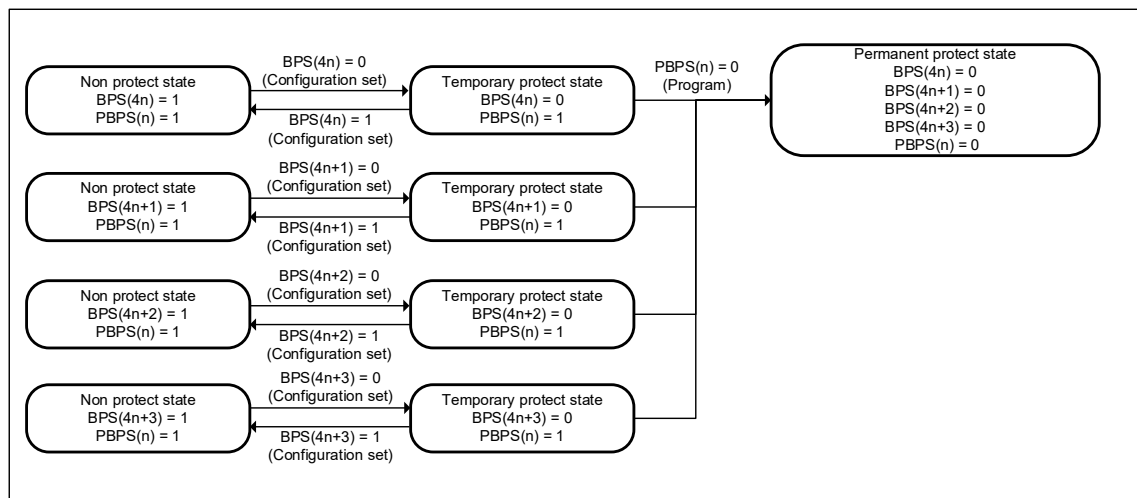


Figure 60.19 Status transition of extra MRAM sequencer by BPS[4n, 4n+1, 4n+2, 4n+3] and PBPS[n]

Table 60.18 Write/clear protection of BPS[4n] and PBPS[n]

Current state		Updatable state by configuration set command	
BPS[4n]	PBPS[n]	BPS[4n] = 1	BPS[4n] = 0
1	1	O	O
1	0	—	—
0	1	O	O
0	0	X	O

Table 60.19 Write/clear protection of BPS[4n+1] and PBPS[n]

Current state		Updatable state by configuration set command	
BPS[4n+1]	PBPS[n]	BPS[4n+1] = 1	BPS[4n+1] = 0
1	1	O	O
1	0	—	—
0	1	O	O
0	0	X	O

Table 60.20 Write/clear protection of BPS[4n+2] and PBPS[n]

Current state		Updatable state by configuration set command	
BPS[4n+2]	PBPS[n]	BPS[4n+2] = 1	BPS[4n+2] = 0
1	1	O	O
1	0	—	—
0	1	O	O
0	0	X	O

Table 60.21 Write/clear protection of BPS[4n+3] and PBPS[n] (1 of 2)

Current state		Updatable state by configuration set command	
BPS[4n+3]	PBPS[n]	BPS[4n+3] = 1	BPS[4n+3] = 0
1	1	O	O

Table 60.21 Write/clear protection of BPS[4n+3] and PBPS[n] (2 of 2)

Current state		Updatable state by configuration set command	
BPS[4n+3]	PBPS[n]	BPS[4n+3] = 1	BPS[4n+3] = 0
1	0	—	—
0	1	O	O
0	0	X	O

Remarks of [Table 60.18](#), [Table 60.19](#), [Table 60.20](#) and [Table 60.21](#):

O: Updatable by Configuration set command.

X: Not updatable by Configuration set command. (error does not occur.)

—: Not reach to this state.

Table 60.22 Write/clear protection of PBPS[n]

Current state					Updatable state by program command	
BPS[4n]	BPS[4n+1]	BPS[4n+2]	BPS[4n+3]	PBPS[n]	PBPS[n] = 1	PBPS[n] = 0
0	0	0	0	1	O	O
1	1 or 0	1 or 0	1 or 0	0	—	—
1 or 0	1	1 or 0	1 or 0	0	—	—
1 or 0	1 or 0	1	1 or 0	0	—	—
1 or 0	1 or 0	1 or 0	1	0	—	—
1	1 or 0	1 or 0	1 or 0	1	O	X
1 or 0	1	1 or 0	1 or 0	1	O	X
1 or 0	1 or 0	1	1 or 0	1	O	X
1 or 0	1 or 0	1 or 0	1	1	O	X
0	0	0	0	0	X	O

Remarks:

O: Updatable by Configuration set command.

X: Not updatable by Configuration set command. (error does not occur.)

—: Not reach to this state.

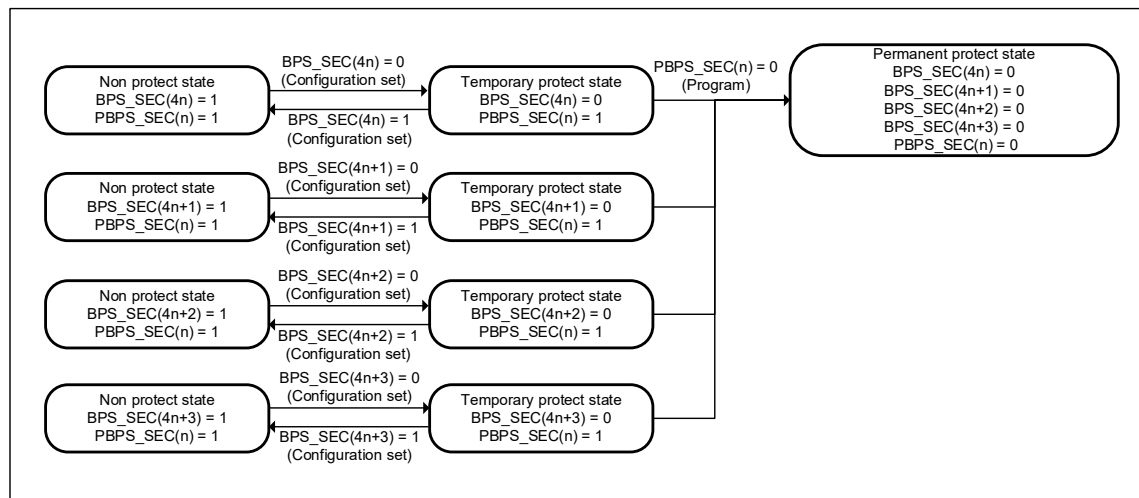


Figure 60.20 Status transition of extra MRAM sequencer by BPS_SEC[4n, 4n+1, 4n+2, 4n+3] and PBPS_SEC[n]

Table 60.23 Write/clear protection of BPS_SEC[4n] and PBPS_SEC[n]

Current state		Updatable state by configuration set command	
BPS_SEC[4n]	PBPS_SEC[n]	BPS_SEC[4n] = 1	BPS_SEC[4n] = 0
1	1	O	O
1	0	—	—
0	1	O	O
0	0	X	O

Table 60.24 Write/clear protection of BPS_SEC[4n+1] and PBPS_SEC[n]

Current state		Updatable state by configuration set command	
BPS_SEC[4n+1]	PBPS_SEC[n]	BPS_SEC[4n+1] = 1	BPS_SEC[4n+1] = 0
1	1	O	O
1	0	—	—
0	1	O	O
0	0	X	O

Table 60.25 Write/clear protection of BPS_SEC[4n+2] and PBPS_SEC[n]

Current state		Updatable state by configuration set command	
BPS_SEC[4n+2]	PBPS_SEC[n]	BPS_SEC[4n+2] = 1	BPS_SEC[4n+2] = 0
1	1	O	O
1	0	—	—
0	1	O	O
0	0	X	O

Table 60.26 Write/clear protection of BPS_SEC[4n+3] and PBPS_SEC[n]

Current state		Updatable state by configuration set command	
BPS_SEC[4n+3]	PBPS_SEC[n]	BPS_SEC[4n+3]=1	BPS_SEC[4n+3]=0
1	1	O	O
1	0	—	—
0	1	O	O
0	0	X	O

Remarks of [Table 60.23](#), [Table 60.24](#), [Table 60.25](#) and [Table 60.26](#):

O: Updatable by Configuration set command.

X: Not updatable by Configuration set command. (error does not occur.)

—: Not reach to this state.

Table 60.27 Write/clear protection of PBPS_SEC[n]

Current state					Updatable state by program command	
BPS_SEC [4n]	BPS_SEC [4n+1]	BPS_SEC [4n+2]	BPS_SEC [4n+3]	PBPS_SEC [n]	PBPS_SEC[n] = 1	PBPS_SEC[n] = 0
0	0	0	0	1	O	O
1	1 or 0	1 or 0	1 or 0	0	—	—
1 or 0	1	1 or 0	1 or 0	0	—	—
1 or 0	1 or 0	1	1 or 0	0	—	—
1 or 0	1 or 0	1 or 0	1	0	—	—
1	1 or 0	1 or 0	1 or 0	1	O	X
1 or 0	1	1 or 0	1 or 0	1	O	X
1 or 0	1 or 0	1	1 or 0	1	O	X
1 or 0	1 or 0	1 or 0	1	1	O	X
0	0	0	0	0	X	O

Remarks:

O: Updatable by Configuration set command.

X: Not updatable by Configuration set command. (error does not occur.)

—: Not reach to this state.

60.8.4 MRAM Protection for TrustZone

The MRAM protection for TrustZone is only available for the security-enabled product.

MRAM has the following four types of protect function against non-secure access to support the TrustZone specification for MRAM. Descriptions of this section are focused on extra MRAM sequencer operation.

- Protection for code MRAM area (program)
- Protection for extra MRAM area (program)
- Protection for MRAM area (read)
- Protection for register

60.8.4.1 Protection for Code MRAM Area (Program)

This function protects code MRAM from bus access such as CPU, DMAC. For details of protection, see [section 52, Security Features](#).

60.8.4.2 Protection for Extra MRAM Area (Program)

This function protects secure area of extra MRAM from MRAM sequencer command of non-secure developer. The condition of protection depends on extra MRAM sequencer command, access attribution, and product specification.

See [Table 60.28](#) for details of protection for extra MRAM area (program).

Extra MRAM area only allows privileged access.

Table 60.28 Protection for extra MRAM area (program)

MACI command		Target area		Issuing MACI command by non-secure access	Issuing MACI command by secure access
Program	Extra MRAM	MRAM and OTP area (Non-secure area)	Secure alias ^{*1}	X	X
			Non-secure alias ^{*2}	O	O
		MRAM and OTP area (Secure area)	Secure alias ^{*1}	X	O
			Non-secure alias ^{*2}	X	X
Configuration Set	Extra MRAM	Configuration area (Non-secure area)	Secure alias ^{*1}	X	X
			Non-secure alias ^{*2}	O	O
		Configuration area (Secure area)	Secure alias ^{*1}	X	O
			Non-secure alias ^{*2}	X	X
Increment Counter Read Counter	Extra MRAM	Anti-rollback counter area (Secure area only)	—	X	O

Note 1. The secure alias is determined when the address of MSADDR[28] is 0.

Note 2. The non-secure alias is determined when the address of MSADDR[28] is 1.

Remarks:

O: MACI command operation is not prohibited.

X: MACI command operation is prohibited.

Error occurs when the area is selected, and MACI command is executed.

The Configuration area has two types of secure region mapping for security-enabled product and security-disabled product. See [Figure 60.21](#) for details of non-secure/secure region of configuration area.

The extra MRAM sequencer determines that the target area is a secure region from the MSADDR register setting and product information.

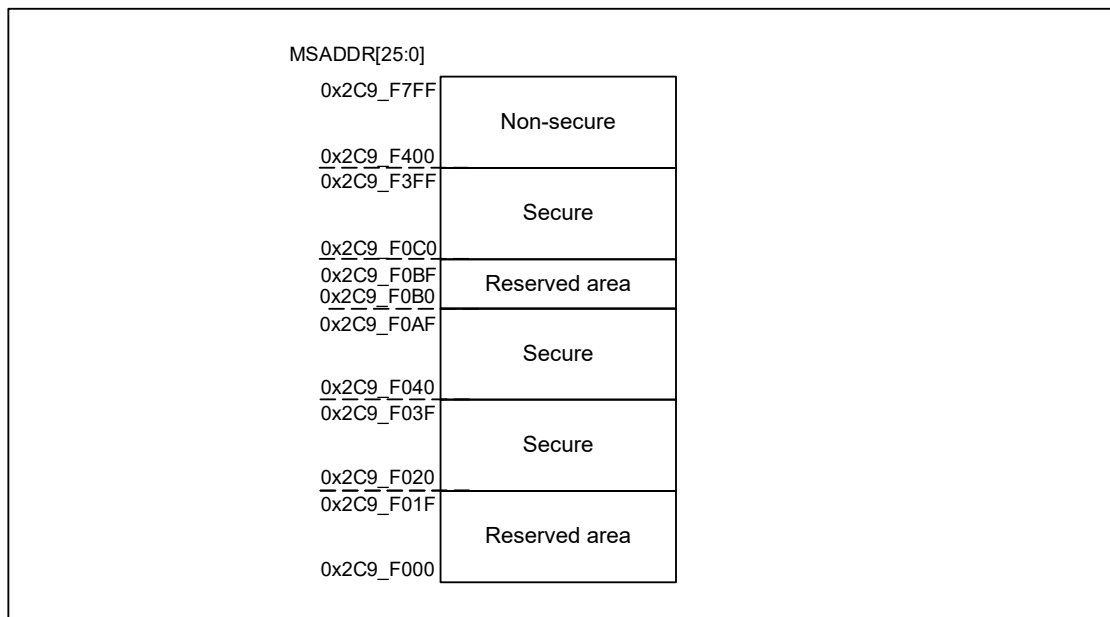


Figure 60.21 Secure/non-secure region in configuration area

60.8.4.3 Protection for MRAM Area (Read)

This function protects code MRAM and extra MRAM from bus access (such as CPU, DMAC). For details of protection, see the User's Manual: Section xx, Security Features.

60.8.4.4 Protection for Register

The MRAM registers have the write access protection against the non-secure access. See [Table 60.29](#) for details of the protected MRAM registers. MRAM registers can be accessed with privileges.

Table 60.29 Protected MRAM registers for TrustZone (1 of 2)

Protection target registers	Secure attribute setting	Remarks
MRCEECC	SA register setting (MSAR.MRBCPSA)	See section 60.5.40. MRCEECC : Code MRAM ECC Encoder Control Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MRPSC and MREEECC	SA register setting (MSAR.MRBEPSCA)	See section 60.5.31. MRPSC : MRAM Program Speed Control Register and MREEECC : Extra MRAM ECC Encoder Control Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MRCPC0, MRCBPROT0, MRCPS, MRCPEAINT, and MRCPEA	SA register setting (MSAR.MRCPSA)	See section 60.5.32. MRCPC0 : Code MRAM Program Control Register , section 60.5.34. MRCBPROT0 : Code MRAM Block Protection Register , section 60.5.36. MRCPS : Code MRAM Program Status Register , section 60.5.37. MRCPEAINT : Code MRAM Program Access Error Interrupt Enable Register and section 60.5.38. MRCPEA : Code MRAM Program Error Address Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MSAR	Always Secure	See section 60.5.13. MSAR : MRAM Security Attribution Register .

Table 60.29 Protected MRAM registers for TrustZone (2 of 2)

Protection target registers	Secure attribute setting	Remarks
MCTRCNTR, MCTRLSR, and MCTRSTATR	SA register setting (MSAR.MACITRSA)	See section 60.5.22. MCTRCNTR : MRAM Configuration Update Transfer Control Register , section 60.5.23. MCTRLSR : MRAM Configuration Update Transfer List Select Register and section 60.5.24. MCTRSTATR : MRAM Configuration Update Transfer Status Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MRAM access area	SA register setting (MSAR.MACICOMISA)	For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MACI*1	SA register setting (MSAR.MACICOMRSA)	For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MRCPFB	SA register setting (MSAR.MPFBENSA)	See section 60.5.1. MRCPFB : Code MRAM Prefetch Buffer Enable Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MRCFREQ	SA register setting (MSAR.MRCFREQSA)	See section 60.5.2. MRCFREQ : Code MRAM Frequency Notifications Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MRCDECC, MRCRAEINT, MRCRAES, MRCRTEA, and MRCRDEA	SA register setting (MSAR.MRCECCSA)	See section 60.5.4. MRCDECC : Code MRAM ECC Decoder Control Register , section 60.5.5. MRCRAEINT : Code MRAM Read Access Error Interrupt Enable Register , section 60.5.6. MRCRAES : Code MRAM Read Access Error Status Register , section 60.5.7. MRCRTEA : Code MRAM TED Error Address Register , and section 60.5.8. MRCRDEA : Code MRAM DEC Error Address Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MREFREQ	SA register setting (MSAR.MREFREQSA)	See section 60.5.3. MREFREQ : Extra MRAM Frequency Notifications Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MREDECC, MRERAEINT, MRERAES, MRERTEA, and MRERDEA	SA register setting (MSAR.MREECCSA)	See MREDECC : Extra MRAM ECC Decoder Control Register , section 60.5.9. MRERAEINT : Extra MRAM Read Access Error Interrupt Enable Register , section 60.5.10. MRERAES : Extra MRAM Read Access Error Status Register , section 60.5.11. MRERTEA : Extra MRAM TED Error Address Register , and section 60.5.12. MRERDEA : Extra MRAM DEC Error Address Register . For details of MSAR register, see section 60.5.13. MSAR : MRAM Security Attribution Register .
MREZS, MREZC, MCNTSELR, MCNTDTR0 to 1, MRCPC1, MRCBPROT1, and MSUACR	Always Secure	See section 60.5.14. MREZS : Extra MRAM Zeroization Status Register , section 60.5.15. MREZC : Extra MRAM Zeroization Control Register , section 60.5.20. MCNTSELR : MRAM Counter Select Register , section 60.5.21. MCNTDTRn : MRAM Counter Data Register (n = 0, 1) , section 60.5.33. MRCPC1 : Code MRAM Program Control Register for Secure , section 60.5.35. MRCBPROT1 : Code MRAM Block Protection Register for Secure , and section 60.5.30. MSUACR : MRAM Startup Area Control Register .

Note 1. The target MACI registers are MASTAT, MPAEINT, MRDYIE, MSADDR, MSTATR, MENTRYR, MSUINITR, MCMDR, and MSUASMON.

60.8.5 Anti-Rollback Counter

The anti-rollback counter cannot be decremented and is used in the firmware update. Only secure applications are available. Anti-rollback counter shifts one bit to the left with the Increment Counter command from the initial value of 0. The least significant bit (bit 0) is set to 1. When the anti-rollback counter is of maximum value, the extra MRAM sequencer does not change the counter by the Increment Counter command (error does not occur).

There are three commands that control the counters. See [section 60.7.4.9. Increment Counter Command](#) and [section 60.7.4.10. Read Counter Command](#) for details.

Figure 60.22 shows anti-rollback counter increment image when increment counter command is issued.

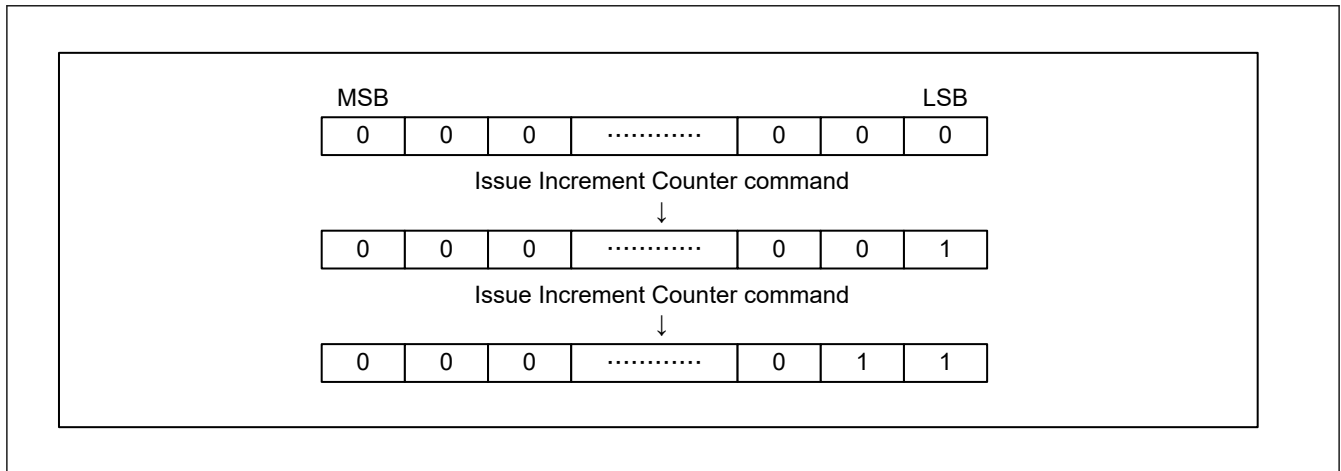


Figure 60.22 Increment image of anti-rollback counter

Anti-rollback counter has three types of counters (ARC_SEC, ARC_NSEC, and ARC_OEMBL) and lock bit (ARCSEC_LK, ARCNSEC_LK0 to 3, and ARCOEMBL_LK) for protection of each counter.

ARC_NSEC can be configured as 1×256 or 4×64 bits by the configuration area setting (CNF_ARCNS1 and CNF_ARCNS0) and cannot be updated in invalid setting.

ARC_OEMBL is protected by the security engine and cannot be updated until the update validation of the OEM boot loader is successful. After successful validation, ARC_OEMBL can be updated to the current OEM boot loader version (OEM BL Ver.) output by the security engine. ARC_OEMBL cannot be updated to a value that exceeds the OEM boot loader version (OEM BL Ver.).

60.8.6 HUK Protection

Zeroization of W-HUK is triggered when one of ZHUKEN[3:0] bits (zeroization enable bit) is 0 and anti-tamper is detected or 0x07 is set to WHUKWE[2:0] bits in MREZC register. Zeroing W-HUK disables the generation of HUK. Table 60.30 shows the combinations of enable bits and request signals. If a zeroing request is received during code MRAM or extra MRAM program, the executing program is forcibly terminated, and the zeroing process is started. If a code MRAM program request is received during zeroing, the code MRAM program is started after zeroing is completed. MACI commands are not accepted during zeroing. During W-HUK zeroing, all MACI commands are not accepted and extra MRAM cannot be read.

Table 60.30 Combinations of request signals and enable bits (1 of 2)

Zeroization request signal				Zeroization enable bit				Function
WHUKZE [2:0]	RTCIC			ZHUKEN				
	2	1	0	3	2	1	0	
0x7	—	—	—	0	—	—	—	O
0x7	—	—	—	1	—	—	—	Oe
Except 0x7	—	—	—	—	—	—	—	X
—	1	—	—	—	0	—	—	O
—	1	—	—	—	1	—	—	Oe
—	0	—	—	—	—	—	—	X
—	—	1	—	—	—	0	—	O
—	—	1	—	—	—	1	—	Oe
—	—	0	—	—	—	—	—	X
—	—	—	1	—	—	—	0	O
—	—	—	1	—	—	—	1	Oe

Table 60.30 Combinations of request signals and enable bits (2 of 2)

Zeroization request signal				Zeroization enable bit				Function
WHUKZE [2:0]	RTCIC			ZHUKEN				
	2	1	0	3	2	1	0	
—	—	—	0	—	—	—	—	X

Remarks:

O: W-HUK zeroing

Oe: W-HUK Pseudo-zeroing

X: Zeroization of W-HUK does not execute

—: Don't care

[Table 60.31](#) shows differences of W-HUK pseudo-zeroing and W-HUK zeroing described in [Table 60.30](#). If the zeroization enable bit is 1 and tamper is detected, the MREZS.WHUKZF bit is set to 0 to complete zeroing without physically zeroing W-HUK (W-HUK pseudo-zeroing).

Pseudo means the zeroization tamper event has happened, but the W-HUK zeroization response/action was not enabled and therefore not physically executed. The flag of zeroized W-HUK in the MREZS register is set, but because it is a pseudo tamper, it is restored at the next power cycle. W-HUK usage is disabled until the next power cycle. If the zeroization enable bit is 0 and a tamper is detected, physical zeroization of W-HUK is performed and the Zeroized Monitor register is set to 0 (W-HUK zeroing). When the zeroization function is enabled and after a tamper or software trigger, it is permanent (not pseudo). The flag of zeroized W-HUK in the MREZS register is cleared (set to 0 and remain at 0 permanently). See [section 12.1.7. Tamper Detection Function](#) for details of the tamper detection.

Table 60.31 Internal state at the time of zeroing request

Table 60.30 function	Operation name	Zeroization enable bits	W-HUK	Zeroized monitor register (MREZS)
Oe	W-HUK Pseudo-zeroing	Disable(1)	Not zeroed	Clear
O	W-HUK zeroing	Enable(0)	Zeroed	Clear flag after zeroization finished

60.8.7 OEM ROOT KEY Protection

OEM_ROOT_PKn can be revoked by the REVOKE setting in Option-Setting Memory. See [section 52, Security Features](#) for details of how OEM_ROOT_PKs are used when updating OEM_BL.

60.9 Protection Function

60.9.1 Software Protection

Software protection function disables MRAM programming operation according to register and configuration area settings.

60.9.1.1 Code MRAM Program Protection

When the MRCPC0 register is set to 0x0000, programming of non-secure aliases in code MRAM is prohibited. When the MRCPC1 register is set to 0x0000, programming of secure aliases in code MRAM is prohibited. Code MRAM program protection has a higher priority than block protection and cannot be canceled by MRCBPROT0 or MRCBPROT1 register.

60.9.1.2 Protection by Block Protect Setting

Each block in the user area has the block protect setting (BPS or BPS_SEC). When the MRCBPROT0 or MRCBPROT1 register is 0x0000 and the block protect bit is 0, program access to the non-secure user area of code MRAM is prohibited. When the MRCBPROT1 register is 0x0000 and the block protect bit is 0, program access to the secure user area of code MRAM is prohibited. To program the sector whose block protect bit is 0, set MRCBPROT0 or MRCBPROT1 register to 0x0001.

The block protect setting can be locked by the permanent block protect setting (PBPS or PBPS_SEC). When the permanent block protect setting and the block protect setting are 0, program access cannot be performed.

Effective block protect setting (BPS or BPS_SEC) and permanent block protect setting (PBPS or PBPS_SEC) depends on code MRAM boundary setting.

See [section 60.8.3. Permanent Block Protect Setting](#) for details of the permanent block protect setting and block protect setting.

See [section 60.5.34. MRCBPROT0 : Code MRAM Block Protection Register](#) and [section 60.5.35. MRCBPROT1 : Code MRAM Block Protection Register for Secure](#) details of the MRCBPROT0 and MRCBPROT1 registers.

For details of block protect setting (BPS or BPS_SEC), permanent block protect setting (PBPS or PBPS_SEC), and code MRAM boundary setting, see the User's Manual: Section xx, Option-Setting Memory.

[Table 60.32](#) to [Table 60.35](#) show the relation of user area and the block protect setting in each function setting.

- BPS[0] to BPS[n] or BPS_SEC[0] to BPS_SEC[n] are assigned to the block of user area (for example, address is BASE_MC + 0x0000 0000 to the last block address).
- BPS[0]/BPS_SEC[0] and BPS[1]/BPS_SEC[1] are assigned to the block of user area depending on the startup area select setting (SAS.BTFLG bit) and startup area size select setting (SAS.BTSIZE bits). See [section 60.9.3. Startup Program Protection](#) for details.

[Table 60.32](#) to [Table 60.35](#) show example of the block protect setting.

Table 60.32 Example of block protect setting when SAS.BTFLG is 1 and all sizes

Address (offset address: BASE_MC)	Block size	Block protect setting	User area block number	Remarks
The last block address	2 KB	BPS[n] or BPS_SEC[n]	Block n	
⋮	⋮	⋮	⋮	
0x0000 3800 to 0x0000 3FFF	2 KB	BPS[7] or BPS_SEC[7]	Block 7	
0x0000 3000 to 0x0000 37FF	2 KB	BPS[6] or BPS_SEC[6]	Block 6	
0x0000 2800 to 0x0000 2FFF	2 KB	BPS[5] or BPS_SEC[5]	Block 5	
0x0000 2000 to 0x0000 27FF	2 KB	BPS[4] or BPS_SEC[4]	Block 4	
0x0000 1800 to 0x0000 1FFF	2 KB	BPS[3] or BPS_SEC[3]	Block 3	
0x0000 1000 to 0x0000 17FF	2 KB	BPS[2] or BPS_SEC[2]	Block 2	
0x0000 0800 to 0x0000 0FFF	2 KB	BPS[1] or BPS_SEC[1]	Block 1	
0x0000 0000 to 0x0000 07FF	2 KB	BPS[0] or BPS_SEC[0]	Block 0	

Table 60.33 Example of block protect setting when SAS.BTFLG is 0 and SAS.BTSIZE is 11 (1 of 2)

Address (offset address: BASE_MC)	Block size	Block protect setting	User area block number	Remarks
The last block address	2 KB	BPS[n] or BPS_SEC[n]	Block n	—
⋮	⋮	⋮	⋮	—
0x0000 4000 to 0x0000 47FF	2 KB	BPS[8] or BPS_SEC[8]	Block 8	—
0x0000 3800 to 0x0000 3FFF	2 KB	BPS[7] or BPS_SEC[7]	Block 3	Swap block 3 and block 7 in this startup area select setting
0x0000 3000 to 0x0000 37FF	2 KB	BPS[6] or BPS_SEC[6]	Block 2	Swap block 2 and block 6 in this startup area select setting
0x0000 2800 to 0x0000 2FFF	2 KB	BPS[5] or BPS_SEC[5]	Block 1	Swap block 1 and block 5 in this startup area select setting
0x0000 2000 to 0x0000 27FF	2 KB	BPS[4] or BPS_SEC[4]	Block 0	Swap block 0 and block 4 in this startup area select setting
0x0000 1800 to 0x0000 1FFF	2 KB	BPS[3] or BPS_SEC[3]	Block 7	Swap block 3 and block 7 in this startup area select setting
0x0000 1000 to 0x0000 17FF	2 KB	BPS[2] or BPS_SEC[2]	Block 6	Swap block 2 and block 6 in this startup area select setting

Table 60.33 Example of block protect setting when SAS.BTFLG is 0 and SAS.BTSIZE is 11 (2 of 2)

Address (offset address: BASE_MC)	Block size	Block protect setting	User area block number	Remarks
0x0000 0800 to 0x0000 0FFF	2 KB	BPS[1] or BPS_SEC[1]	Block 5	Swap block 1 and block 5 in this startup area select setting
0x0000 0000 to 0x0000 07FF	2 KB	BPS[0] or BPS_SEC[0]	Block 4	Swap block 0 and block 4 in this startup area select setting

Table 60.34 Example of block protect setting when SAS.BTFLG is 0 and SAS.BTSIZE is 10

Address (offset address: BASE_MC)	Block size	Block protect setting	User area block number	Remarks
The last block address	2 KB	BPS[n] or BPS_SEC[n]	Block n	—
⋮	⋮	⋮	⋮	—
0x0000 8000 to 0x0000 87FF	2 KB	BPS[16] or BPS_SEC[16]	Block 16	—
0x0000 7800 to 0x0000 7FFF	2 KB	BPS[15] or BPS_SEC[15]	Block 7	Swap block 7 and block 15 in this startup area select setting
0x0000 7000 to 0x0000 77FF	2 KB	BPS[14] or BPS_SEC[14]	Block 6	Swap block 6 and block 14 in this startup area select setting
⋮	⋮	⋮	⋮	—
0x0000 4800 to 0x0000 4FFF	2 KB	BPS[9] or BPS_SEC[9]	Block 1	Swap block 1 and block 9 in this startup area select setting
0x0000 4000 to 0x0000 47FF	2 KB	BPS[8] or BPS_SEC[8]	Block 0	Swap block 0 and block 8 in this startup area select setting
0x0000 3800 to 0x0000 3FFF	2 KB	BPS[7] or BPS_SEC[7]	Block 15	Swap block 7 and block 15 in this startup area select setting
0x0000 3000 to 0x0000 37FF	2 KB	BPS[6] or BPS_SEC[6]	Block 14	Swap block 6 and block 14 in this startup area select setting
⋮	⋮	⋮	⋮	—
0x0000 0800 to 0x0000 0FFF	2 KB	BPS[1] or BPS_SEC[1]	Block 9	Swap block 1 and block 9 in this startup area select setting
0x0000 0000 to 0x0000 07FF	2 KB	BPS[0] or BPS_SEC[0]	Block 8	Swap block 0 and block 8 in this startup area select setting

Table 60.35 Example of block protect setting when SAS.BTFLG is 0 and SAS.BTSIZE is 01 (1 of 2)

Address (offset address: BASE_MC)	Block size	Block protect setting	User area block number	Remarks
The last block address	2 KB	BPS[n] or BPS_SEC[n]	Block n	—
⋮	⋮	⋮	⋮	—
0x0001 0000 to 0x0001 07FF	2 KB	BPS[32] or BPS_SEC[32]	Block 32	—
0x0000 F800 to 0x0000 FFFF	2 KB	BPS[31] or BPS_SEC[31]	Block 15	Swap block 15 and block 31 in this startup area select setting
0x0000 F000 to 0x0000 F7FF	2 KB	BPS[30] or BPS_SEC[30]	Block 14	Swap block 14 and block 30 in this startup area select setting
⋮	⋮	⋮	⋮	—
0x0000 8800 to 0x0000 8FFF	2 KB	BPS[17] or BPS_SEC[17]	Block 1	Swap block 1 and block 17 in this startup area select setting
0x0000 8000 to 0x0000 87FF	2 KB	BPS[16] or BPS_SEC[16]	Block 0	Swap block 0 and block 16 in this startup area select setting
0x0000 7800 to 0x0000 7FFF	2 KB	BPS[15] or BPS_SEC[15]	Block 31	Swap block 15 and block 31 in this startup area select setting
0x0000 7000 to 0x0000 77FF	2 KB	BPS[14] or BPS_SEC[14]	Block 30	Swap block 14 and block 30 in this startup area select setting

Table 60.35 Example of block protect setting when SAS.BTFLG is 0 and SAS.BTSIZE is 01 (2 of 2)

Address (offset address: BASE_MC)	Block size	Block protect setting	User area block number	Remarks
⋮	⋮	⋮	⋮	—
0x0000 0800 to 0x0000 0FFF	2 KB	BPS[1] or BPS_SEC[1]	Block 17	Swap block 1 and block 17 in this startup area select setting
0x0000 0000 to 0x0000 07FF	2 KB	BPS[0] or BPS_SEC[0]	Block 16	Swap block 0 and block 16 in this startup area select setting

60.9.2 Error Protection

Error protection detects the issuing of illegal MACI commands, illegal access, and extra MRAM sequencer malfunction. MACI command acceptance is disabled (command-locked state) in response to the detection of these errors. The extra MRAM cannot be programmed while the extra MRAM sequencer is in the command-locked state. For release from the command-locked state, issue the Status Clear or Forced Stop command. The Status Clear command can only be used while the MRDY bit in the MSTATR register is 1. The Forced Stop command can be used regardless of the value of the MRDY bit. While the CMDLKIE bit in the MPAEINT register is 1, an extra MRAM access error (MRAM_MREPR) interrupt is generated if the extra MRAM sequencer enters the command-locked state (the CMDLK bit in the MASTAT register is set to 1).

If the extra MRAM sequencer enters the command-locked state in response to a command, the extra MRAM sequencer continues the processing for programming. If a command is issued in the command-locked state, the ILGLERR bit becomes 1 and the other bits retain the values set due to previous error detection.

Table 60.36 shows error protection types and status bit values after error detection.

Table 60.36 Error protection type (1 of 2)

Error type	Description	ILGCO MERR	SECE R	OTERR	TZFER R	ILGLE RR	PRGER R	CFGPR G	MREA E ^{*1}
Illegal command error	An undefined size has been specified in the first cycle of command (not byte write)	1	0	0	0	1	0	0	0
	An undefined code is written in the first access of the MACI command	1	0	0	0	1	0	0	0
	The value specified in the last access of the multiple access MACI command is not 0xD0	1	0	0	0	1	0	0	0
	The value (N) specified in the second write access of the MACI command in the Program command or Configuration Set command is wrong	1	0	0	0	1	0	0	0
	Increment Counter or Read Counter command has been issued with an inconsistent MCNTSELR[2:0] setting. (Include CNF_ARCNS[1:0] setting is the mismatch).	1	0	0	0	1	0	0	0
	The MACI command not acceptable in each mode has been issued (see Table 60.13)	1	0	0	0	1	0	0	0
	The MACI command has been issued when command acceptance conditions are not satisfied (see Table 60.14)	0/1	0/1	0/1	0	1	0/1	0/1	0/1
	Increment Counter command has been issued against the anti-rollback area protected by lock bit setting	1	0	0	0	1	0	0	0
	Increment Counter command against ARC_OEMBL is issued when the OEM boot loader version (OEM BL Ver.) output from the security engine is the same or smaller than the ARC_OEMBL value (see section 60.8.5. Anti-Rollback Counter)	1	0	0	0	1	0	0	0
	Increment Counter command against ARC_OEMBL is issued when the previous command is not the Read Counter command against ARC_OEMBL (see section 60.8.5. Anti-Rollback Counter)	1	0	0	0	1	0	0	0
	A command has been issued against extra MRAM sequencer command illegal status (see Table 60.11)	0/1	0/1	0/1	0	1	0/1	0/1	0/1
Program error	An error occurs during programming (when the OTP area data changes from 0 to 1 or verifies the error)	0	0	0	0	0	1	0	0
Configuration set error	An error has occurred during the configuration set	0	0	0	0	0	0	1	0
Access error	Program command has been issued to reserved area in extra MRAM program mode	0/1	0	0	0	1	0	0	1
	Configuration Set command has been issued to reserved configuration area in program mode	0/1	0	0	0	1	0	0	1
	Program command or Configuration Set command has been issued to the TrustZone protection area in extra MRAM program mode	0/1	0	0	1	1	0	0	1
	Configuration Set command has been issued to the operation mode protection area in extra MRAM program mode	0/1	0	0	0	1	0	0	1

Table 60.36 Error protection type (2 of 2)

Error type	Description	ILGCO MERR	SECE R	OTERR	TZFER R	ILGLE RR	PRGER R	CFGPR G	MREA E ^{*1}
Security error	Configuration Set command for the BTFLG bit setting has been issued when the FSPR bit is 0 (see section 60.7.4.8. Configuration Set Command)	0	1	0	0	1	0	0	0
Others	The MACI command issuing area has been accessed in read mode	0	0	1	0	1	0	0	0
	The MACI command issuing area has been read in program mode	0	0	1	0	1	0	0	0

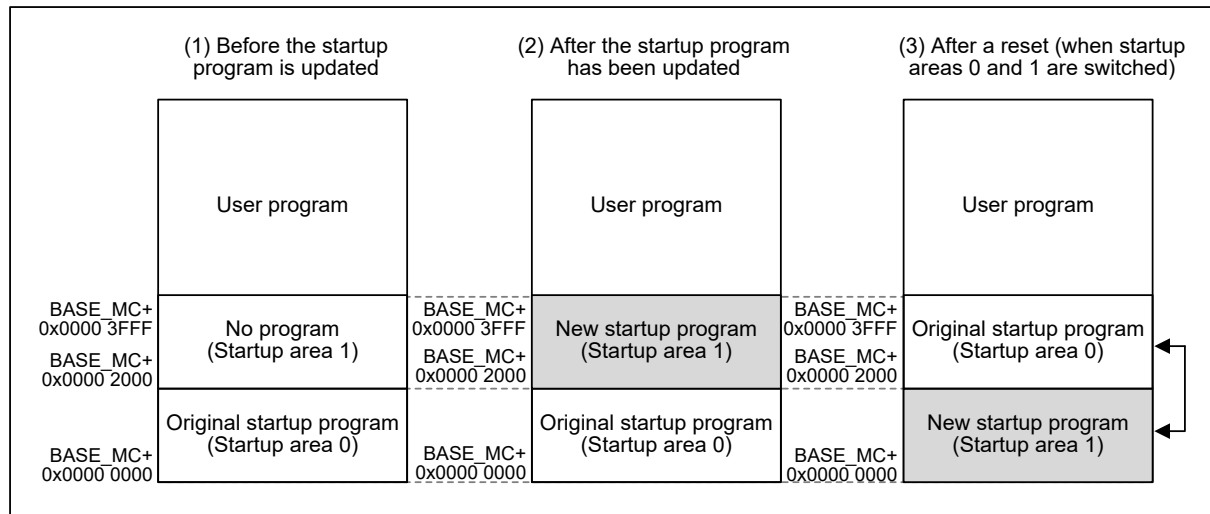
Note 1. MREA value depends on the MSADDR setting.

60.9.3 Startup Program Protection

Protection of the startup program is for protection of the program to be started after a reset (the startup program). This function provides a way to safely update the startup program when rewriting is suspended during a reset.

The startup area is 8 KB, 16 KB, or 32 KB in size and is assigned to the user area in the code MRAM. The size of the startup area can be changed by the BTSIZE bit. When resizing the startup area, be careful not to destroy the software in use. This function uses the values of the SAS.BTFLG bit, the SAS.BTSIZE bits, and the MSUACR.SAS[1:0] bits to change the area where the startup program is stored in block units (see [Figure 60.23](#) to [Figure 60.26](#)).

In protection of the startup program, the state of the selection of the startup area can be fixed by the POFSPS.POFSPS[13] bit. However, the POFSPS.POFSPS[13] bit is not restored to 1 once the flag is set to 0. Exercise extra caution when handling the POFSPS.POFSPS[13] bit.



Note: The addresses are used for reading, programming, and erasure. Startup area 0 is erased when the Block Erase command is issued for the address BASE_CF + 0x0000_0000 with the allocation shown under (3).

- Before the startup program is updated
When SAS.BTFLG = 1 and MSUACR.SAS[1:0] = 0xb, the address range from BASE_MC + 0x0000_2000 to BASE_MC + 0x0000_3FFF is selected as startup area 1 and the address range from BASE_MC + 0x0000_0000 to BASE_MC + 0x0000_1FFF is selected as startup area 0.
- After the startup program has been updated
A new startup program can be written to startup area 1. At this time, startup area 0 still contains the previous startup program, so rewriting by starting up from startup area 0 remains possible if rewriting to the startup area 1 fails.
- After a reset (when startup areas 0 and 1 are switched)
When rewriting startup area 1 is successfully completed, switch the assignment of startup areas 0 and 1 by rewriting the startup area selection bit (the SAS.BTFLG bit) with the Configuration Set command. Afterwards, the program is started up from startup area 1 after a reset.

Figure 60.23 Concept of protection of the startup program

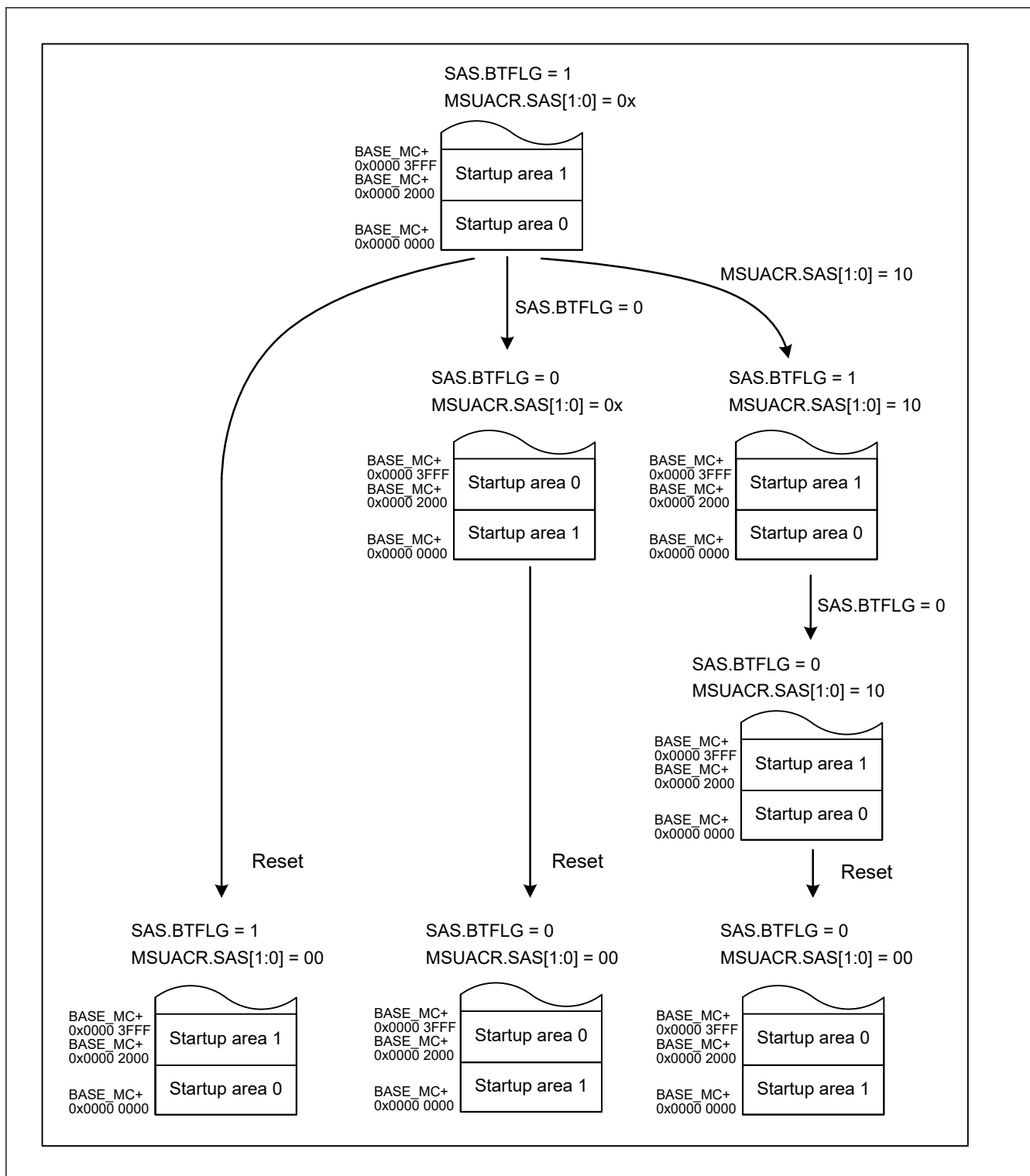


Figure 60.24 Example 1 of transitions for startup program protection settings

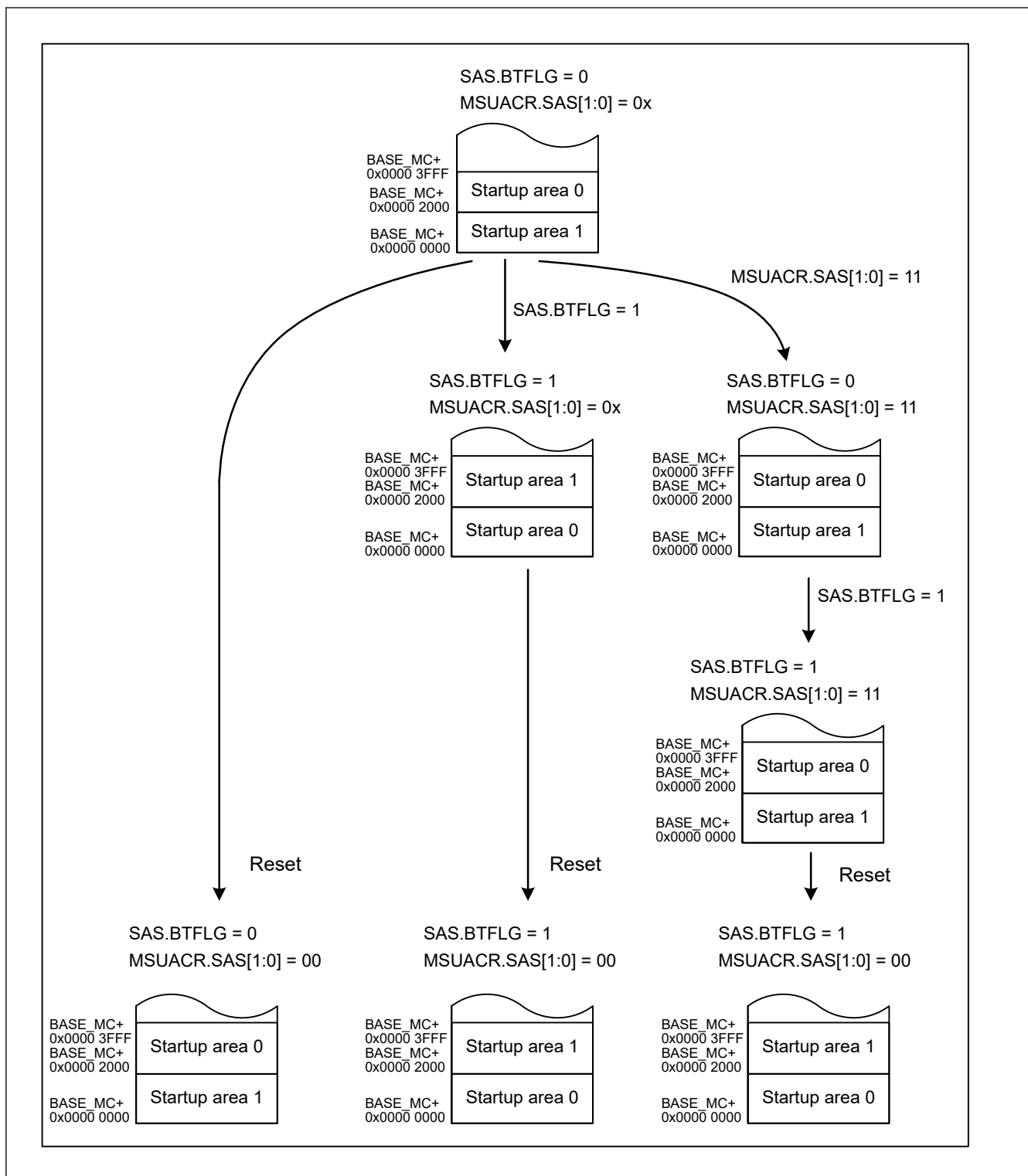


Figure 60.25 Example 2 of transitions for startup program protection settings

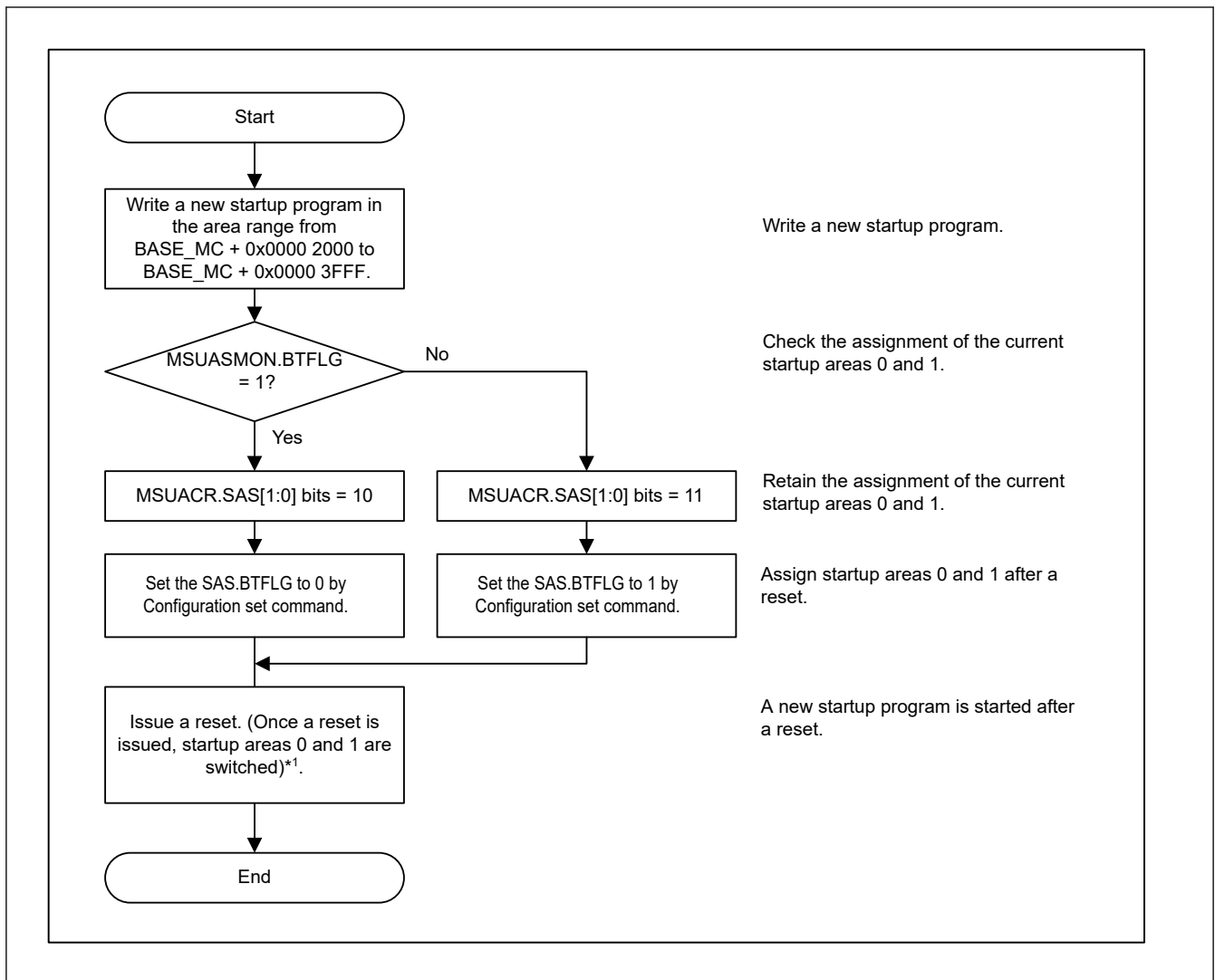


Figure 60.26 Concept of protection of the startup program

60.9.4 Fault Correction and Detection

MRAM systems are equipped with ECC circuitry to correct and detect errors. DEC-TED (double-bit error correction and triple-bit error detection) ECC corrects errors in 128-bit data units. Code MRAM uses two ECC circuits for 256-bit readout, and extra MRAM uses one ECC circuit for 128-bit readout.

If there is a DEC error in the code MRAM reading, error correction is performed and the MRCRAES.DECERRC bit is set to 1. If there is a TED error, the MRCRAES.TEDERRC bit is set to 1, but error correction is not possible. When an error is detected, the error address is stored in the MRCRDEA register for DEC error and in the MRCRTEA register for TED error. If a DEC error occurs, read the data at the error address, and write back. If a TED error occurs, update all 32 bytes of data and overwrite with the correct data.

If there is a DEC error in the extra MRAM reading, error correction is performed and the MRERAES.DECERRE bit is set to 1. If there is a TED error, the MRERAES.TEDERRE bit is set to 1, but error correction is not possible. When an error is detected, the error address is stored in the MRERDEA register for DEC errors and in the MRERTEA register for TED errors. If a DEC error occurs, read the data at the error address, and write back. If a TED error occurs, update all 32 bytes of data and overwrite with the correct data.

60.9.4.1 Test Method for ECC Circuits

The ECC circuit can be tested by setting the program data and ECC data to desired values. A block diagram of the ECC test circuit is shown in [Figure 60.27](#).

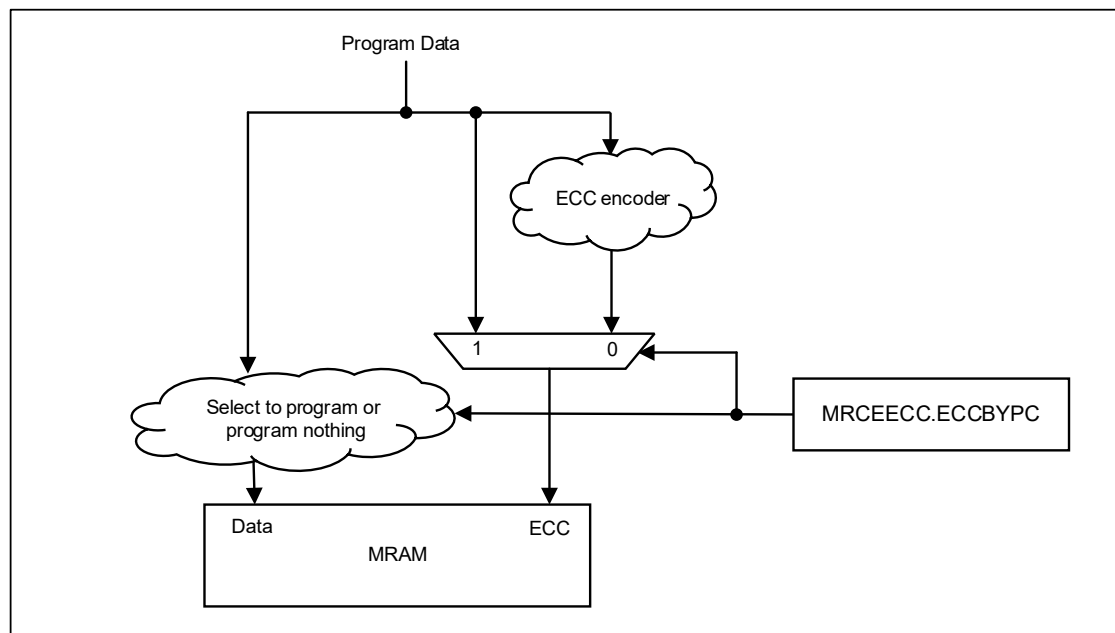


Figure 60.27 Block diagram of the ECC test circuit

Table 60.37 Changes in data allocation depending on the value of MRCECC.ECCBYPC

MRCECC.ECCBYPC	Code MRAM	
	Data[255:0]	ECC[33:0]
0	Program data[255:0]	ECC code generated from program data[127:0]
1	Program nothing	{Program data[144:128], Program data[16:0]}

Figure 60.28 shows the ECC decoder test flow of the code MRAM. The code MRAM has two ECC decoders and ECC errors are merged, so when testing the ECC decoder, test the ECC decoder circuits individually.

[Notation of Figure 60.28]

Note 1. To eliminate ECC errors in instruction fetching.

Note 2. If you get an error, try the test again at a different address.

Note 3. Invert the bits of the code according to the error condition you want to check (1 to 3)

Replace the bits shown in Table 60.37 with the code you created and write to the same address as DATA.

Note 4. See Table 60.38.

Table 60.38 Check results

Check results	Expected value		
	1-bit invert	2-bit invert	3-bit invert
MRCRAES.DECERRC	0	1	0
MRCRAES.TEDERRC	0	0	1

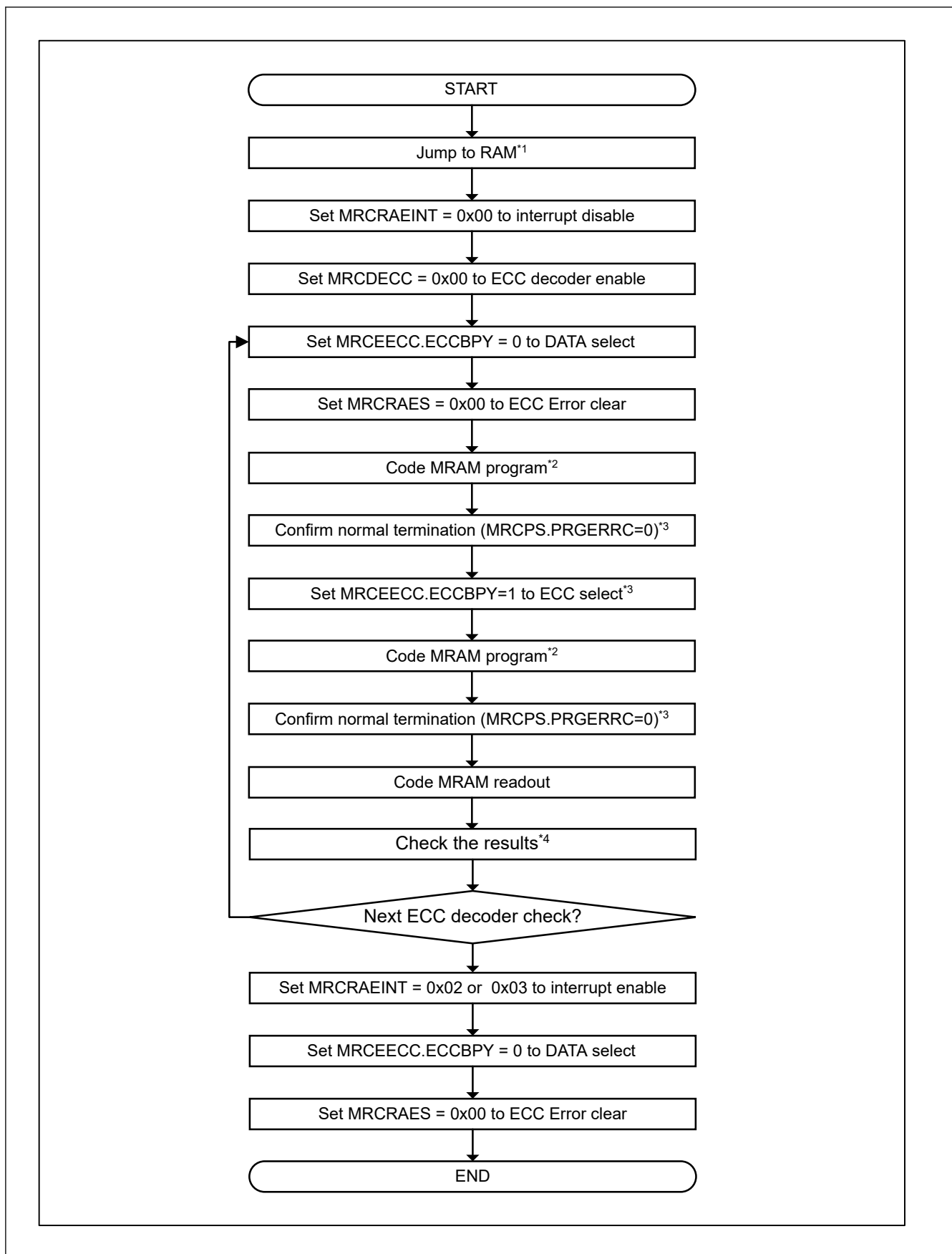


Figure 60.28 Code MRAM ECC decoder test flow

60.10 Configuration Update Transfer

After Program or Configuration Set command operation, the system status can be updated by configuration update transfer without system reset. MACI automatically determines a valid area and executes multidata transfer with the MCTRLSR register setting. With the TRTRG bit in MTRCNTR written as 1 in the read mode (MENTRYR is 0x0000 and the WHUKEXE bit in MREZS register is 0), MACI transfers data from extra MRAM. MCTRSTATR register can monitor the processing status of configuration update transfer. Extra MRAM cannot be read during configuration data transfer.

Figure 60.29 shows configuration update transfer usage.

If the WHUKEXE bit in the MREZS register is set to 1 in the configuration update transfer flow, the configuration update transfer must be reexecuted after the W-HUK zeroing is finished.

See [section 60.5.22. MTRCNTR : MRAM Configuration Update Transfer Control Register](#), [section 60.5.23. MCTRLSR : MRAM Configuration Update Transfer List Select Register](#), and [section 60.5.24. MCTRSTATR : MRAM Configuration Update Transfer Status Register](#) for details of the associated registers specifications (MTRCNTR, MCTRLSR, and MCTRSTATR registers).

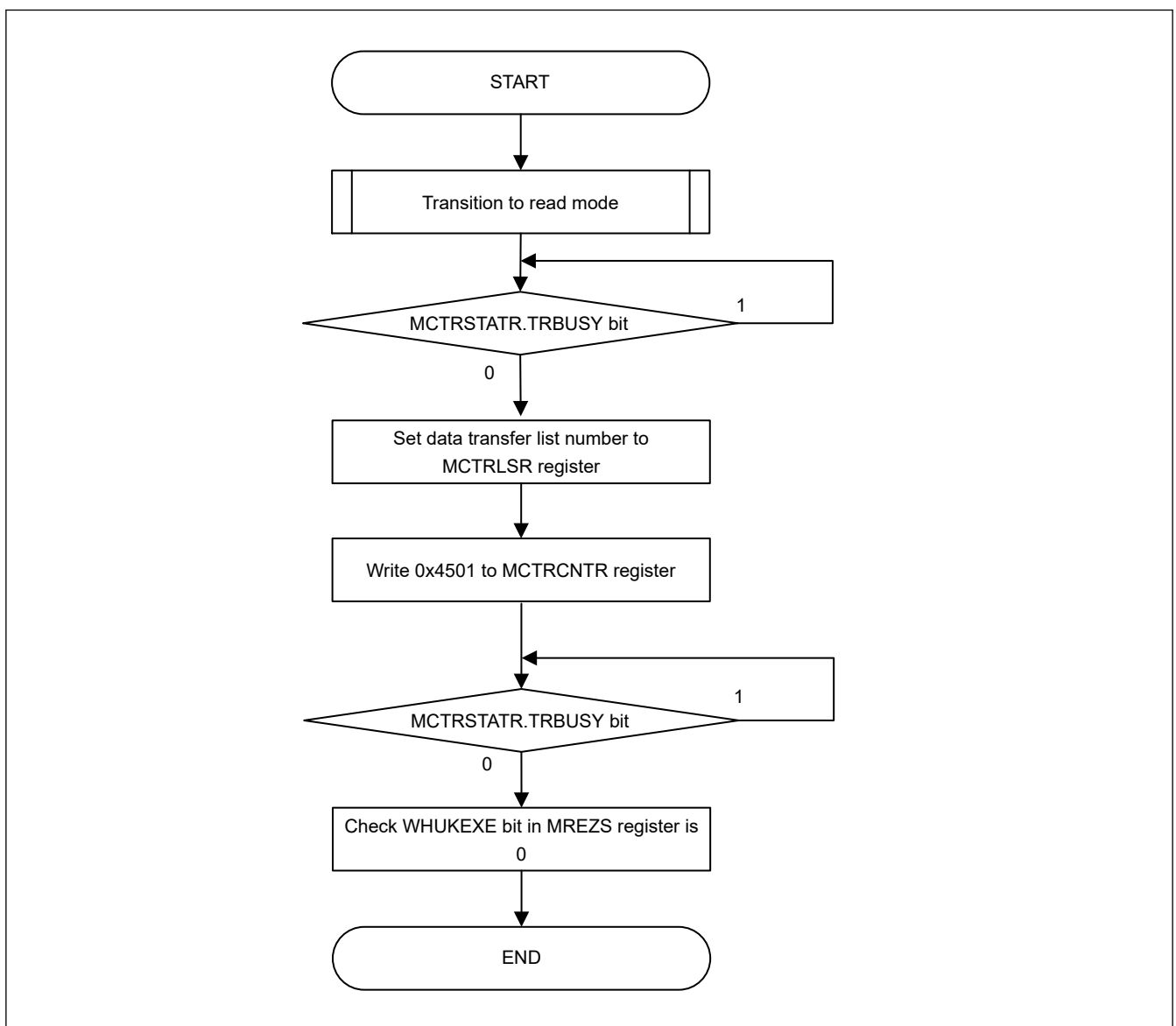


Figure 60.29 Configuration update transfer usage

60.11 Boot Mode

There are two serial programming modes:

- The boot mode (for the SCI interface) with SCI9
- The boot mode (for the USB interface) with USBFS.

Table 60.39 lists the I/O pins used in boot mode. Table 60.40 lists the available communication interface used in the boot mode.

Table 60.39 I/O pins used in boot mode

Pin name	I/O	Mode to be used	Use
MD	Input	Boot mode (for the SCI interface) Boot mode (for the USB interface)	Selection of operating mode
P208/RXD9	Input	Boot mode (for the SCI interface)	For host communication (to receive data through SCI)
P209/TXD9	Output		For host communication (to transmit data through SCI)
USB_DP, USB_DM	I/O	Boot mode (for the USB interface)	Data input/output of USB
USB_VBUS	Input		Detection of connection and disconnection of USB cables

Table 60.40 Available communication interface used in boot mode

Main clock oscillator or external clock is connected	Yes	No	No
Sub-clock oscillator is connected*1	Yes or No	Yes	No
Available interface	SCI or USB	SCI or USB	SCI
Tool connection time*2	Up to 1 second	Up to 2 seconds	Up to 3 seconds

Note 1. The drive capability of the sub-clock oscillator is set to standard by the SOMCR.SODRV bit. If you use the crystal that corresponds to the low drive capability on your board, the crystal may not oscillate in the boot mode.

Note 2. See the boot firmware manual for detail of the connection time.

60.11.1 Boot Mode (for the SCI Interface)

In boot mode (for the SCI interface), the host sends control commands and data for programming, and the MRAM is programmed accordingly. An on-chip SCI handles transfer between the host and this MCU in asynchronous mode. Tools for transmission of control commands and the data for programming must be prepared in the host.

When this MCU is activated in boot mode (for the SCI interface), the program on the dedicated area of the MCU is executed. The boot program automatically adjusts the bit rate of the SCI and controls programming/erasure by receiving control commands from the host.

Figure 60.30 shows the system configuration for operations in boot mode (for the SCI interface).

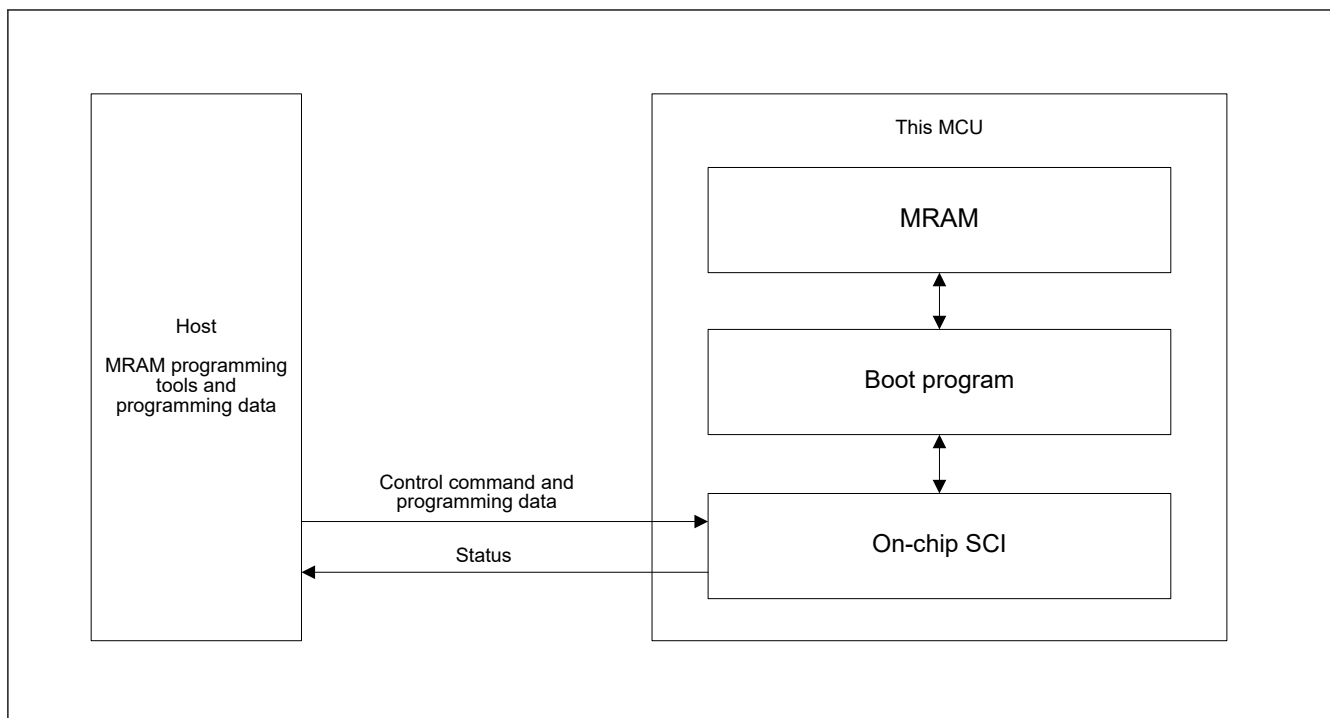


Figure 60.30 System configuration for operations in boot mode (for the SCI interface)

60.11.2 Boot Mode (for the USB Interface)

In boot mode (with the USB interface), the MRAM can be programmed by sending control commands and program data from the host. An on-chip USB is used for communications between the host and this MCU. The host requires tools for sending control commands and data for programming. [Figure 60.31](#) shows the configuration of a system for use in boot mode (for the USB interface). The USB cable must be connected on reset release.

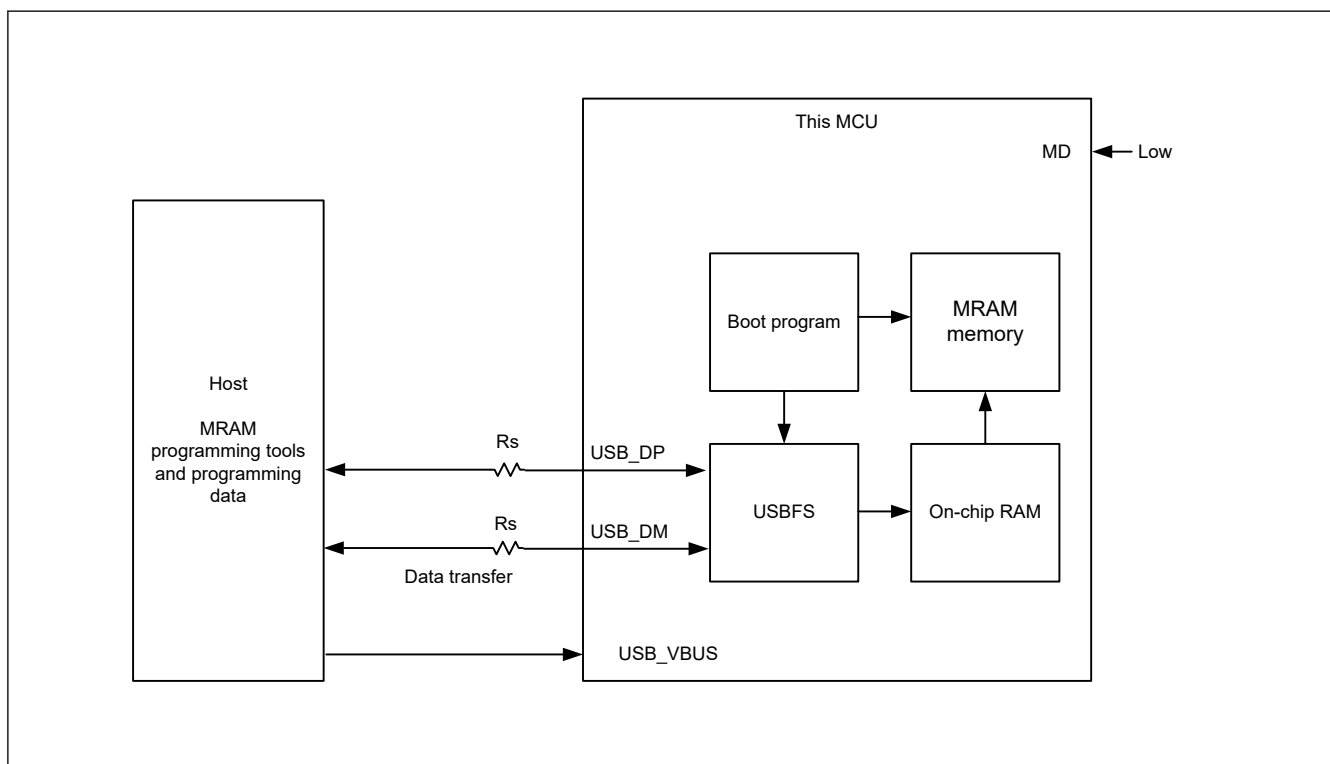


Figure 60.31 System configuration in boot mode (for the USB interface)

60.12 Using the Serial Programmer for Rewriting

A serial programmer can be used to rewrite MRAM in boot mode.

(1) Serial Programming

This MCU is mounted on the system board at the time of serial programming. Providing a connector to the board enables rewriting of this MCU by the serial programmer to proceed.

60.12.1 Environments for Serial Programming

The recommended environments for rewriting the MRAM of the MCU with data are described in [Figure 60.32](#).

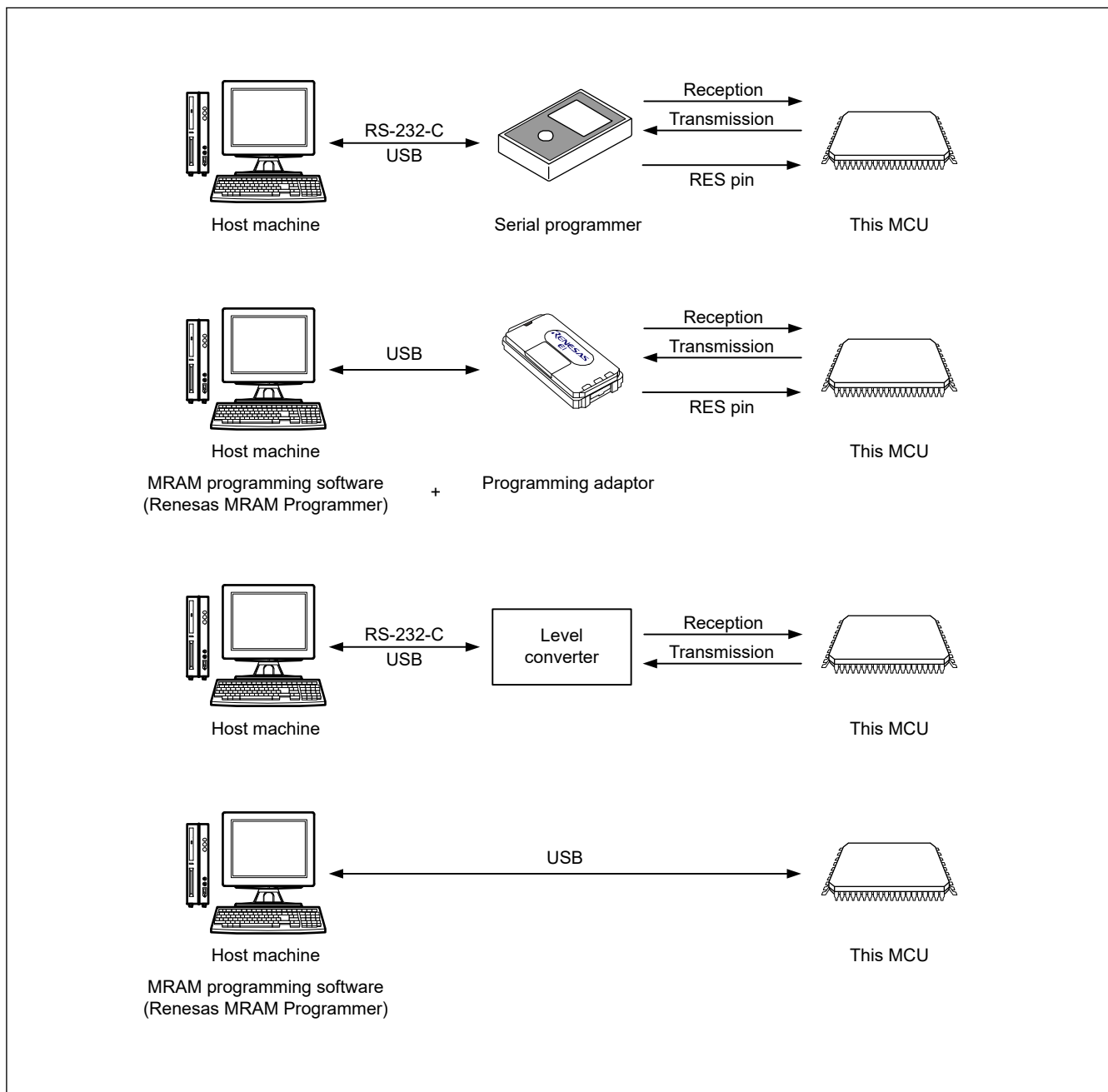


Figure 60.32 Environments for rewriting the MRAM

60.13 Programming Through Self-Programming

60.13.1 Overview

This MCU supports programming of the MRAM by the user program itself. The write page buffer or the MACI commands can be used with user programs for writing to the MRAM. This allows upgrading of user programs and rewriting of constant data fields.

The program for rewriting must be transferred to the internal RAM or external memory in advance when the BGO is not available or when rewriting the option-setting memory.

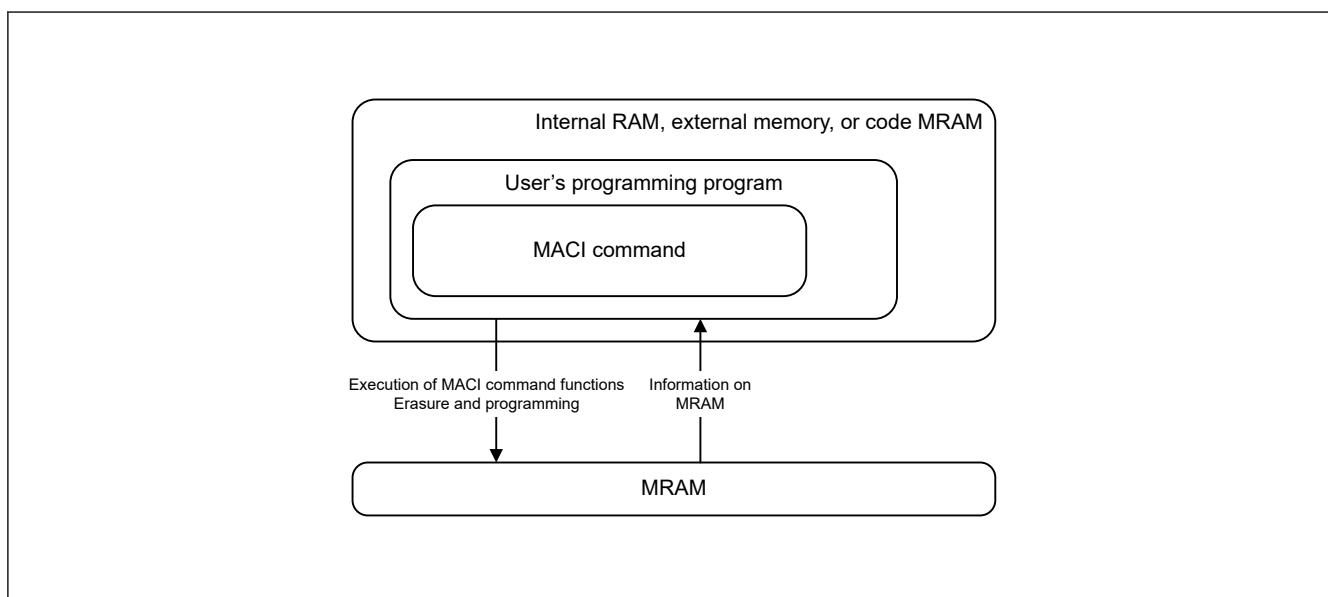


Figure 60.33 Schematic view of self-programming

For comprehensive information on the self-programming, see [section 60.7. MACI Commands](#).

60.13.2 Parallel Accessibility

Parallel read and program accesses are possible in conditions shown in [Table 60.41](#).

Read while program (Background operation: BGO) is possible between different MRAM banks. Parallel read to different MRAM banks is possible. Only one MRAM bank can be programmed at the same time. W-HUK zeroing has the highest priority among all MRAM programming requests. See [section 60.8.6. HUK Protection](#).

Programming the code MRAM and the Extra MRAM cannot be executed at the same time, so they are arbitrated. The program executed later is forced to wait until the program executed first is completed. If the program for the code MRAM is started while `MSTATR.MRDY = 0`, the program will start after `MSTATR.MRDY = 1`. And if the program for the extra MRAM (issue the Program, Configuration set, or Increment Counter command) is started while `MRCPS.PRGBSYC = 1`, the program will start after `MRCPS.PRGBSYC = 0`. Therefore, in this case, the program time is the sum of the code and the extra MRAM program time.

There is also arbitration between read and program operations for the same MRAM macro, and they cannot be executed simultaneously. The values listed in the MRAM characteristics are for independent operation time without arbitration.

Table 60.41 Parallel accessibility

		Code MRAM		Extra MRAM	
		Read	Program	Read	Program
Code MRAM	Read	X	X	O*1	O
	Program	X	X	O	X
Extra MRAM	Read	O*1	O	X	X
	Program	O	X	X	X

Note: O: Executable
X: Not executable

Note 1. Bus masters cannot read extra MRAM during configuration update transfer. See [section 60.10. Configuration Update Transfer](#).

60.13.3 Reading MRAM

Special settings are not required to read MRAM after release from the reset state. Data can simply be read out through access to addresses in the MRAM.

When reading MRAM that has not yet been programmed (that is, in the nonprogrammed state), all bits are read as 1.

60.14 Usage Notes

(1) Area Where Programming was Interrupted by W-HUK Zeroing

The data stored in the area where programming was suspended due to zeroing of the W-HUK is undefined. To prevent malfunctions due to reading undefined data, care should be taken not to fetch instructions or read data from the area where writing has been suspended.

(2) Actions Prohibited During Programming

- Do not reduce the power to less than the operating voltage.

(3) Prohibition of Additional Programming

An OTP area that uses ECC cannot be programmed twice with different data.

(4) Resets During Programming

For a reset due to the signal on the RES# pin during programming of the MRAM, wait for at least t_{RESW} (see [section 70, Electrical Characteristics](#)) of the reset input period once the operating voltage is within the range stipulated in the electrical characteristics, then release the device from the reset state.

(5) Allocation of Vectors for Interrupts and Other Exceptions During Programming

Generation of an interrupt or other exception during programming may lead to fetching of the vector from the code MRAM. The vectors cannot be read from code MRAM until the program is finished. Code MRAM requires t_{PMC} (see [section 70, Electrical Characteristics](#)) to program. Under conditions where BGO cannot be used, set the address of the vector to an address that is not in the code MRAM. Alternatively, make sure that no handling of interrupts or exceptions proceeds during programming.

(6) Synchronization of Code MRAM Program Completion and Subsequent Instruction Generation

When the CPU writes to code MRAM, there is a time lag from execution of MRAM programming by code MRAM program sequencer. For a barrier instruction that requires memory access completion, appropriate processing for synchronization is required to execute subsequent instructions. The processing for synchronization is shown as follows:

1. Code MRAM programming (for example, STR instruction).
2. Barrier instruction (for example, DMB instruction).
3. Flush the code MRAM programming data (see [Figure 60.2](#)).
4. Synchronization (wait for the flushing process to finish).
5. Subsequent instruction.

If it is necessary to avoid CPU stall in Step 4., completion of code MRAM programming should be checked by the PRGBSYC bit in the MRCPS register instead of dummy reading in Step 3..

(7) Constraints During Programming of MRAM

Frequency change, transition to software standby, and transition to deep standby are prohibited while MRAM is being programmed.

(8) Emulator Connection

Renesas provides the emulator which supports both debugging using SWD or JTAG communication and serial programming using SCI communication.

Table 60.42 shows the pinout of 10-pin or 20-pin socket pinouts when using this emulator. The pinout of SWD and JTAG is ARM standard, and MD, TXD, RXD pins are added for the serial programming using SCI communication.

Table 60.42 Pin assignment for emulator

Pin No.	SWD	JTAG	Serial programming using SCI
1	VCC	VCC	VCC
2	P210/SWDIO	P210/TMS	NC
4	P211/SWCLK	P211/TCK	P201/MD
6	P209/SWO	P209/TDO	P209/TXD9
8	P208	P208/TDI	P208/RXD9
9	GNDdetect	GNDdetect	GNDdetect
10	nRESET	nRESET	nRESET
12	P308/TCLK	P308/TCLK	NC
14	P307/TDATA[0]	P307/TDATA[0]	NC
16	P306/TDATA[1]	P306/TDATA[1]	NC
18	P305/TDATA[2]	P305/TDATA[2]	NC
20	P304/TDATA[3]	P304/TDATA[3]	NC
3, 5, 15, 17, 19	GND	GND	GND
7	NC	NC	NC
11, 13	NC	NC	NC